



AI105

Computing Fundamentals



AI105 COMPUTING FUNDAMENTALS

ABET eCOURSEBOOK

AY2021-2026

TABLE OF CONTENTS

SECTION I. COURSE INFORMATION

TAB A.....	AY2026 Course Assessment
TAB B.....	AY2025 Course Assessment
TAB C.....	AY2024 Course Assessment
TAB D.....	AY2023 Course Assessment
TAB E.....	AY2022 Course Assessment
TAB F.....	AY2021 Course Assessment

SECTION II. EXAMINATIONS

TAB G.....	WPR1
TAB H	WPR2
TAB I	WPR3
TAB J	Term End Examination

SECTION III. ASSIGNMENTS

TAB K.....	Problem Set 3
TAB L	Problem Set 4
TAB M	Problem Set 5
TAB N	Problem Set 6
TAB O	Problem Set 7
TAB P.....	Problem Set 8
TAB Q	Problem Set 9



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION I

COURSE INFORMATION



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB A

AY2026 COURSE ASSESSMENT (EXSUM)



REPLY TO
ATTENTION OF

DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
ELECTRICAL ENGINEERING AND COMPUTER SCIENCE
WEST POINT, NY 10996

MADN-EEC-C

19 May 2026

MEMORANDUM THRU

Core Program Director, Department of Electrical Engineering & Computer Science,
United States Military Academy, West Point, New York 10996
Professor and Head, Department of Electrical Engineering & Computer Science, United
States Military Academy, West Point, New York 10996

FOR Dean of the Academic Board, USMA, West Point, NY 10996

SUBJECT: Annual Executive Summary for AI105, Computing Fundamentals, AY26

1. AI105 met its learning outcomes in AY2026 as measured through a formal assessment process. The EECS Core Program performs an annual assessment of all student learning outcomes (SLOs) and their mappings to the higher-level academy outcomes to continually monitor and improve AI105. This year, additional mappings were performed against the AI Task Force's KSIs to establish baselines for future revisions.
2. The attached report summarizes the annual assessment and curriculum change actions for AI105 (Computing Fundamentals).
3. POC for this is Dr. Suzanne J. Matthews, suzanne.matthews@westpoint.edu, x5577.

Encl

Suzanne J. Matthews, Ph.D.
Professor of Computer Science
AI105 Course Director

Enclosure: Executive Summary on CY105 Annual Assessment for AY25

1. DEPARTMENT-LEVEL ASSESSMENT

1.1) AI105 Student Learning Outcomes (SLOs):

SLO1: Examine the legal, ethical, professional, and security issues associated with the cyber domain.

SLO2: Function as an ethical, secure, and competent user in West Point's immersive cyber environment.

SLO3: Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process.

SLO4: Describe how data flows from data sources across air, land, sea and cyberspace to form a network of information that is constantly being tracked and exploited by allies, adversaries, and third parties.

SLO5: Describe and explore the opportunities and challenges presented by artificial intelligence in the modern world.

SLO6: Practice teaming with AI in various contexts, while maintaining a critical and privacy-conscious mindset.

1.2) Summary of Previous AY Assessment:

AY26 is the first AY that the Computing Fundamentals course is AI105; prior to this year, the course was CY105. The AY25 assessment concluded that CY105 met the standard with respect to both its learning outcomes and What Graduates Can Do (WCGD) statements. Table 1 summarizes student achievement of learning objectives in prior years. Please note that while the AY21, AY22, and AY24-2 numbers below only included the TEE, AY23 included all the WPRs, because students performed comparatively better on the WPRs. We do not include SLO 4 or SLO 5 in previous years in the table below, because they were revised in AY24-2 to fit the updated focus of the course. SLO 6 is not included, as it was introduced in AY26.

Table 1. Percentage of cadets meeting the standard with respect to learning objectives.

% of Cadets Meeting Standard on TEE					
LO	AY21	AY22	AY23*	AY24-2	AY25
1	91.9%	80.9%	87.4%	98%	98.2%
2	90.7%	80.9%	82.8%	98.5%	98.2%
3 R/W	88.0%	81.7%	92.7%	81.5%	85.6%/56.4%
4	-	-	-	96%	94.6%
5	-	-	-	87.4%	94.8%

1.3) Status of Previous AY Recommendations:

The AY25 EXSUM made several recommendations, several of which we chose to adopt. To prepare for the launch of AI105 in AY26-1, we radically increased the amount of AI material in the course. The recommendations and the status of their adoptions are summarized in Table 3 below.

Table 2. AY25 Recommendation and Adoption Status

AY25 Recommendation	Status of Adoption
Encourage students to use AI as a partner, not a shortcut.	(Complete) As part of the launch of AI105, I developed detailed studying with GenAI and citing GenAI documents (Enclosure 1) that gave explicit guidance on how to leverage GenAI tools to help scaffold learning.
Increase infusion of Generative AI in curriculum.	(Complete) The course launched as AI105 has 33% of all the lessons touching on AI in some matter, and approximately 60% of graded events.
Increase number of (GenAI) labs.	(Complete) Six GenAI labs were piloted successfully in AY26. These include the Microsoft365 GenAI Lab, the GenAI Create-A-Game Programming Lab, the GenAI Wireshark Lab, the GenAI Web Tracking Lab, and the GenAI Data Analysis Lab. We also included a new GenAI Reverse Engineering component to the Robotics Lab.
Have students build their own GenAI Tutor.	(Complete) A in-class ICE was created that walked students through using the Agent Building feature in Microsoft365 Copilot Chat to build their own assistant to aid them in the programming unit of the course. The instructors (Enclosure 2) are widely adaptable to almost any course; these instructions were shared widely with the faculty in a CEP TIP session in April 2026.
Work with Canvas POCs to Identify Efficiencies in Assessment	(Partially Completed) In AY25, it took 8 hours to pull and clean Canvas data for assessment. For our XOs, setting up the assessment pieces in Canvas remains an arduous task. Hayden add more here.

Commented [SM1]: Hayden, please add more.

1.4) Additional Changes and Improvements:

In addition to the changes outlined above, I spent considerable time updating the textbook and available resources to ensure students had high quality material on which to learn from the AI transformation. Furthermore, I worked with a student to transition the contents of the textbook out of zyBooks and into Runestone, an open-source interactive textbook framework. This book is now hosted internally on the WREN: <https://code.westpoint.edu/ns/books/published/ai105text/>. In addition to being expensive, students lose access to the Runestone text approximately two weeks after the course ends, which means they are no longer able to refer to the material throughout the remainder of their careers at West Point. Transitioning the book to the WREN reduces student cost down to zero, and ensure they have long-term access to the material.

In addition to the changes to the textbook, I spent considerable time creating lesson summary videos using NotebookLM and a podcast based on the book readings, to provide additional modalities for students to learn from. It took considerable time, because, while NotebookLM would create a 95% perfect solution, their were subtle errors in the generated content that would have to be edited out. In the beginning, I would try regenerating the videos and podcasts, but this was time consuming and produced inconsistent results. I then transitioned to manually editing episodes and videos using ClipChamp to excise out the incorrect pieces and stitch together the correct pieces.

Lastly, I instituted daily reading checks, mimicking practices in CY305 and PL100. These were unpopular, both among students and faculty, but judged grimly by the faculty to be a necessity. A key frustration that was consistently expressed by faculty teaching the course is that students would do better, if they just did the readings. The reading checks forced students to be consistently accountable for the readings. Reading check points were created through the elimination of instructor points.

Overall, the transformation to AI105 was a success, with faculty teaching the course for the first time enjoying teaching it, and faculty who taught CY105 reporting that they felt it a significant improvement over the previous course. Table 3 below summarizes course end statistics for CY105 prior to the revision in AY24-2, while Table 4 summarizes the course end statistics for CY105 in 24-2 through 25-2. Table 5 shows the AI105 course-end statistics for comparison. Despite the uptick in AI content and reduced coverage in programming content, the course averages remained remarkably stable, with few Ds and Fs. This suggests that the revision was largely a success.

Table 3. CY105 TEE and Course-End Grades pre-revision (AY23-1, AY23-2, and AY24-1)

Semester	23-1	23-2	24-1
Total Cadets		471	526
TEE Average	81.68% (B-)	79.58% (C+)	79.47 (C+)
#Fs, #Ds	51 Fs, 24 Ds	62 Fs, 37 Ds	88 Fs, 28 Ds

Final Grade	86.77% (B)	86.67% (B)	86.99 (B)
#Fs, #Ds	3Fs, 7 Ds	3 Fs, 3 Ds	5 Fs, 18 Ds
Final QPA	3.18	3.20	3.22

Table 4. CY105 TEE and Course-End Grades post-revision (AY24-2, AY25-1, and AY25-2)

Semester	24-2	25-1	25-2
Total Cadets	544	574	556
TEE Average	86.74 (B+)	83.16 (B)	84.49 (B)
#Fs, #Ds	2 Fs, 10 Ds	15 Fs, 43 Ds	6 Fs, 21 Ds
Final Grade	88.86 (B+)	88.16 (B+)	87.80 (B+)
#Fs, #Ds	2 Fs, 0 Ds	1 F, 2 Ds	0 Fs, 1 D
Final QPA	3.40	3.42	3.36

Table 5. AI105 TEE and Course-End Grades

Semester	26-1	26-2
Total Cadets	526	388
TEE Average	85.34 (B)	83.78 (B)
#Fs, #Ds	8 Fs, 22 Ds	6 Fs, 20 Ds
Final Grade	86.83 (B+)	86.95 (B)
#Fs, #Ds	2 Fs, 3 Ds	0 Fs, 2 Ds
Final QPA	3.25	3.26

1.5) Current Year Assessment:

Prior to 24-2, the assessment of SLOs was rotated annually. The adoption of Canvas as the LMS of record led to the recommendation in 24-1 that CY105 assess all its SLOs annually. Since then, we have tried to leverage the full capabilities of Canvas to create a detailed assessment. Canvas allows us to align particular questions in an exam (or components of an assignment) to individual outcomes via a rubric interface. Canvas then generates an Outcomes Analysis by student (Outcome mastery based on all alignments) and even by exam (WPR 2 and TEE, for example). This year, with the launch of AI105, we also assessed the course against the West Point AI Task Force's Knowledge, Skills and Attitudes (KSAs).

The evidence used to assess each SLO is listed in Table 6:

1.5.a) Evidence Table (SLOs):

Table 6. Summary of Evidence (SLOs)

Course Outcome	Direct Evidence	Indirect Evidence	Total # of Indicators
-----------------------	------------------------	--------------------------	------------------------------

SLO 1: Examine legal, ethical, professional, and security issues associated with the cyber domain.	Digital Footprint Audit (data access limitations, data vulnerabilities, security clearance risks, adjusting social media habits) WPR 3 Questions (Cybersecurity, Cyber Hygiene, Online Tracking and Ethics, Web Tracking Lab, Open-Ended Problem [Attack and Actor Threat Goals]) AI Final Project (ethical issues related to AI/copyright, potential misuse by malicious actors) TEE Questions (Identifying Malware, Cyberspace Layers, Instagram photo sharing, AI Use, Online Tracking, Copyright Law (2), Robotics, Public Key Infrastructure, Cyber Hygiene, Internet, Data and Ethics), Free Response Question on Ethical Use of Robots	Course-End Feedback Survey	4
SLO 2 Function as an ethical, secure, and competent user in West Point's immersive cyber environment.	WPR 3 Questions (Robotics [3], Cybersecurity [2], Cyber Hygiene, Online Tracking and Ethics) Digital Footprint Audit (data sources, data vulnerabilities, character, adjusting social media habits) TEE Question (Identifying Malware, Instagram photo sharing, Online Tracking, Copyright Law (2), Microsoft 365 (2), Public Key	Course-End Feedback Survey	3

	Infrastructure, Cyber Hygiene, I/O Devices, Computer Components), Free Response Question on Data and Ethics, Free Response Question on Ethical Use of Robots		
SLO 3 Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process.	WPR 3 Questions (Robotics Code Tracing [2], Modules [2], Boolean Expressions) TEE Questions (Code Tracing (11), Python Basics (3), Algorithms (2), Conditional Statements) TEE Code Writing Questions	Course-End Feedback Survey	3
SLO 4 Describe how data flows from data sources across air, land, sea and cyberspace to form a network of information that is constantly being tracked and exploited by allies, adversaries, and third parties.	Digital Footprint Audit (data sources, surprises in data, data access limitations, data vulnerabilities) WPR 3 Questions (Computers and Data, Internet and the Web, Online Tracking and Ethics, Cyberspace and Threat Actors, Robotics, Open-Ended Problem [Cyber Attacks & Threat Actor Goals]) TEE Questions (Internet/Networks (3), I/O Devices, Data Storage, Data Sources and Representation (2)),	Course-End Feedback Survey	3
SLO 5 Describe and explore the opportunities and challenges presented by	AI Final Project (AI's role in project, AI's societal benefits/risks, AI's future development) TEE Questions (AI/ML/Deep Learning,	Course-End Feedback Survey	2

artificial intelligence in the modern world.	ANI/ASI/AGI, AI Uses, Ethical Use of AI, Social Media AI training)		
--	--	--	--

Table 6. Summary of Evidence (KSAs).

Course Outcome	Direct Evidence	Indirect Evidence	Total # of Indicators
Knowledge 1: Identify various types of AI along with their capabilities and use cases	AIFP (AI Use) WPS 2 Questions (LLMs for CUI processing, LLM Use at West Point		2
Knowledge 2 Identify and understand the implications of potential risks for AI-generated content from factors such as bias, adversarial manipulation, misinformation, and hallucination.	AI Final Project (Use and Abuse of GenAI) WPS 2 (Data Literacy, Evaluating Content Accuracy)		2
Knowledge 3 Explain key considerations for the trustworthy use of AI. (AI is a tool but not always an authority on information.)	WPS 2 (Data Literacy, Evaluating Content Accuracy, LLM use at West Point)		1
Knowledge 4 Explain the legal, ethical, privacy, and security considerations of digital actions	AIFP (AI Tool Copyright Infringement, Is it Copyrightable) WPS 2 (US Copyright, AI & Privacy, LLMs for CUI Processing)		2

including handling sensitive or controlled information.			
Skills 2 Select appropriate digital tools, including AI, and justify their use based on task requirements, context, risks, and intended outcomes.	AI Final Project (AI Contributions, Expectation vs Reality, AI Use)		1
Skills 3 Team with AI-based tools to effectively solve problems.	WPS 5 (Part B), WPS 5 (Part B Grade) AIFP (Final Project Grade)		3
Skills 4 Apply techniques to improve AI performance, such as problem decomposition, contextualization, progressive refinement, and effective prompting strategies.	WPS 3 (Part B Grade), WPS 3 (Part B Rubric), AIFP (Prompt Engineering, Cadet Contribution)		3
Skills 5 Critically assess the veracity of AI generated output.	AIFP (AI Use),		1
Skill 6 Disclose and appropriately document the use of AI tools in the creation of work.	WPS 5 (Part B), AIFP (Progress Photos), WPS 3, WPS 5		4

Attitude 1 Be open to collaborating with AI while maintaining a critical mindset	Course End Survey Q: "I feel aware how to use AI effectively and securely, while maintaining a privacy focused mindset" Q: "I feel more open to collaborate with AI"		2
Attitude 2 Posses Optimism that teaming with AI can enable faster and further progress.	Course End Survey Q: "I am optimistic that teaming with AI can enable faster and further progress"		1

1.5.b) **Rubric Definition:** To facilitate a quantitative assessment, each individual data point of evidence from each embedded indicator was classified into one of the five below categories. For AY24, no unique rubrics were used, so the following rubric was uniformly applied to all SLOs and indicators.

1	2	3	4	5
Failed to meet the standard	Marginally met the standard	Met the standard	Slightly above standard	Greatly exceeded standard
0 – 64.9%	65 – 72.9%	73 - 79%	80 - 89%	90 - 100%

Pre-Observation identification of the overall acceptable standard. At least 80% of the students meet the standard. No more than 15% can completely fail to meet the standard.

1.5.c) **Quantitative Assessment of SLOs:** Figures 1-6 below summarize the data from the rubric scoring of the embedded indicators for each of the SLOs.

SLO1: Examine legal, ethical, professional, and security issues associated with the cyber domain. On the DFA, 98.3% of the students met or exceeded the standard. On WPR 3, 90.8% of the students met the standard or higher. On the AI Final Project, 97.3% of the students met or exceeded the standard. On the TEE, 98.8% of the students met the standard or higher. Based on these indicators, AI105 is greatly exceeded the standard on this outcome with strong performance on all indicators.

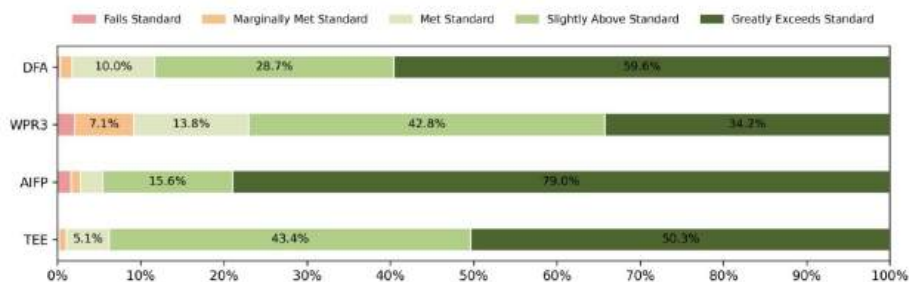


Figure 1. Summary of Assessment for SLO #1

SLO2: Function as an ethical, secure, and competent user in West Point's immersive cyber environment. On the DFA, 98.3% of the students met or exceeded the standard. On WPR3, 98.7% of the students met or exceeded the standard. On the TEE, 99.1% of the students met the standard or higher. Based on these indicators, AI105 greatly exceeded the standard on this outcome with very strong performance on all three indicators.

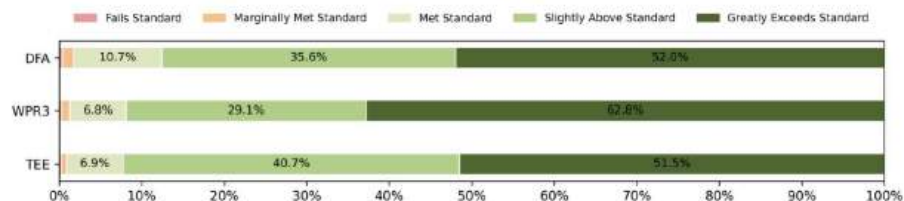


Figure 2. Summary of Assessment for SLO #2

SLO3: Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process. On WPR3 code tracing, only 66.2% of the students met the standard or higher, with 16.7% completely failing to meet the standard, a significant drop from last year. Similarly, on WPR 3 code writing, only 65.2% of students met the standard or higher, with 24.7% completely failing to meet the standard. However, there is much stronger performance on the TEE for code tracing. On the TEE code tracing component, 92.8% of the students met the standard or higher. However, on the code writing component of the TEE, only 57.1% met or exceeded the standard, with 30.7% completely failing to meet the standard. Based on these indicators, AI105 marginally met this outcome on code tracing, but did not meet this outcome on code writing.

Commented [SM2]: Check this. Is WPR 3 really correct?



Figure 3. Summary of Assessment for SLO #3

SLO4: Describe how data flows from data sources across air, land, sea and cyberspace to form a network of information that is constantly being tracked and exploited by allies, adversaries, and third parties. On the DFA, 98.1% of the students met or exceeded the standard. On WPR3, 63.2% of the students met the standard or higher, with 11.6% completely failing the standard. On the TEE, 94.4% of the students met the standard or higher. Based on these indicators, CY105 met this outcome with strong performance on two indicators, but with weak performance on one indicator.



Figure 4. Summary of Assessment for SLO #4

SLO5: Describe and explore the opportunities and challenges presented by artificial intelligence in the modern world. On the AI Final Project, 98.7% of the students met or exceeded the standard. On the TEE, 96.1% of the students met or exceeded the standard. Based on these indicators, CY105 greatly exceeded this outcome with strong performance across both indicators.

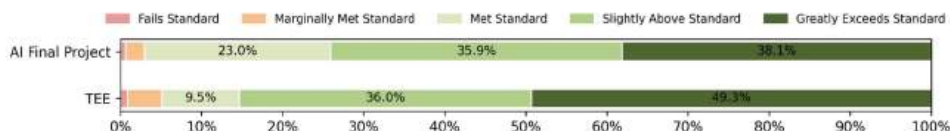


Figure 5. Summary of Assessment for SLO #5

SLO6: Describe and explore the opportunities and challenges presented by artificial intelligence in the modern world. On WPS3, 91.2% of the students met or exceeded the standard. On WPS 5, 90.9% of the students met or exceeded the standard. On the Robotics Lab GenAI activity, 86.8% of the students met or exceeded the standard. Lastly, on the AIFP, 97.6% of the students met or exceeded the standard. Based on these indicators, CY105 exceeded this outcome with very strong performance across three indicators, and strong performance on one indicator.

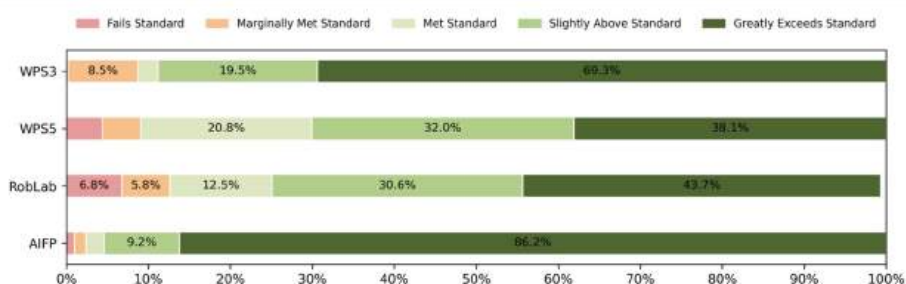


Figure 6. Summary of Assessment for SLO #6

1.5.d) Qualitative Analysis of SLOs: In AY26, we asked students to assess how well they felt they met the course objectives. We use this as another point of comparison for how well AI105 met the course objectives in AY26. Figure 6 below summarizes student responses on the course end surveys for AY26 ($n=650$). In general, the vast majority of students agreed ($\geq 89\%$) that they have met standards for SLO 1, 2, 4, 5 and 6. However, only 68.6% of the students agreed that they “can effectively use Python and algorithmic thinking to solve problems” (SLO 3). We note that this is *higher* than AY25, even though cadet performance on code writing and reading tasks in AY26 are *lower* than AY25. This discrepancy is notable, because there is literature that suggests that AI tools falsely increase student understanding of their knowledge.

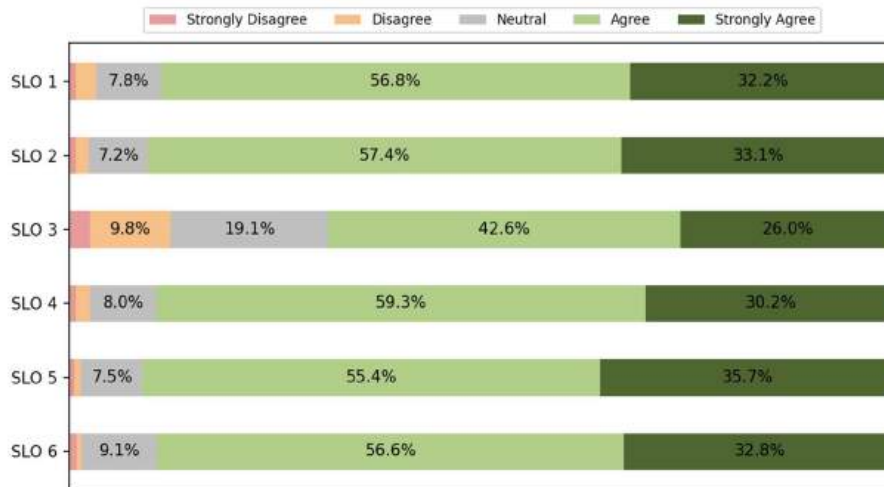


Figure 7. Student agreement on if they met different course objectives (course end survey).

Table 6 lists question text associated with the individual SLOs in the above survey. The “% Agreement” column is the total percentage of students that indicated either “Agree” or “Strongly Agree” on the survey above:

Table 7. Course-end survey questions mapped to individual SLOs

SLO	Question Text	% Agreement
1.	My understanding of the legal, ethical, and professional issues in the cyber domain has increased	89%
2.	I am more aware of the ethical principles and security best practices to employ when navigating cyberspace	90.5%
3.	I can effectively use Python and algorithmic thinking to solve problems.	68.6%
4.	I am more aware of how data and networks can be exploited by various parties.	89.6%
5.	I better understand the opportunities and challenges presented by AI today	91.1%
6.	I feel aware how to use AI effectively and securely, while maintaining a privacy-focused mindset	89.4%

1.5.e) Assessment Analysis of AI Task Force KSAs:

In recognition of the transition of the course from CY105 to AI105, we analyzed AI105 against the Knowledge Skills and Attitudes (KSAs) established by the AI Task Force. Figures 8-X summarize the data from the rubric scoring of the knowledge, skills and attitudes:

Knowledge: AI105's performance on the knowledge identifiers are as follows:

- (1) Identify various types of AI along with their capabilities and use cases: 99.2% of the students met or exceeded the standard.
- (2) Identify and understand the implications of potential risks for AI-generated content from factors such as bias, adversarial manipulation, misinformation, and hallucination: 98.2% met or exceeded the standard.
- (3) Explain key considerations for the trustworthy use of AI. (AI is a tool but not always an authority on information.): 95.3% of the students met or exceeded the standard.
- (4) Explain the legal, ethical, privacy, and security considerations of digital actions including handling sensitive or controlled information: 93.9% of the students met or exceeded the standard.

Based on these indicators, AI105 is greatly exceeding the standard on all AI Knowledge identifiers.

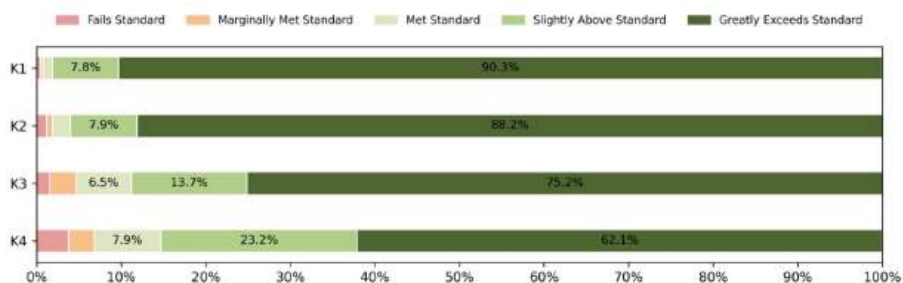


Figure 6. Summary of Assessment for Knowledge identifiers.

Skills: AI105's performance on the Skills identifiers are as follows. Please note that I (the course director) could not figure out how to directly assess Skill 1 given the graded events in the course. As a result, it is omitted from analysis:

- (2) Select appropriate digital tools, including AI, and justify their use based on task requirements, context, risks, and intended outcomes: 98.9% of students met or exceeded the standard.
- (3) Team with AI-based tools to effectively solve problems. 80.2% of students met or exceeded the standard.
- (4) Apply techniques to improve AI performance, such as problem decomposition, contextualization, progressive refinement, and effective prompting strategies: 72.0% of students met or exceeded the standard, with 15.8% failing to meet the standard.
- (5) Critically assess the veracity of AI generated output: 99.2% of the students met or exceeded the standard.
- (6) Disclose and appropriately document the use of AI tools in the creation of work: 72.3% of the students met or exceeded the standard, with 21.6% completely failing to meet the standard.

Based on these indicators, AI105 is greatly exceeding the standard on all AI Knowledge identifiers.

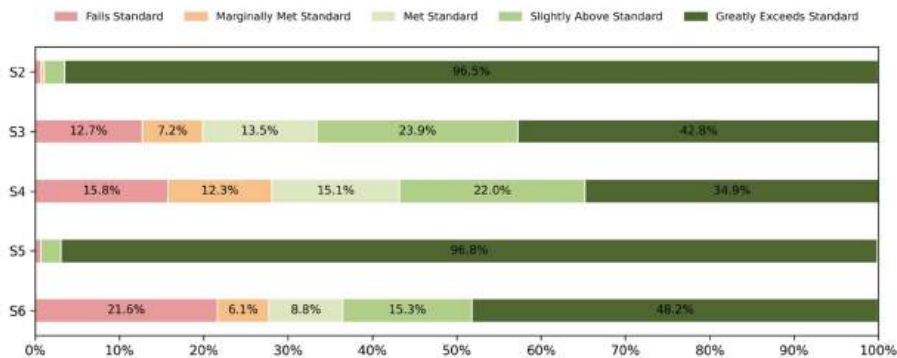


Figure 6. Summary of Assessment for Skill identifiers.

Attitudes: AI105's performance on the Attitudes identifiers were derived from course end survey questions, not assignments. We used three questions on the course end survey to capture cadet attitudes with regards to GenAI. Below are the questions and the percentage of students who either agreed or strongly agreed.

- (1) Be open to collaborating with AI while maintaining a critical mindset
 - (Attitude 1a) "I feel aware how to use AI effectively and securely, while maintaining a privacy-focused mindset": 89.4%
 - (Attitude 1b) "I feel more open to collaborate with AI": 79.2%
- (2) Posses Optimism that teaming with AI can enable faster and further progress.
 - (Attitude 2) "I am optimistic that teaming with AI can enable faster and further progress": 88.7%

Summary of results: Based on these results, AI105 met the standard for both Attitude 1 and Attitude 2.



Figure 6. Summary of Assessment for Attitude identifiers.

1.5.f) Assessment Analysis:

Commented [SM3]: Need Maria's help, I think.

a. Areas for Sustainment:

Based on the evidence, we believe the following should be sustained for AI105 in AY 2027:

- **Exposure to AI topics positive.** The opportunity to learn about AI in an academically sanctioned environment was immensely popular and motivating to students, and our students reported increased confidence in using AI tools. Our labs were likewise positive and positively rated by students and faculty.
- **Use of Copilot generally positive.** It is a good idea to ensure students learn how to use genAI in a data-aware mindset. Microsoft 365 Copilot chat has data protections that other third-party LLMs do not currently have. Especially as Microsoft365 Copilot's capabilities continue to increase, it is important that students get into the habit of using it over other LLMs. Establishing this habit is possible only if it is introduced repeatedly in AI105 and then again in follow-on classes.
- **Learning to code is important.** AI105 is going to undergo revision as it merges with AI105x in AY27. In order to teach students to be equal partners with AI, it is my belief as the outgoing course director that either students be taught to code before using AI to generate it themselves, or that coding is not covered at all in AI105. Otherwise, it gives the false impression that professions who deal with code do not actually need to understand it. Already the warnings about "vibe slop" are starting to sound¹, and West Point needs to decide how best to navigate these waters.

Commented [LC4]: Use these sub-bullets to discuss the assessment data, including any qualitative analysis you may have. List any actions you plan to take based on this assessment to improve the course.

b. Areas for Improvement:

Despite the successes in the revision, there are still more improvements that can be made. Specifically:

- **Encourage students to actually practice and read book outside of class.** Students who read the book and practice outside of class tend to do well in this course. Students who do not read the book and practice do poorly. We are going to explore ways to keep students more accountable for doing the requisite prep work for the course by incorporating a class participation grade (most likely enforced by daily short quizzes). While SLO 3 performance suffered, we believe this is mainly due to the elimination of note sheets (students got one double-sided sheet per exam, and three sheets for the TEE in 24-2 vs no note sheets in AY25), rather than students learning worse than they did in

¹ [Vibe-Coded AI 'Slop' Is Flooding the World With Potentially Dangerous Code—'Eventually It Will Catch Up to Us' - WSJ](#)

previous semesters.

- **Encourage students to use AI as a partner, not a shortcut.** There still exists an incongruity in homework grades and code writing grades, suggesting that many students are getting too much external help and/or use ChatGPT to avoid opportunities to practice. In AY2026, we plan on continuing to explore ways to teach students to how to use AI *help them* learn material, rather than as a tool to circumvent learning. The USMA Duck that was piloted was a good steppingstone toward this, but more work needs to be done to make the duck available to all sections.

c. Any additional comments, analysis, and insights:

- **We are still seeing the effects of COVID based on cadet study skills.** I wrote about this in last year's EXSUM. In support of articles written in the Chronicle of Higher Education² and other sources³, there is increasing evidence that cadets do not have the attention span to complete readings or summarize material on their own. I have heard directly from students that they use ChatGPT to summarize long readings in other courses. It remains an open question on how to continue to challenge these expectations and perspectives.
- **Misuse of ChatGPT and other AI tools is hurting critical thinking across the board.** A recent report⁴ highlighted that independent studies suggest that students are increasingly using AI to offload important critical thinking processes. Many students are coming to West Point with some prior knowledge of AI. The challenge now is quickly becoming how to get students to *unlearn* unhelpful practices with AI and convince them that critical thinking is actually useful.

d. Significant planned or implemented curricular changes.

Based on our assessment data, feedback from instructors and experiential data, the following curricular changes are planned for AY 2026:

- **Increase the infusion of generative AI in CY105 curriculum.** CY105's primary goal with respect to generative AI is to ensure students learn how to team with generative AI without it replacing their learning. Given our success with the AI final project and some inklings of success with the duck, we plan to increase the level of exposure students have to generative AI in AY26. The updated version of CY105 have students exposed to generative AI concepts in every unit, with 33% of the content touching on generative AI in some manner. Likewise, 66% of our out-of-class assignments now touch upon generative AI.

² [Students Are Less Able and Less Willing to Read. Professors Are Stymied. \(chronicle.com\)](https://www.chronicle.com/article/students-are-less-able-and-less-willing-to-read-professors-are-stymied)

³ [The Pandemic's Babies and Toddlers Are Struggling in School - The New York Times \(nytimes.com\)](https://www.nytimes.com/2020/09/01/us/politics/coronavirus-schools-struggling.html)

⁴ [University students offload critical thinking, other hard work to AI](https://www.technologyreview.com/2020/09/01/1005854/university-students-offload-critical-thinking-other-hard-work-to-ai/)

- **Increase number of labs.** For AY 2026, we plan on replacing the existing exam review periods and the copyright lecture with in-class laboratory sessions. Based on the assessment results above, SLO 3 (code writing) is where our students struggle the most. By deliberately creating more opportunities for them to practice their skills, we hope to see improved performance on this SLO in the upcoming year. Our labs are also opportunities for students to get exposure to generative AI. In AY26, there are three novel labs related to generative AI: (1) leverage Microsoft 365 copilot for a number of every tasks; (2) using generative AI to create a website and learn about tracking elements; and (3) using generative AI to illustrate how malicious parties can use LLMs to assist them analyze unencrypted network traffic.
- **Have students build their own AI Virtual Tutor.** In AY2025, CY105 piloted the USMA Duck, a custom AI virtual assistant that is trained on our course materials. While the duck certainly met many of our intents, it started experiencing latency and model-switching issues when we tried to scale the number of sections up. In AY26, students will use Microsoft 365 Copilot Chat's "Build an Agent" feature to build their own custom Python Tutor agent that will act as an allowed tutor in the course. Having students build their own tutor achieves two purposes: 1) it solves our latency and model switching issues, and 2) it demystifies AI and how agents work.

e. Significant planned or implemented assessment process changes.

The following assessment changes will be made in AY 2025:

- **Work with Canvas POCs to identify efficiencies in assessment.**
We completely depended on Canvas for data for this assessment cycle and the last year's assessment cycle. While we made considerable effort to improve our ability to use Canvas in a meaningful manner to collect assessment data, it is still difficult for us to pull and analyze data from Canvas. The pulling and cleaning of Canvas data can take up to 8 hours to do. We are meeting with the Canvas POCs at West Point to see if additional efficiencies can be identified, especially as far as the ability to export SLO outcomes with Canvas exams.

2. HIGHER-LEVEL ASSESSMENT (ACADEMIC PROGRAM)

2.1) AI105 supports WGCD outcomes 5.1, 5.4, and 5.5. As of AY24, WGCD 4.2 was also assessed, given the curriculum of CY105 supported (and AI105 continues to support) this outcome. After further assessment, we may offer to the Curriculum and Assessment Committee that this supporting outcome be formally added as a contribution of AI105 to the Academic Program.

2.2) Course Outcomes Mapped to WGCD Statements: The table below shows the mapping of AI105 course outcomes to the assessed WGCD supporting objectives.

			APG4. Ethical Reasoning	APG5. STEM			
			WGCD 4.2	WGCD 5.1	WGCD 5.4	WGCD 5.5	
			Recognize ethical components of problems and situations	Apply mathematics, science, and computing to model devices, processes, or behaviors	Apply an engineering design process to create effective and adaptable solutions. [3CES]	Explain and apply computing and information technology concepts and practices in the context of the Cyber Domain. [IT/CYBER]	
Course Outcomes							
CY105	1	Examine legal, ethical, professional, and security issues associated with the cyber domain.	X			X	
	2	Function as an ethical, secure, and competent user in West Point's immersive cyber environment.	X			X	
	3	Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process.		X	X	X	
	4	Describe how data flows from data sources across air, land, sea and cyberspace to form a network of information that is constantly being tracked and exploited by allies, adversaries, and third parties.		X		X	
	5	Describe and explore the opportunities and challenges presented by artificial intelligence in the modern world.	X		x	X	

Commented [LCJ5]: Please confirm the mappings below for your course. This was the draft used during the assessment kickoff. To keep it simple, please only use X's to indicate an assessed connection, rather than symbols to show strength of connection.

2.3) WGCD Outcome Assessment. The WGCD outcomes were assessed using a combination of the SLO assessments from Part 1 and the mapping chart above.

WGCD	Assessment Grade	Degree of Confidence
4.2	"Greatly exceeded the standard"	High
5.1	"Marginally met the standard"	Moderate
5.4	"Met the standard"	Moderate
5.5	"Slightly exceeded the standard"	High

Commented [LC6]: The precise means of your assessment can be abstracted in the EXSUM, but you are free to combine the SLO scores in a way that makes sense to you (e.g., maybe a weighted average makes the most sense based on one SLO being more closely tied to a WGCD outcome than another). You can also use more qualitative (indirect evidence) to determine your assessment grade.

2.3.a.) Strengths: AI105 represents one of the first introductions to 4.2, 5.1, 5.4 and 5.5 at West Point. We are especially strong at supporting 4.2 and 5.5, especially ensuring that students are aware of modern technology in the context of the cyber domain. Compared to AY25, we have seen improvements in cadet attainment of our WGCD statements.

2.3.b.) Areas for improvement: Based on the indicators we collected, we believe we met the standard for 5.1, albeit marginally. While we met the standard for 5.4, performance on SLO 3 limited our ability to exceed the standard. Working on strategies to increase the performance of students in SLO 3 will be a prime focus of AY 2026.

2.3.c) Additional Elements:

Support to Character Thread.

- (1) The Digital Footprint Audit (DFA) is the primary way that AI105 supports West Point's character thread. Most plebes are unaware that commercial third parties actively track and store their data. In AI105, students learn that every action they take online generates a *digital footprint* that projects a particular online persona. Students are required to request data from three social media sources, and study and reflect on whether the online persona projected by their data actually represents their own personal character values. This is an eye-opening assignment for our cadets that highlights the vulnerabilities that exist in third-party curated data and ultimately leads many to change their social media habits. This assignment is supported by in-class lessons on online tracking and good cyber hygiene practices.
- (2) Connections to other courses: The academic advising and registrar services (AARS) office scheduled PL100 and AI105 in a way that enables both courses to support each other. Typically, students enrolled in PL100 in one semester take CY105 in the other, and vice versa. In the Fall Semester, PL100 hosts the annual PL100 lecture, while in the spring semester, CY105 holds the annual Castle Lecture. To ensure that the entire plebe class can attend both lectures, AI105 has a drop to support the PL100 lecture in the fall semester, while PL100 has a drop in the spring to support the Castle Lecture. In recent years, both lectures have focused on topics technology, such as AI and social media. There is much value in working together to support common yearly themes and work together to identify speakers who can help us reinforce each other's messages.



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB B

AY2025 COURSE ASSESSMENT (EXSUM)



REPLY TO
ATTENTION OF

DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
ELECTRICAL ENGINEERING AND COMPUTER SCIENCE
WEST POINT, NY 10996

MADN-EEC-C

20 May 2025

MEMORANDUM THRU

Core Program Director, Department of Electrical Engineering & Computer Science,
United States Military Academy, West Point, New York 10996
Professor and Head, Department of Electrical Engineering & Computer Science, United
States Military Academy, West Point, New York 10996

FOR Dean of the Academic Board, USMA, West Point, NY 10996

SUBJECT: Annual Executive Summary for CY105, Computing Fundamentals, AY24

1. CY105 met its learning outcomes in AY2025 as measured through a formal assessment process. The EECS Core Program performs an annual assessment of all student learning outcomes (SLOs) and their mappings to the higher-level academy outcomes to continually monitor and improve CY105.
2. The attached report summarizes the annual assessment and curriculum change actions for CY105 (Computing Fundamentals).
3. POC for this is Dr. Suzanne J. Matthews, suzanne.matthews@westpoint.edu, x5577.

Encl

Suzanne J. Matthews, Ph.D.
Professor of Computer Science
CY105 Course Director

Enclosure: Executive Summary on CY105 Annual Assessment for AY25

1. DEPARTMENT-LEVEL ASSESSMENT

1.1) CY105 Student Learning Outcomes (SLOs):

SLO1: Examine legal, ethical, professional, and security issues associated with the cyber domain.

SLO2: Function as an ethical, secure, and competent user in West Point's immersive cyber environment.

SLO3: Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process.

SLO4: Describe how data flows from data sources across air, land, sea and cyberspace to form a network of information that is constantly being tracked and exploited by allies, adversaries, and third parties.

SLO5: Describe and explore the opportunities and challenges presented by artificial intelligence in the modern world.

1.2) Summary of Previous AY Assessment:

The AY24 assessment concluded that CY105 met the standard with respect to both its learning outcomes and What Graduates Can Do (WCGD) statements. Table 1 and Table 2 summarize student achievement of learning objectives and WCGD statements in prior years. Please note that while the AY21, AY22, and AY24-2 numbers below only included the TEE, AY23 included all the WPRs, because students performed comparatively better on the WPRs. We do not include SLO 4 or SLO 5 in previous years in the table below, because they were revised in AY24-2 to fit the updated focus of the course.

Table 1. Percentage of cadets meeting the standard with respect to learning objectives.

% of Cadets Meeting Standard on TEE				
LO	AY21	AY22	AY23*	AY24-2
1	91.9%	80.9%	87.4%	98%
2	90.7%	80.9%	82.8%	98.5%
3	88.0%	81.7%	92.7%	81.5%
4	-	-	-	96%
5	-	-	-	87.4%

Commented [SM1]: I need some help with this number. How were the numbers derived for the AY21, AY22, and AY23 tables for SLO3? Was it just the total grade on the code portion of the TEE? Can we compute something similar? I can see it for the AY25 EXSUM data, but can't find the equivalent number for AY24-2. Can you help me figure out if this data exists somewhere or compute it?

Commented [SM1R2]: [@Flick, Daniel MA](#)

Commented [DF1R3]: For AY21, AY22, and AY23, SLO3 was assessed in varied ways (see notes below).

-AY21: SLO3 (sequential logic SLO, which was actually SLO2 in AY21) was assessed based on four code writing problems on the TEE
AY21 EXSUM: [exsum_CY105_AY21.pdf](#)

-AY22: SLO3 was assessed for AY221 based on four code writing on the TEE, and for AY222 it was assessed based on 15 code tracing questions and 1 code writing problem on TEE

(for context, code tracing questions were introduced for the first time in AY222 based on the removal of code writing using an IDE on exams, as had been done previously with a homegrown IDE in AY221 before the adoption of Replit)
AY22 EXSUM: [exsum_CY105_AY22.pdf](#)

-AY23: SLO3 was assessed based on both WPRs and TEE (code tracing and code writing)
AY23 EXSUM [Signed_EECS_EXSUM_AY23_b_CY105.pdf](#)

Archive of CY105 EXSUMs from AY17-AY24: [EXSUMs \(CY105-->Archive-->EXSUM\)](#)

Commented [DF1R4]: So the most equivalent AY24-2 data would probably be from the TEE code tracing and code writing. This percentage is 81.5%.
This percentage comes from the TEE AY24-2 data here: [AY24-2_EXSUM_Data_CY105_5bins.xlsx](#). Which also matches the writeup's language for SLO3 in the EXSUM itself on page 8: [exsum_CY105_AY24.pdf](#).

I must admit it makes me sad that AY24-1 data is just never accounted for anywhere...my one semester as course director 😞 It seems like it would have made sense to include AY24-1 in last year's EXSUM despite the difference between semesters, similar to how there was a change in embedded indicators for SLO3 in AY22. But I understand 😊

Commented [MS1R5]: I'm ok with adding a column to this table for AY24-1 if you are willing to create it! We would just need the numbers for SLO 1,2 and 3

Commented [DF1R6]: It's ok, I need something to keep fueling my impostor syndrome anyway 😊

Commented [SM1R7]: ❤️

1.3) Status of Previous AY Recommendations:

The AY24 EXSUM made several recommendations, some of which we chose to adopt. These recommendations and the status of their adoptions are summarized in Table 3 below. As a reminder, last year's EXSUM only included data from AY24-2 due to radical revision that occurred between AY24-1 and AY24-2.

Table 2. AY24-2 Recommendation and Adoption Status

AY24-2 Recommendation	Status of Adoption
Switch from 30-lesson format to 40-lesson format.	(Complete) CY105 was run as a 40-lesson course in both AY25-1 and 25-2.
Eliminate note-sheets for exams; instead, expand the CY105 reference card to a 4-page reference booklet.	(Complete) New reference booklet introduced. This was only authorized resource for exams in AY25-1 and 25-2.
Explore novel ways to incorporate generative AI that motivate learning and not shortcuts.	(In Progress) The USMA Duck Virtual AI Assistant was piloted in CY105 in both AY25-1 and AY25-2. The AI Final project first introduced in AY24-2 was sustained. We also added more AI-related in-class exercises to give students opportunities to explore generative AI for different applications.
Increase transferability of knowledge to upper-level courses by having cadets run Python locally on laptops.	(Partially successful) Lesson plans were updated to ask instructors to use Python locally on laptops. It is unclear if all instructors followed these instructions though. Students still needed to use zyBooks for homework and exams.
Increase emphasis on problem solving/encourage students to go to boards more.	(Adopted, but uncertain success) In AY25, we increased the amount of time cadets spent at boards in class and increased our emphasis on problem solving by devoting a page of our reference card to the CY105 problem solving process and asking students to write algorithms for each problem. However, it is very unclear if this actually made any type of real difference in student performance.

1.4) Additional Changes and Improvements:

In addition to the changes outlined above, I spent considerable time updating the textbook that students used for the course. One of the advantages of zyBooks is that it enables us to customize content as much as we'd like. Therefore, I removed all the

zyBooks pre-populated content and replaced it with either better sources that I identified from across the web or my own custom writing. We also added additional exercises at the end of every chapter to give students more opportunities to practice, while cutting down the amount that we require students to complete outside of class. My updates had the added benefit of allowing me to negotiate down the price of the textbook from \$99/student to \$65/student.

The initial transition from 30 lessons to 40 lessons unintentionally clustered the programming content closer together. In AY25-2, effort was made to more evenly spread out programming through the course. I also had graded review sessions in AY25-2 where students worked in teams to earn the last 5 points toward their next WPR. The number of graded events in the course has been further reduced from 26 in AY24-2 to 19 in AY25-2 (a 26% reduction).

I believe the changes had a net-positive effect on the course, though more refinement is needed. Table 4 below summarizes course end statistics for CY105 prior to the revision in AY24-2, while Table 5 summarizes the course end statistics in 24-2 and each semester after. The revision produced significantly less failures on the TEE and led to consistent performance on the TEE and the course. These numbers were taken from prior course memo records. Please note that between 24-2 and 25-1 the use of note sheets for CY105 exams was completely eliminated. We believe that is the reason that there is an uptick in the number of TEE failures in AY25-1 and AY25-2 compared to AY24-2. However, we believe that overall, this data illustrates that the revision was successful in producing better performance in the course and the TEE overall.

Table 3. TEE and Course-End Grades pre-revision (AY23-1, AY23-2, and AY24-1)

Semester	23-1	23-2	24-1
Total Cadets		471	526
TEE Average	81.68% (B-)	79.58% (C+)	79.47 (C+)
#Fs, #Ds	51 Fs, 24 Ds	62 Fs, 37 Ds	88 Fs, 28 Ds
Final Grade	86.77% (B)	86.67% (B)	86.99 (B)
#Fs, #Ds	3Fs, 7 Ds	3 Fs, 3 Ds	5 Fs, 18 Ds
Final QPA	3.18	3.20	3.22

Commented [MS2]: @Filcik, Daniel MA 24-1 data is here!

Commented [FD2R2]: let's goooo!

Table 4. TEE and Course-End Grades post-revision (AY24-2, AY25-1, and AY25-2)

Semester	24-2	25-1	25-2
Total Cadets	544	574	556
TEE Average	86.74 (B+)	83.16 (B)	84.49 (B)
#Fs, #Ds	2 Fs, 10 Ds	15 Fs, 43 Ds	6 Fs, 21 Ds
Final Grade	88.86 (B+)	88.16 (B+)	87.80 (B+)
#Fs, #Ds	2 Fs, 0 Ds	1 F, 2 Ds	0 Fs, 1 D
Final QPA	3.40	3.42	3.36

1.5) Current Year Assessment:

The adoption of Canvas as the LMS of record led to the recommendation that CY105 assess all its SLOs annually. In prior years, the assessment of SLOs was rotated annually. For AY24-2 and onward, we try to leverage the full capabilities of Canvas to create a detailed assessment. Canvas allows us to align particular questions in an exam (or components of an assignment) to individual outcomes via a rubric interface. Canvas then generates an Outcomes Analysis by student (Outcome mastery based on all alignments) and even by exam (WPR 2 and TEE, for example). However, there isn't an easy way to get an Outcomes Analysis of a *subset* of questions from an exam (e.g., only the auto-graded questions or only a particular question), or an Outcomes Analysis by assignment. We reduced the number of manual assessment events that instructors had to complete to reduce assessment fatigue and thus increase the quality of the assessments that faculty produce.

The evidence used to assess each SLO is listed in Table 6:

1.5.a) Evidence Table:

Table 5. Summary of Evidence.

Course Outcome	Direct Evidence	Indirect Evidence	Total # of Indicators
SLO 1: Examine legal, ethical, professional, and security issues associated with the cyber domain.	Digital Footprint Audit (data access limitations, data vulnerabilities, security clearance risks, adjusting social media habits) WPR 3 Questions (Cybersecurity, Cyber Hygiene, Online Tracking and Ethics, Open-Ended Problem [Attack and Actor Threat Goals]) AI Final Project (ethical issues related to AI/copyright, potential misuse by malicious actors) TEE Questions (Identifying Malware, Cyberspace Layers, Instagram photo sharing, AI Use, Online	Course-End Feedback Survey	4

Commented [SM3]: I need help filling in this table (parts in Red)

Commented [SM3R2]: @Tippet, Hayden J CP1

Commented [CL4]: Add SLO text here for each SLO

	Tracking, Copyright Law (2), Robotics, Public Key Infrastructure, Cyber Hygiene, Internet, Data and Ethics), Free Response Question on Ethical Use of Robots		
SLO 2 Function as an ethical, secure, and competent user in West Point's immersive cyber environment.	WPR 3 Questions (Robotics [3], Cybersecurity [2], Cyber Hygiene, Online Tracking and Ethics) Digital Footprint Audit (data sources, data vulnerabilities, character, adjusting social media habits) TEE Question (Identifying Malware, Instagram photo sharing, Online Tracking, Copyright Law (2), Microsoft 365 (2), Public Key Infrastructure, Cyber Hygiene, I/O Devices, Computer Components), Free Response Question on Data and Ethics, Free Response Question on Ethical Use of Robots	Course-End Feedback Survey	3
SLO 3 Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process.	WPR 3 Questions (Robotics Code Tracing [2], Modules [2], Boolean Expressions) TEE Questions (Code Tracing (11), Python Basics (3), Algorithms (2), Conditional Statements) TEE Code Writing Questions	Course-End Feedback Survey	3
SLO 4 Describe how data flows from data	Digital Footprint Audit (data sources, surprises in data, data access limitations, data vulnerabilities)	Course-End Feedback Survey	3

sources across air, land, sea and cyberspace to form a network of information that is constantly being tracked and exploited by allies, adversaries, and third parties.	WPR 3 Questions (Computers and Data, Internet and the Web, Online Tracking and Ethics, Cyberspace and Threat Actors, Robotics, Open-Ended Problem [Cyber Attacks & Threat Actor Goals]) TEE Questions (Internet/Networks (3), I/O Devices, Data Storage, Data Sources and Representation (2)),		
SLO 5 Describe and explore the opportunities and challenges presented by artificial intelligence in the modern world.	AI Final Project (AI's role in project, AI's societal benefits/risks, AI's future development) TEE Questions (AI/ML/Deep Learning, ANI/ASI/AGI, AI Uses, Ethical Use of AI, Social Media AI training)	Course-End Feedback Survey	2

1.5.b) Rubric Definition: To facilitate a quantitative assessment, each individual data point of evidence from each embedded indicator was classified into one of the five below categories. For AY24, no unique rubrics were used, so the following rubric was uniformly applied to all SLOs and indicators.

1	2	3	4	5
Failed to meet the standard	Marginally met the standard	Met the standard	Slightly above standard	Greatly exceeded standard
0 – 64.9%	65 – 72.9%	73 - 79%	80 - 89%	90 - 100%

Pre-Observation identification of the overall acceptable standard. At least 80% of the students meet the standard. No more than 15% can completely fail to meet the standard.

1.5.c) **Quantitative Assessment:** Figures 2-6 below summarize the data from the rubric scoring of the embedded indicators for each of the SLOs.

SLO1: Examine legal, ethical, professional, and security issues associated with the cyber domain. On the DFA, 96.5% of the students met or exceeded the standard. On WPR 3, 89.8% of the students met the standard or higher. On the AI Final Project, 96.3% of the students met or exceeded the standard. On the TEE, 98.3% of the students met the standard or higher. Based on these indicators, CY105 is slightly above standard on this outcome with generally strong performance on all indicators.

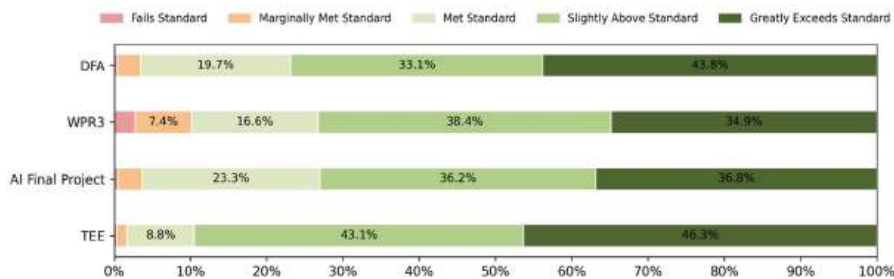


Figure 1. Summary of Assessment for SLO #1

SLO2: Function as an ethical, secure, and competent user in West Point's immersive cyber environment. On the DFA, 97.0% of the students met or exceeded the standard. On WPR3, 95.5% of the students met or exceeded the standard. On the TEE, 98.2% of the students met the standard or higher. Based on these indicators, CY105 greatly exceeded the standard on this outcome with very strong performance on all three indicators.



Figure 2. Summary of Assessment for SLO #2

SLO3: Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process. On WPR3 code tracing, 85.2% of the students met the standard or

higher. In contrast, on WPR 3 code writing, only 66.7% of students met the standard or higher, with 25.4% completing failing to meet the standard. A similar story is evidence on the TEE. For the TEE, only 61.4% of the students met the standard, with 25.2% completely failing to meet the standard. There is a clear dichotomy between the code tracing and code writing. On the TEE code tracing component, 85.6% of the students met the standard or higher. However, on the code writing component of the TEE, only 56.4% met or exceeded the standard, with 33.7% completely failing to meet the standard. Based on these indicators, CY105 marginally met this outcome with generally unacceptable performance in code writing, but with generally good performance on code tracing.

Commented [SM5]: MUCH lower than last year. It was 81.5% last year.

Commented [SM6]: This is lower than last year. Last year, it was 94.4%

Commented [SM7]: 68.4% last year.



Figure 3. Summary of Assessment for SLO #3

SLO4: Describe how data flows from data sources across air, land, sea and cyberspace to form a network of information that is constantly being tracked and exploited by allies, adversaries, and third parties. On the DFA, 96.2% of the students met or exceeded the standard. On WPR3, 77.7% of the students met the standard or higher, with 8.9% completely failing the standard. On the TEE, 94.6% of the students met the standard or higher. Based on these indicators, CY105 met this outcome with strong performance on two indicators, but with weak performance on one indicator.

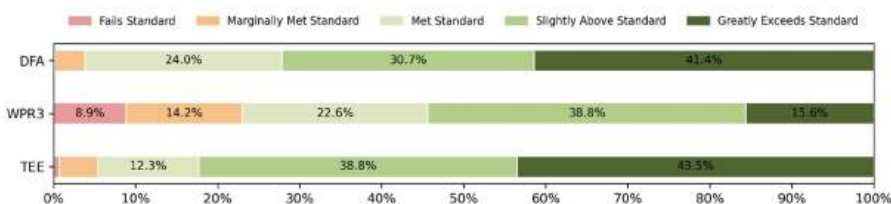


Figure 4. Summary of Assessment for SLO #4

SLO5: Describe and explore the opportunities and challenges presented by artificial intelligence in the modern world. On the AI Final Project, 97.1% of the students met or exceeded the standard. On the TEE, 94.9% of the students met or exceeded the standard. Based on these indicators, CY105 greatly exceeded this outcome with strong performance across both indicators.

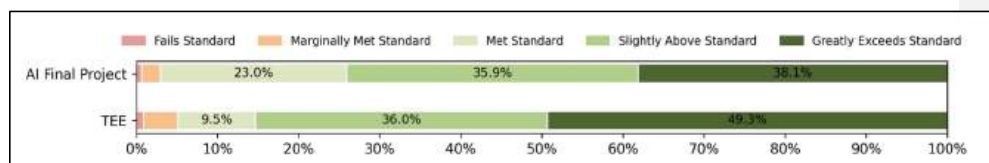


Figure 5. Summary of Assessment for SLO #5

1.5.d) **Qualitative Analysis:** In AY25, we asked students to assess how well they felt they met the course objectives. We use this as another point of comparison for how well CY105 met the course objectives in AY25. Figure 6 below summarizes student responses on the course end surveys for AY25 ($n=883$). In general, a majority of students agreed (> 80%) that they have met standards for SLO 1, 2, 4, and 5. However, only 66.9% of the students agreed that they “can effectively use Python and algorithmic thinking to solve problems” (SLO 3). We also asked students about their confidence related to “employing AI tools to solve problems”, with 84.7% of students reporting increased confidence in using such tools.

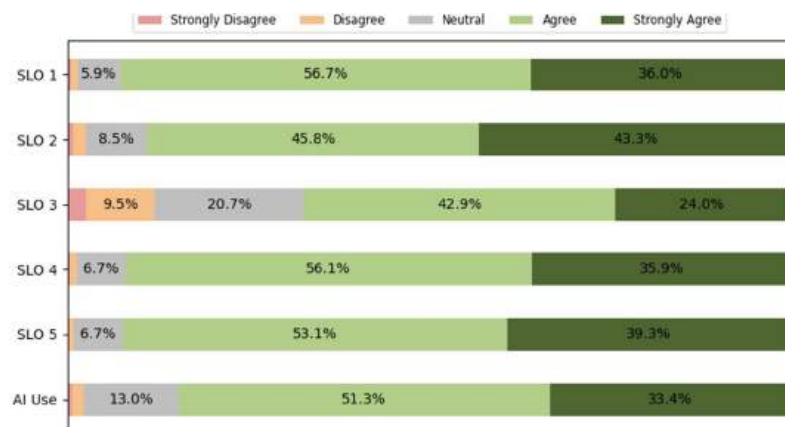


Figure 6. Student agreement on if they met different course objectives (course end survey).

Table 6 lists question text associated with the individual SLOs in the above survey. The “% Agreement” column is the total percentage of students that indicated either “Agree” or “Strongly Agree” on the survey above:

Table 6. Course-end survey questions mapped to individual SLOs

SLO	Question Text	% Agreement
1.	My understanding of the legal, ethical, and professional issues in the cyber domain has increased	92.7%

2.	I feel competent in using basic Microsoft 365 products (Teams, Excel, OneDrive, Notes, etc.) to aid my success as a student at West Point.	89.1%
3.	I can effectively use Python and algorithmic thinking to solve problems.	66.9%
4.	I am more aware of how data and networks can be exploited by various parties.	92.0%
5.	I better understand the opportunities and challenges presented by artificial intelligence in the modern world.	92.4%
AI Use	I am more confident in employing artificial intelligence tools to solve problems.	84.7%

1.5.d) Assessment Analysis:

a. Areas for Sustainment:

Based on the evidence, we believe the following should be sustained for CY105 in AY 2026:

- **The 40-lesson format a success.** The switch to a 40-lesson format for CY105 increased our ability to achieve our SLOs much better than the 30-lesson format.
- **Exposure to AI topics positive.** The opportunity to learn about AI in an academically sanctioned environment was immensely popular and motivating to students, and our students reported increased confidence in using AI tools. We plan to figure out ways to continue to augment existing exercises with generative AI so that students see how it can be used in additional applications.
- **Reduction of graded events a net positive.** We were able to achieve good performance with fewer homework assignments.
- **Increased emphasis on code tracing in homework and in class.** In AY25-2, we modified our homework assignments to have more code tracing and general knowledge check questions. We saw an improvement in code tracing ability in our students' exams, though less so on the TEE. We plan to sustain this for future semesters, and explore additional ways to emphasize code tracing.

b. Areas for Improvement:

Despite the successes in the revision, there are still more improvements that can be made. Specifically:

- **Encourage students to actually practice and read book outside of class.** Students who read the book and practice outside of class tend to do well in this course. Students who do not read the book and practice do poorly. We are going to explore ways to keep students more accountable for doing the requisite prep work for the course by incorporating a class participation grade (most likely enforced by daily short quizzes). While SLO 3 performance suffered, we believe this is mainly due to the elimination of note sheets (students got one double-

Commented [LC8]: Use these sub-bullets to discuss the assessment data, including any qualitative analysis you may have. List any actions you plan to take based on this assessment to improve the course.

sided sheet per exam, and three sheets for the TEE in 24-2 vs no note sheets in AY25), rather than students learning worse than they did in previous semesters.

- **Encourage students to use AI as a partner, not a shortcut.** There still exists an incongruity in homework grades and code writing grades, suggesting that many students are getting too much external help and/or use ChatGPT to avoid opportunities to practice. In AY2026, we plan on continuing to explore ways to teach students to how to use AI *help them* learn material, rather than as a tool to circumvent learning. The USMA Duck that was piloted was a good steppingstone toward this, but more work needs to be done to make the duck available to all sections.

c. Any additional comments, analysis, and insights:

- **We are still seeing the effects of COVID based on cadet study skills.** I wrote about this in last year's EXSUM. In support of articles written in the Chronicle of Higher Education¹ and other sources², there is increasing evidence that cadets do not have the attention span to complete readings or summarize material on their own. I have heard directly from students that they use ChatGPT to summarize long readings in other courses. It remains an open question on how to continue to challenge these expectations and perspectives.
- **Misuse of ChatGPT and other AI tools is hurting critical thinking across the board.** A recent report³ highlighted that independent studies suggest that students are increasingly using AI to offload important critical thinking processes. Many students are coming to West Point with some prior knowledge of AI. The challenge now is quickly becoming how to get students to *unlearn* unhelpful practices with AI and convince them that critical thinking is actually useful.

d. Significant planned or implemented curricular changes.

Based on our assessment data, feedback from instructors and experiential data, the following curricular changes are planned for AY 2026:

- **Increase the infusion of generative AI in CY105 curriculum.** CY105's primary goal with respect to generative AI is to ensure students learn how to team with generative AI without it replacing their learning. Given our success with the AI final project and some inklings of success with the duck, we plan to increase the level of exposure students have to generative AI in AY26. The updated version of CY105 have students exposed to generative AI concepts in every unit, with 33% of the content touching on generative AI in some manner.

¹ [Students Are Less Able and Less Willing to Read. Professors Are Stymied. \(chronicle.com\)](https://www.chronicle.com/article/students-are-less-able-and-less-willing-to-read-professors-are-stymied)

² [The Pandemic's Babies and Toddlers Are Struggling in School - The New York Times \(nytimes.com\)](https://www.nytimes.com/2020/09/01/us/politics/coronavirus-schools.html)

³ [University students offload critical thinking, other hard work to AI](https://www.technologyreview.com/2020/09/01/1005841/university-students-offload-critical-thinking-other-hard-work-to-ai/)

Likewise, 66% of our out-of-class assignments now touch upon generative AI.

- **Increase number of labs.** For AY 2026, we plan on replacing the existing exam review periods and the copyright lecture with in-class laboratory sessions. Based on the assessment results above, SLO 3 (code writing) is where our students struggle the most. By deliberately creating more opportunities for them to practice their skills, we hope to see improved performance on this SLO in the upcoming year. Our labs are also opportunities for students to get exposure to generative AI. In AY26, there are three novel labs related to generative AI: (1) leverage Microsoft 365 copilot for a number of every tasks; (2) using generative AI to create a website and learn about tracking elements; and (3) using generative AI to illustrate how malicious parties can use LLMs to assist them analyze unencrypted network traffic.
 - **Have students build their own AI Virtual Tutor.** In AY2025, CY105 piloted the USMA Duck, a custom AI virtual assistant that is trained on our course materials. While the duck certainly met many of our intents, it started experiencing latency and model-switching issues when we tried to scale the number of sections up. In AY26, students will use Microsoft 365 Copilot Chat's "Build an Agent" feature to build their own custom Python Tutor agent that will act as an allowed tutor in the course. Having students build their own tutor achieves two purposes: 1) it solves our latency and model switching issues, and 2) it demystifies AI and how agents work.
- e. Significant planned or implemented assessment process changes.
The following assessment changes will be made in AY 2025:
- **Work with Canvas POCs to identify efficiencies in assessment.**
We completely depended on Canvas for data for this assessment cycle and the last year's assessment cycle. While we made considerable effort to improve our ability to use Canvas in a meaningful manner to collect assessment data, it is still difficult for us to pull and analyze data from Canvas. The pulling and cleaning of Canvas data can take up to 8 hours to do. We are meeting with the Canvas POCs at West Point to see if additional efficiencies can be identified, especially as far as the ability to export SLO outcomes with Canvas exams.

2. HIGHER-LEVEL ASSESSMENT (ACADEMIC PROGRAM)

2.1) CY105 supports WGCD outcomes 5.1, 5.4, and 5.5. As of AY24, WGCD 4.2 was also assessed, given the curriculum of CY105 supports this outcome. After further assessment, we may offer to the Curriculum and Assessment Committee that this supporting outcome be formally added as a contribution of CY105 to the Academic Program.

2.2) Course Outcomes Mapped to WGCD Statements: The table below shows the mapping of CY105 course outcomes to the assessed WGCD supporting objectives.

			APG4. Ethical Reasoning	APG5. STEM		
			WGCD 4.2	WGCD 5.1	WGCD 5.4	WGCD 5.5
			Recognize ethical components of problems and situations	Apply mathematics, science, and computing to model devices, processes, or behaviors	Apply an engineering design process to create effective and adaptable solutions. [3CES]	Explain and apply computing and information technology concepts and practices in the context of the Cyber Domain. [IT/CYBER]
Course Outcomes						
CY105	1	Examine legal, ethical, professional, and security issues associated with the cyber domain.	X			X
	2	Function as an ethical, secure, and competent user in West Point's immersive cyber environment.	X			X
	3	Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process.		X	X	X
	4	Describe how data flows from data sources across air, land, sea and cyberspace to form a network of information that is constantly being tracked and exploited by allies, adversaries, and third parties.		X		X
	5	Describe and explore the opportunities and challenges presented by artificial intelligence in the modern world.	X		x	X

Commented [LCJC9]: Please confirm the mappings below for your course. This was the draft used during the assessment kickoff. To keep it simple, please only use X's to indicate an assessed connection, rather than symbols to show strength of connection.

2.3) WGCD Outcome Assessment. The WGCD outcomes were assessed using a combination of the SLO assessments from Part 1 and the mapping chart above.

WGCD	Assessment Grade	Degree of Confidence
4.2	"Slightly exceeded the standard"	High
5.1	"Marginally met the standard"	Moderate
5.4	"Met the standard"	Moderate
5.5	"Slightly exceeded the standard"	High

Commented [LC10]: The precise means of your assessment can be abstracted in the EXSUM, but you are free to combine the SLO scores in a way that makes sense to you (e.g., maybe a weighted average makes the most sense based on one SLO being more closely tied to a WGCD outcome than another). You can also use more qualitative (indirect evidence) to determine your assessment grade.

2.3.a.) Strengths: CY105 represents one of the first introductions to 4.2, 5.1, 5.4 and 5.5 at West Point. We are especially strong at supporting 4.2 and 5.5, especially ensuring that students are aware of modern technology in the context of the cyber domain. Compared to AY24, we have seen improvements in cadet attainment of our

WGCD statements.

2.3.b.) Areas for improvement: Based on the indicators we collected, we believe we met the standard for 5.1, albeit marginally. While we met the standard for 5.4, performance on SLO 3 limited our ability to exceed the standard. Working on strategies to increase the performance of students in SLO 3 will be a prime focus of AY 2026.

2.3.c) Additional Elements:

Support to Character Thread.

- (1) The Digital Footprint Audit (DFA) is the primary way that CY105 supports West Point's character thread. Most plebes are unaware that commercial third parties actively track and store their data. In CY105, students learn that every action they take online generates a *digital footprint* that projects a particular online persona. Students are required to request data from three social media sources, and study and reflect on whether the online persona projected by their data actually represents their own personal character values. This is an eye-opening assignment for our cadets that highlights the vulnerabilities that exist in third-party curated data and ultimately leads many to change their social media habits. This assignment is supported by in-class lessons on online tracking and good cyber hygiene practices.
- (2) Connections to other courses: The academic advising and registrar services (AARS) office scheduled PL100 and CY105 in a way that enables both courses to support each other. Typically, students enrolled in PL100 in one semester take CY105 in the other, and vice versa. In the Fall Semester, PL100 hosts the annual PL100 lecture, while in the spring semester, CY105 holds the annual Castle Lecture. To ensure that the entire plebe class can attend both lectures, CY105 has a drop to support the PL100 lecture in the fall semester, while PL100 has a drop in the spring to support the Castle Lecture. In recent years, both lectures have focused on topics technology, such as AI and social media. There is much value in working together to support common yearly themes and work together to identify speakers who can help us reinforce each other's messages.



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB C

AY2024 COURSE ASSESSMENT (EXSUM)



REPLY TO
ATTENTION OF

**DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
ELECTRICAL ENGINEERING AND COMPUTER SCIENCE
WEST POINT, NY 10996**

MADN-EEC

20 September 2024

MEMORANDUM THRU

Core Program Director, Department of Electrical Engineering & Computer Science,
United States Military Academy, West Point, New York 10996
Professor and Head, Department of Electrical Engineering & Computer Science, United
States Military Academy, West Point, New York 10996

FOR Dean of the Academic Board, USMA, West Point, NY 10996

SUBJECT: Annual Executive Summary for CY105, Computing Fundamentals, AY24

1. CY105 met its learning outcomes in AY2024 as measured through a formal assessment process. The EECS Core Program performs an annual assessment of all student learning outcomes (SLOs) and their mappings to the higher-level academy outcomes to continually monitor and improve CY105.
2. The attached report summarizes the annual assessment and curriculum change actions for CY105 (Computing Fundamentals).
3. POC for this is Dr. Suzanne J. Matthews, suzanne.matthews@westpoint.edu, x5577.

Encl

Suzanne J. Matthews, Ph.D.
Professor of Computer Science
CY105 Course Director

Enclosure: Executive Summary on CY105 Annual Assessment for AY24

1. DEPARTMENT-LEVEL ASSESSMENT

1.1) CY105 Student Learning Outcomes (SLOs):

SLO1: Examine legal, ethical, professional, and security issues associated with the cyber domain.

SLO2: Function as an ethical, secure, and competent user in West Point's immersive cyber environment.

SLO3: Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process.

SLO4: Describe how data flows from data sources across air, land, sea and cyberspace to form a network of information that is constantly being tracked and exploited by allies, adversaries, and third parties.

SLO5: Describe and explore the opportunities and challenges presented by artificial intelligence in the modern world.

1.2) Summary of Previous AY Assessment:

The AY23 assessment concluded that CY105 met the standard with respect to both its learning outcomes and What Graduates Can Do (WCGD) statements.

1.3) Status of Previous AY Recommendations:

The AY23 EXSUM made several recommendations, some of which we chose to adopt. These recommendations and the status of their adoptions are summarized in Table 3 below.

Table 3. AY23 Recommendation and Adoption Status

AY23 Recommendation	Status of Adoption
Adopt a new interactive digital textbook. Replace daily quizzes with pre-class activities within the textbook. The goal is that cadets will come to class better prepared to discuss course material.	(Complete) Adopted zyBooks, which included pre-class participation activities. However, it is unclear that the zyBook resulted in increased cadet preparation.
Plan to explore AI-assisted programming, specifically generative AI to assist programmer with developing code.	(Complete) Created the AI Summative Project, which requires students to team with generative AI.

Weave the character integration thread more explicitly into the course material, especially the digital footprint audit.	(Complete) The digital footprint audit was modified to ensure cadets reflected on the character values reflected in the data collected on them by third parties.
Adopt Canvas as the LMS of record to assess all course learning outcomes per year.	(Complete) We relied solely on Canvas to help us collect assessment data. However, technical difficulties limited the number of indicators that could be collected.
Add WPRs and other assignments as assessment evidence instead of relying solely on the TEE.	(Partially Adopted) Instructors were asked to manually assess outcomes for all graded events, leading to “assessment fatigue” and instructor inattentiveness to completing assessments. Data in the AY24 EXSUM focused on data from select graded events, and primarily those in the latter part of the course.

1.4) Additional Changes and Improvements:

I was also tasked to become the new course director of CY105 in AY24-2, and COL Christopher Lowrance was made the new Core program director in AY24-2. The leadership change presented the opportunity for identifying new areas of improvement. Those immediate areas identified were reducing the number of graded events, increasing time for students to practice the content during class, increasing the depth of some lesson material, improving the quality of the textbook, refactoring the TEE so that it better reflects the learning outcomes, and increasing the coverage of data-centric topics and artificial intelligence.

My primary goal as a new course director was not to radically redesign CY105, but to strengthen the existing content and create more opportunities for students to practice. For AY24-2, I revised the lesson schedule, consolidating some of the lessons lacking technical depth into fewer lessons, to create WPR review sessions and practice labs. I also increased the number of problems on the weekly problem sets (from 1 to 3) to give students more focused opportunities to practice programming outside of class. The programming content was also spread out throughout the semester, and we heavily revised the readings. To increase the focus on data, I began to deemphasize the turtle graphics module in favor of other, more analysis-focused modules, such as the math module and random module.

From a process standpoint, I reduced the number of graded events by 48%, largely by eliminating out-of-class reading assignments. We were forced to transition the programming integrated development environment (IDE) from repl.it because it started using artificial intelligence (AI) to autocomplete coding assignments, circumventing the learning process. We have since switched to zyIDE, the built-in IDE that comes with

zyBooks. I also started returning all exam questions to the students so they could study better. Additionally, I created new lesson plans and improved the quality of the slides to promote consistent teaching across the various sections of CY105. Lastly, I revised the SLOs to better reflect the increased focus on data and artificial intelligence.

I believe the changes had a net-positive effect on the course. Table 4 below summarizes course end statistics for CY105 prior to the revision in AY24-2 and after the revision. The revision produced significantly less failures on the TEE and led to consistent performance on the TEE and the course.

Table 4. TEE and Course-End Grades pre-revision (AY23, AY24-1) vs post-revision (AY24-2)

Semester	23-1	23-2	24-1	24-2
Total Cadets		471	526	544
TEE Average #Fs, #Ds	81.68% (B-) 51 Fs, 24 Ds	79.58% (C+) 62 Fs, 37 Ds	79.47 (C+) 88 Fs, 28 Ds	86.74 (B+) 2 Fs, 10 Ds
Final Grade #Fs, #Ds	86.77% (B) 3Fs, 7 Ds	86.67% (B) 3 Fs, 3 Ds	86.99 (B) 5 Fs, 18 Ds	88.86 (B+) 2 Fs, 0 Ds
Final QPA	3.18	3.20	3.22	3.42

Figure 1 provides another view of the impact of the CY105 revision. Each data point represents a single CY105 student (plebes only) who took the course in AY24. The x-axis represents their CQPA at the end of their plebe year. The y-axis represents the final grade that they earned in CY105, either in AY24-1 or AY24-2. The hypothesis is that generally stronger students (those with higher CQPAs) should perform better in a core course like CY105, while weaker students perform worse. For AY24-1, a weak positive correlation (0.25) was found, while for AY24-2, a moderately positive correlation (0.65) was found¹. Thus, the changes in AY24-2 seem to support the hypothesis above.

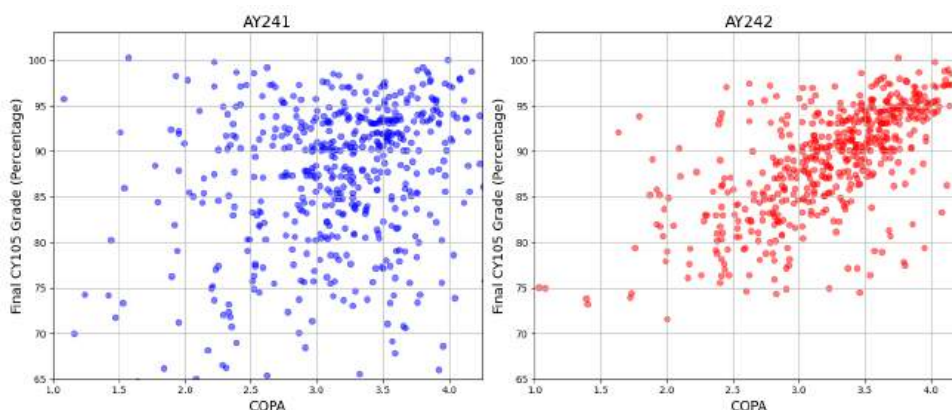


Figure 1. CY105 course grades compared to CQPA (plebes only)

¹ Categorizations of “weak” and “moderate” based on accepted interpretations of Pearson’s and Spearman’s correlation statistic, as outlined in Table 1 of [User’s guide to correlation coefficients](#) by H. Akoglu (2018)

Given the level of revisions (and the revisions to the SLOs), the underlying data used for SLO and WGCD analysis for AY2024 is from AY24-2 only.

1.5) Current Year Assessment:

The adoption of Canvas as the LMS of record led to the recommendation that CY105 assess all its SLOs annually. In prior years, the assessment of SLOs was rotated annually. For AY24, we tried to leverage the full capabilities of Canvas to create a detailed assessment. Canvas allows us to align particular questions in an exam to Outcomes, as well as aligning an assignment to Outcomes via a rubric. Canvas then generates an Outcomes Analysis by student (Outcome mastery based on all alignments) and even by exam (WPR 2 and TEE, for example). However, there appears to be no way to get an Outcomes Analysis of a *subset* of questions from an exam (e.g, only the auto-graded questions or only a particular question), or an Outcomes Analysis by assignment. Rather, the Outcome alignments are rolled into the holistic Outcomes Analysis by student or the holistic Outcomes Analysis by exam. Being unable to export learning mastery reports by assignment meant we could not assess the number of indicators we originally wished to in AY24.

The evidence used to assess each SLO are listed in Table 5:

1.5.a) Evidence Table:

Table 5. Summary of Evidence.

Course Outcome	Direct Evidence	Indirect Evidence	Total # of Indicators
SLO 1	WPR 2 Questions (Denial of Service, Cyber Personas) TEE Questions (Identifying Malware, Cyberspace Layers, Instagram photo sharing, AI Use, Online Tracking, Copyright Law (2), Robotics, Public Key Infrastructure, Cyber Hygiene, Internet, Data and Ethics), Free Response Question on Ethical Use of Robots	N/A	2
SLO 2	WPR 2 Questions (Embedded Computers, Malware, Encryption, Cyber Hygiene, CIA+ triad) TEE Question (Identifying Malware, Instagram photo sharing, Online Tracking,	N/A	2

	Copyright Law (2), Microsoft 365 (2), Public Key Infrastructure, Cyber Hygiene, I/O Devices, Computer Components), Free Response Question on Data and Ethics, Free Response Question on Ethical Use of Robots WPS 2 (Excel)		
SLO 3	WPR 2 Questions (Iteration (2), Conditional Statements (3), Writing Functions (3), Python Basics) TEE Questions (Code Tracing (11), Python Basics (3), Algorithms (2), Conditional Statements) TEE Code Writing Questions	Instructor feedback	3
SLO 4	WPR 2 Questions (Binary/Decimal, Networks (4)) , Free Response Question on Network Protocols TEE Questions (Internet/Networks (3), I/O Devices, Data Storage, Data Sources and Representation (2)), Digital Footprint Audit (Grade) Digital Footprint Audit (Grade)		3
SLO 5	TEE Questions (AI/ML/Deep Learning, ANI/ASI/AGI, AI Uses, Ethical Use of AI, Social Media AI training) AI Summative Project (Grade)	Instructor/Student Feedback	2

1.5.b) Rubric Definition: To facilitate a quantitative assessment, each individual data point of evidence from each embedded indicator was classified into one of the five below categories. For AY2024, no unique rubrics were used, so the following rubric was uniformly applied to all SLOs and indicators.

1	2	3	4	5
Failed to meet the standard	Marginally met the standard	Met the standard	Slightly above standard	Greatly exceeded standard
0 – 64.9%	65 – 72.9%	73 - 79%	80 - 89%	90 - 100%

Pre-Observation identification of the overall acceptable standard. At least 80% of the students meet the standard. No more than 15% can completely fail to meet the standard.

1.5.c) Quantitative Assessment: Figures 2-6 below summarize the data from the rubric scoring of the embedded indicators for each of the SLOs.

SLO1: Examine legal, ethical, professional, and security issues associated with the cyber domain. On WPR2, only 68.6% of the students met the standard or higher, with 23% completely failing the standard. On the TEE, 98% of the students met the standard or higher. Based on these indicators, CY105 met this outcome with generally acceptable performance, but with weak performance on one indicator.



Figure 2. Summary of Assessment for SLO #1

SLO2: Function as an ethical, secure, and competent user in West Point's immersive cyber environment. On WPR2, only 58.5% of the students met the standard or higher, with 19.5% completely failing the standard. On the TEE, 98.5% of the students met the standard or higher. On Weekly Problem Set 2 (which focused on EXCEL skills) 92.8% of students met or exceeded the standard. Based on these indicators, CY105 met this outcome with generally acceptable performance, but with weak performance on one indicator.



Figure 3. Summary of Assessment for SLO #2

SLO3: Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process. On WPR2, only 74% of the students met the standard or higher. On the TEE, 81.5% of the students met the standard or higher. On the code writing component of the TEE, 68.4% met or exceeded the standard, but 22.4% completely failed to meet the standard. Based on these indicators, CY105 marginally met this outcome with generally unacceptable performance, but with marginal performance on two indicators.



Figure 4. Summary of Assessment for SLO #3

SLO4: Describe how data flows from data sources across air, land, sea and cyberspace to form a network of information that is constantly being tracked and

exploited by allies, adversaries, and third parties. On WPR2, only 69.3% of the students met the standard or higher. On the TEE, 96% of the students met the standard or higher. On the Digital Footprint Audit, 98% of the students met or exceeded the standard. Based on these indicators, CY105 met this outcome with generally acceptable performance, but with weak performance on one indicator.

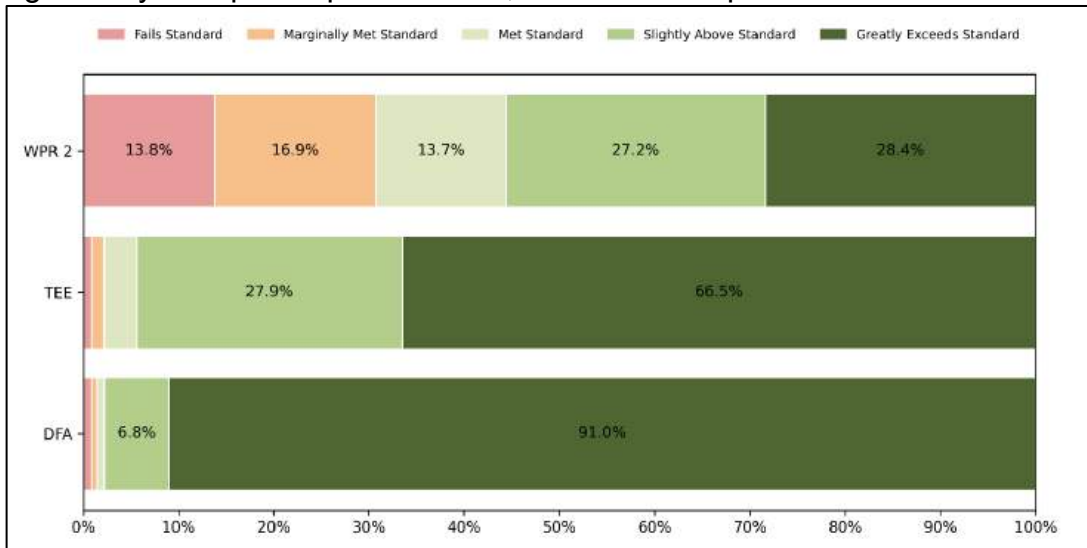


Figure 5. Summary of Assessment for SLO #4

SLO5: Describe and explore the opportunities and challenges presented by artificial intelligence in the modern world. On the TEE, 87.4% of the students met the standard or higher. On the AI summative presentation, 97.3% of the students met or exceeded the standard. Based on these indicators, CY105 met this outcome with generally acceptable performance.

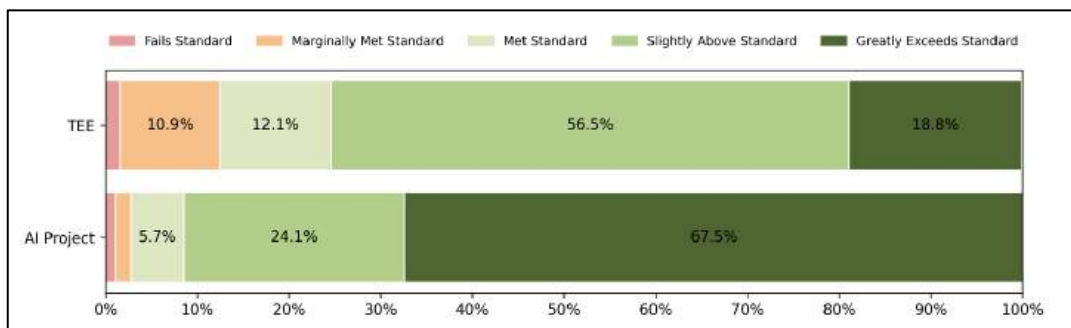


Figure 6. Summary of Assessment for SLO #5

1.5.d) Assessment Analysis:

a. Areas for Sustainment:

Based on the evidence, we believe the following should be sustained for CY105 in AY 2025:

- **AI Summative Project.** Based on course end feedback, the opportunity to learn about AI in an academically sanctioned environment was immensely popular and motivating to students.
- **Character-Focused Digital Footprint Audit.** The addition of character-reflection questions to the digital footprint audit was a positive addition.
- **Reduction of graded events a net-positive.** Despite reducing graded events by 48%, which was largely associated with low-point pre-class reading assignments, performance was not negatively impacted.

b. Areas for Improvement:

Despite the successes in the revision, there are still more improvements that can be made. Specifically:

- **Increase focus on problem solving.** Our results on SLO 3 suggests that students still need more experience with problem solving techniques. We plan to explore this in AY 2025.
- **Switch to 40-lesson format to increase opportunities for contact with students and reduce speed of coverage of programming content.** Despite concerted effort to spread the programming content throughout the semester, our results on SLO 3 and the course-end feedback suggest that the programming content is overly compressed. In AY2025, CY105 will return to its previous 40-lesson/55-minute format.
- **Encourage students to use AI as a tutor, not a shortcut.** There existed an incongruity in homework grades and code writing grades. Our SLO 3 results plus instructor and course-end feedback suggest that students were using ChatGPT to quickly solve problem sets, thus avoiding opportunities to practice. In AY2025, we plan on exploring ways to teach students to how to use AI *help them* learn material, rather than as a tool to circumvent learning.

c. Any additional comments, analysis, and insights:

- **We are still seeing the effects of COVID based on cadet study skills.** In support of articles written in the Chronicle of Higher Education² and other sources³, there is quite a bit of evidence that cadets do not seem to know how to complete readings or summarize material properly. Students also have an expectation that a minimal amount of effort should earn them a good grade, and a very transactional perspective when it comes to graded events. It remains an open question on how to continue to challenge these expectations and perspectives.

² [Students Are Less Able and Less Willing to Read. Professors Are Stymied. \(chronicle.com\)](https://www.chronicle.com/article/students-are-less-able-and-less-willing-to-read)

³ [The Pandemic's Babies and Toddlers Are Struggling in School - The New York Times \(nytimes.com\)](https://www.nytimes.com/2020/09/01/us/politics/coronavirus-schools.html)

- d. Significant planned or implemented curricular changes.
- Based on our assessment data, feedback from instructors and experiential data, the following curricular changes are planned for AY 2025:
- **Switch to 40-lesson format.** Switching back to a 40-lesson will allow a greater breadth of coverage and ensure that instructors and students have more face-to-face time per week. We predict that SLO 3 will improve in AY 2025.
 - **Increase emphasis on problem solving.** For AY 2025, a complete lesson on problem solving and algorithms has been re-introduced, with specific focus a four-step problem-solving methodology that was taught in IT105 back in AY 2000. The first few weekly problem sets on programming have been modified so that students practice writing algorithms and implementing them. The aim is to improve performance on SLO 3 in AY 2025.
 - **Increase transferability of knowledge to upper-level courses by having cadets run Python locally on their laptops.** In prior semesters, students completed in-class exercises in cloud-based environments. Consequently, students frequently believed the only way to run Python is to have access to those environments. In AY 2025, we will encourage cadets to program in Python directly on their laptops, so they can more directly reuse the Python skills gained in CY105 in their upper-level courses, regardless of major.
 - **Encourage students to go to boards more.** Students have an opportunity to learn from the expertise of each other. Consequently, lesson plans will be revised to encourage students to go to boards for nearly every programming lesson. The expectation is that students will have more opportunities practicing problem-solving together and increase the productivity of class time. Again, this change is in support of improving SLO 3.
 - **Explore novel ways to incorporate generative AI that motivate learning and not shortcuts.** CY105 is embracing generative AI in a deliberate, yet cautious manner. Our goal is to ensure that generative AI tools augment cadet learning, not replace it. Given our success with the AI summative project, we plan to explore the use of AI assistants to aid students in studying for the course, and how to use generative AI to support data analysis in the digital footprint audit.
 - **Update reference card to be reference booklet, eliminate note sheets.** Previously, cadets could use sheets of hand-written notes on in-class graded events. In AY2025, CY105 will adopt a four-page reference booklet. This reference booklet will be the only authorized reference for in-class graded events.
- e. Significant planned or implemented assessment process changes.
- The following assessment changes will be made in AY 2025:
- **Be selective of what questions we choose to align with SLOs in Canvas.** A weakness of our current assessment cycle is that we

aggregated events as embedded indicators for each SLO. However, this complicates and masks the ability to do assessment well, as not all events strongly support SLOs, and can hide valuable trends. In the future, we are going to be very deliberate on how we assign evidence to SLOs, in an effort to ensure we have higher quality data.

- **Reduce reliance on Canvas for assessment/maintain separate assessment records.** We completely depended on Canvas for data for this assessment cycle. However, this decision was made without fully recognizing some of Canvas' limitations. Since Canvas generates data by SLO (and not by question), it was impossible to tease out free response questions (which were aligned to particular SLOs) from other types of questions (such as autograded) on a common event. To rectify this issue in the future, we will use Canvas for autograded material, but maintain a separate assessment spreadsheet where we manually enter scores for free response and programming questions.
- **Fewer, high-quality embedded indicators.** Even if we could extract the data from Canvas, we doubt the custom learning mastery rubrics for the AY2024 assessment would have yielded much value, due to "assessment fatigue" felt by instructors and poor communication to instructors about the importance of assessing accurately and how to assess accurately. Having fewer targeted indicators with clear guidance should help rectify this issue.

2. HIGHER-LEVEL ASSESSMENT (ACADEMIC PROGRAM)

2.1) CY105 supports WGCD outcomes 5.1, 5.4, and 5.5. As of AY24, WGCD 4.2 was also assessed, given the curriculum of CY105 supports this outcome. After further assessment, we may offer to the Curriculum and Assessment Committee that this supporting outcome be formally added as a contribution of CY105 to the Academic Program.

2.2) Course Outcomes Mapped to WGCD Statements: The table below shows the mapping of CY105 course outcomes to the assessed WGCD supporting objectives.

			APG4. Ethical Reasoning	APG5. STEM		
			WGCD 4.2	WGCD 5.1	WGCD 5.4	WGCD 5.5
			Recognize ethical components of problems and situations	Apply mathematics, science, and computing to model devices, processes, or behaviors	Apply an engineering design process to create effective and adaptable solutions. [3CES]	Explain and apply computing and information technology concepts and practices in the context of the Cyber Domain. [IT/CYBER]
Course Outcomes						
CY105	1	Examine legal, ethical, professional, and security issues associated with the cyber domain.	X			X
	2	Function as an ethical, secure, and competent user in West Point's immersive cyber environment.	X		X	X
	3	Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process.		X	X	X
	4	Describe how data flows from data sources across air, land, sea and cyberspace to form a network of information that is constantly being tracked and exploited by allies, adversaries, and third parties.		X		X
	5	Describe and explore the opportunities and challenges presented by artificial intelligence in the modern world.	X		x	X

2.3) WGCD Outcome Assessment. The WGCD outcomes were assessed using a combination of the SLO assessments from Part 1 and the mapping chart above.

WGCD	Assessment Grade	Degree of Confidence
4.2	"Met the standard"	High
5.1	"Marginally met the standard"	Moderate
5.4	"Met the standard"	Moderate
5.5	"Slightly above the standard"	High

2.3.a.) Strengths: CY105 represents one of the first introductions to 4.2, 5.1, 5.4 and 5.5 at West Point. We are especially strong at supporting 5.5, especially ensuring that students are aware of modern technology in the context of the cyber domain. While this course was previously not assessed against 4.2, we believe it does give students a valuable exposure to the ethical components of technology, which are greatly shaping their lives as both modern soldiers and members in modern society.

2.3.b.) Areas for improvement: Based on the indicators we collected, we believe we met the standard for 5.1, albeit marginally. While we fully met the standard for 5.5,

performance on SLO 3 limited our ability to exceed the standard. The assessment was strengthened by the results of the AI Summative Project, where students collaborated with AI to create something innovative. Informal feedback from both instructors and students indicated that students face challenges in problem solving. To address this, we plan to increase our focus on problem-solving skills in AY25. Additionally, we believe transitioning to a 40-lesson format will provide cadets with more opportunities to engage with the material during class.

2.3.c) Additional Elements:

(1) Use of zyBooks. At the end of AY23, a zyBook was introduced in CY105, providing students with interactive content and coding exercises. In AY25, we plan to customize the zyBook to better suit West Point cadets, as some of the content from AY24 did not engage our students as effectively. We aim to improve the material by curating and creating our own content.

(2) Exploration of generative AI. The introduction of the AI Summative Project in AY 2024 was extremely motivating and gave students an opportunity to learn about generative AI. In AY2025, we will explore how generative AI can help students learn concepts better, including the pilot of an AI-powered virtual assistant. We believe that CY105 leads the way in helping students understand the opportunities and limitations of this novel technology, which is in support of both WCGD 5.1, 5.4 and 5.5.

(3) Support to Character Thread. The Digital Footprint Audit (DFA) was augmented in AY2024 to support the character thread at West Point. We will continue to refine the DFA and plan to explore more ways to incorporate ethics in lessons dealing with technology such as AI.



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB D

AY2023 COURSE ASSESSMENT (EXSUM)



**DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
ELECTRICAL ENGINEERING AND COMPUTER SCIENCE
WEST POINT, NY 10996**

MADN-EECS-C

12 September 2023

MEMORANDUM THRU

Core Program Director, Department of Electrical Engineering & Computer Science,
United States Military Academy, West Point, New York 10996
Professor and Head, Department of Electrical Engineering & Computer Science, United
States Military Academy, West Point, New York 10996

FOR BG Shane R. Reeves, Dean of the Academic Board, United States Military
Academy, West Point, NY 10996

SUBJECT: Annual Executive Summary for CY105, AY23

1. Status of curricular change initiatives completed during the current or previous academic year and results of follow-up assessment.

a. In AY23-2, the department successfully piloted CY155: Explorations in Computing Fundamentals, which is an advanced version of CY105. While all the graded events remained the same between the two courses, the common programming content is covered at an accelerated pace in CY155. As a result, CY155 is able to devote approximately 17% of the course to exploring more advanced topics in computing. These advanced topics included Graphical User Interface (GUI) programming and use of advanced data structures such as unordered collections (i.e., dictionaries) and string manipulation methods. In addition, CY155 cadets complete two additional physical labs. In the computer components lab, cadets assemble and disassemble a desktop computer from base components. In the advanced hardware lab, students use complicated sensors (for example, LIDAR) to control devices attached to a micro-controller. All students in CY155 opted into the course and previously demonstrated aptitude (as evidenced by scores on a placement exam). The CY155 cadets performed significantly better on graded events than CY105 cadets.

b. In AY23, CY105 completed the transition to a new micro-controller used for the hardware lab after first piloting the device in AY22-2. Overall, we believe the new device helped improve cadets' ability to program as we saw a marked improvement in the associated learning outcome (LO); cadets meeting the standard for LO #3 increased by approximately 11%. Further, 35 cadets chose to complete a hardware project for the final summative presentation which represents a marked increase from prior years.

c. In AY23-2, CY105 added a lesson focused on artificial intelligence (AI). The lesson provides a broad overview of AI to include definitions of common terms used in the field, military applications of AI, and concerns regarding AI bias and ethics. To add the AI lesson, CY105 merged two previous lessons (Microsoft 365 and Microsoft Excel) into one lesson. However, an out-of-class assignment was added to ensure cadets have the foundation to work with the Microsoft products used at USMA and in the Army. We saw no change in associated assessment metrics from merging the two lessons.

d. In AY23-1, CY105 added an additional lesson focused on Cyber and the Department of Defense. This culminating lesson provided cadets the opportunity to examine the roles of various federal government departments in our nation's cybersecurity framework, with an in-depth focus on the Department of Defense. Cadets discussed cyberattacks in the context of the Law of Armed Conflict and applied their knowledge of cyberspace to the modern operational environment. To add the lesson, CY105 moved the Intellectual Property lecture to an optional Dean's hour lecture and provided a video recording to cadets unable to attend in-person. The new lesson supports LO #1 and we saw an increase of approximately 7% in cadets meeting the standard for this LO.

2. Summary of results of Student Learning Outcomes. An overview of each Learning Outcome (LO) is discussed in this section. More details are provided in Appendix B. CY105 met the standard for all Learning Outcomes:

Learning Outcomes	CY105	
	Results	Evidence
1	Meets Standard	Strong
2	Meets Standard	Strong
3	Meets Standard	Strong
4	Meets Standard	Strong
5	Meets Standard	Strong

3. Evidence: Evidence is based upon data collected from the TEE and WPRs. LO #3 is further supported by Python programming problems; cadets are required to understand a problem description, determine programming steps to solve the problem, and then implement a solution by writing Python code. For all LOs, multiple choice, true/false, matching, and selection type problems are used for evidence.

a. **Course Learning Outcome #1:** Examine legal, ethical, professional, and security issues associated with the cyber domain. CY105 meets LO #1. A total of 87% of the cadets met the standard.

b. **Course Learning Outcome #2:** Function as an ethical, secure, and competent user in West Point's immersive technology and cyber environment. CY105 meets LO #2. A total of 83% of the cadets met the standard.

c. **Course Learning Outcome #3:** Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process. CY105 meets LO #3. A total of 93% of the cadets met the standard.

d. **Course Learning Outcome #4:** Describe digital data representation, components, and functionality of computer hardware, and computer networks. CY105 meets LO #4. A total of 88% of the cadets met the standard.

e. **Course Learning Outcome #5:** Describe the attributes, opportunities, and challenges of the cyber domain. CY105 meets LO #5. A total of 89% of the cadets met the standard.

f. The percentage of cadets meeting the standards for the learning outcomes has fluctuated over the previous three academic years (summarized in the table below). We believe this is largely an artifact of changing conditions. During AY21, we conducted compassionate grading due to COVID. Normal grading practices were reinstituted in AY22 and AY23. Further, between AY21 and AY22, the TEE was changed from open book/notes to closed book and limited notes. This resulted in cadets performing worse on the TEE. Finally, in AY23, we increased the assessment data points to include WPR results whereas previously we only included TEE performance. Cadets habitually perform better on WPRs than on the TEE.

% of Cadets Meeting Standard			
LO	AY21	AY22	AY23
1	91.9%	80.9%	87.4%
2	90.7%	80.9%	82.8%
3	88.0%	81.7%	92.7%
4	91.9%	80.2%	87.5%
5	90.4%	80.8%	88.8%

4. Summary of results of APG-WGCD responsibility evaluations conducted during the year. An overview of each WGCD is discussed in this section. CY105 met the standard for all supported WGCD elements:

APG-WGCD	CY105	
	Results	Evidence
5.1	Meets Standard	Strong
5.4	Meets Standard	Strong
5.5	Meets Standard	Strong

5. Evidence: Evidence is primarily based upon data collected from the TEE and WPRs.

% of Cadets Meeting Standard	
WGCD	AY23
5.1	81.5%
5.4	82.7%
5.5	90.3%

In addition to the TEEs and WPRs, the hardware lab and summative presentation give cadets the opportunity to exercise and develop competencies related to WCGD 5.1 and 5.4. Further, the entire course supports WCGD 5.5. Finally, an LO-WGCD crosswalk shows how the course WGCD responsibilities are supported by the course LOs. As discussed in paragraphs 2 and 3 above, there is strong evidence that cadets are meeting the standard for the LOs which in turn provides support that cadets are meeting the standard for the associated course WGCD responsibilities.

WGCDs	Supporting Course Outcomes
5.1 Apply mathematics, science, and computing to model devices, systems, processes, or behaviors.	2, 4
5.4 Apply an engineering design process to create effective and adaptable solutions.	2, 3
5.5 Explain and apply computing and information technology concepts and practices in the context of the cyber domain.	1, 2, 3, 4, 5

Overall, our assessments indicate that cadets are developing the skills to be competent users in the cyber domain.

6. Proposed changes to the Course Learning Outcomes. We propose to change LO #3 which is currently “Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process.” We believe the LO is overly specific and attempts to itemize elements of computational thinking. Therefore, we propose to change LO #3 to: “Apply computational thinking to solve problems using a modern programming language.” This LO is also the core of the course and we will rearrange the LO order so that it is LO #1.

7. Significant planned or continuing-into-the-next-academic-year curricular changes.

a. For AY24, CY105 will implement a new interactive digital textbook that will replace the previous printed textbook. The daily in-class quizzes that were administered to assess lesson preparation will be replaced by pre-class participation activities within the digital textbook. The new interactive textbook should result in cadets that come to class

better prepared to discuss the course material and delve deeper into the course topics through in-class discussions and exercises.

b. In AY24, we will explore a new programming paradigm, AI-assisted programming, which uses generative AI to assist the programmer with developing code. We believe AI-assisted programming will allow cadets to focus more on computational thinking principles and less on the specific syntax of the programming language. We have created a working group that will examine how to integrate and assess AI-assisted programming into CY105.

c. In AY24, we plan to weave the character integration thread more explicitly into the course material. For example, over the summer, we worked with CIAG to incorporate character reflection into the existing digital footprint audit assignment which requires cadets to examine their social media presence.

8. Significant planned or implemented assessment process changes.

a. AY23 and beyond: The adoption of Canvas as the LMS of record allows us to assess all Course Learning Outcomes each year. Previously, we rotated the assessed outcomes because it was difficult to collect data.

b. AY23 and beyond: Instead of relying solely on the TEE for the assessment, we now include the assessments from WPRs as well. For AY24 and beyond, we will include data from writs and homework assignments as well. To do this, every question on each assignment and assessment are aligned to a specific learning outcome.

9. Updated current assessment timeline.

AY24	AY25	AY26	AY27
All Course Learning Outcomes	All Course Learning Outcomes	All Course Learning Outcomes	All Course Learning Outcomes

10. Point of contact is the undersigned at daniel.filcik@westpoint.edu or 845-938-5995.

DANIEL J. FILCIK
MAJ, EN
CY105 Course Director

Appendix A: CY105 Course Learning Outcomes

1. Examine legal, ethical, professional, and security issues associated with the cyber domain.
2. Function as an ethical, secure, and competent user in West Point's immersive technology and cyber environment.
3. Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process.
4. Describe digital data representation, components and functionality of computer hardware, and computer networks.
5. Describe the attributes, opportunities, and challenges of the cyber domain.

Appendix B: Assessment Strategy for CY105

1. Course Learning Outcome #1: Examine legal, ethical, professional, and security issues associated with the cyber domain. These metrics come from the WPRs and TEE.

Category	Description	Standard	Result
III	Meets or Exceeds Standard	.751 – 1.00	76.3%
II	Marginally Meets Standard	.651 – .75	11.1%
I	Fails Standard	0.00 – 0.65	12.6%
Course Standard: 70% in Category II & III			

2. Course Learning Outcome #2: Function as an ethical, secure, and competent user in West Point's immersive technology and cyber environment. These metrics come from the WPRs and TEE.

Category	Description	Standard	Result
III	Meets or Exceeds Standard	.751 – 1.00	70.5%
II	Marginally Meets Standard	.651 – .75	12.3%
I	Fails Standard	0.00 – 0.65	17.1%
Course Standard: 70% in Category II & III			

3. Course Learning Outcome #3: Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process. These metrics come from the WPRs and TEE.

Category	Description	Standard	Result
III	Meets or Exceeds Standard	.751 – 1.00	80.6%
II	Marginally Meets Standard	.651 – .75	12.1%
I	Fails Standard	0.00 – 0.65	7.3%
Course Standard: 70% in Category II & III			

4. Course Learning Outcome #4: Describe digital data representation, components and functionality of computer hardware, and computer networks. These metrics come from the WPRs and TEE.

Category	Description	Standard	Result
III	Meets or Exceeds Standard	.751 – 1.00	77.3%
II	Marginally Meets Standard	.651 – .75	10.2%
I	Fails Standard	0.00 – 0.65	12.5%
Course Standard: 70% in Category II & III			

5. Course Learning Outcome #5: Describe the attributes, opportunities, and challenges of the cyber domain. These metrics come from the WPRs and TEE.

Category	Description	Standard	Result
III	Meets or Exceeds Standard	.751 – 1.00	80.8%
II	Marginally Meets Standard	.651 – .75	8.0%
I	Fails Standard	0.00 – 0.65	11.2%
Course Standard: 70% in Category II & III			



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB E

AY2022 COURSE ASSESSMENT (EXSUM)



**DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
ELECTRICAL ENGINEERING AND COMPUTER SCIENCE
WEST POINT, NY 10996**

MADN-EECS-C

14 September 2022

MEMORANDUM THRU

LTC Malcolm Haynes, Core Program Director, Department of Electrical Engineering and Computer Science, USMA, West Point, NY 10996

COL James Raftery Jr., Professor and Head, Department of Electrical Engineering and Computer Science, USMA, West Point, NY 10996

FOR Dean of the Academic Board, USMA, West Point, NY 10996

SUBJECT: Annual Executive Summary for CY105, AY22

1. Status of curricular change initiatives completed during the current or previous academic year and results of follow-up assessment.

a. Each semester, CY105 conducts a hardware lab in which cadets learn to program a microprocessor and attached sensors and actuators. In AY22-2, CY105 piloted two sections using the Circuit Playground Bluefruit micro-controller in lieu of the NUCLEO-F411RE for the hardware lab. The Bluefruit device was selected because it is a more capable board as it has seven built in sensors and 10 RGB LEDs. The kit also comes with a color display. Cadets were able to use the Bluefruit device with no significant issues. They were also able to spend more time experimenting with the device during the lab because it required less soldering. As a result, CY105 will use the Bluefruit micro-controller in all future sections.

b. In AY22-1, CY105 added a lesson focused on spreadsheet fundamentals via Microsoft Excel. Based on the usage of Excel in academia and throughout the Army, cadets benefit from establishing a foundation early in their cadet career.

c. In AY22-1, CY105 transitioned to using a collaborative Integrated Development Environment (IDE) for all programming problems. The environment provides cadets with an up-to-date Python interpreter. The chief benefit of the new environment is it enables faculty to monitor and assist cadets in real time. Additionally, it allows a cadet to display and explain their code to the rest of the class thus facilitating peer learning.

2. Summary of results of Student Learning Outcome. An overview of each Learning Outcome (LO) is discussed in this section. More details are provided in Appendix B. CY105 met the standard for all Learning Outcomes:

Learning Outcomes	CY105	
	Results	Evidence
1	Meets Standard	Strong
2	Meets Standard	Strong
3	Meets Standard	Strong
4	Meets Standard	Strong
5	Meets Standard	Strong

3. Evidence: Evidence is based upon data collected from the TEE. For LO#3, Python code and the standard question types for the other LOs were used. All cadets are required to understand a problem description, determine programming steps to solve the problem, and then implement a solution by writing Python code. For all other LOs, multiple choice, true/false, matching, and selection type problems, implemented in Canvas, were utilized for evidence.

a. **Course Learning Outcome #1:** Examine legal, ethical, professional, and security issues associated with cyber technology. CY105 meets LO#1. A total of 80.2% of the cadets met the standard.

b. **Course Learning Outcome #2:** Function as an ethical, secure, and competent user in West Point's immersive technology and cyber environment. CY105 meets LO#2. A total of 79.4% of the cadets met the standard.

c. **Course Learning Outcome #3:** Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process. CY105 meets LO#3. A total of 78.4% of the cadets met the standard.

d. **Course Learning Outcome #4:** Describe digital data representation, components, and functionality of computer hardware, and computer networks. CY105 meets LO#4. A total of 77.6% of the cadets met the standard.

e. **Course Learning Outcome #5:** Describe the attributes, opportunities, and challenges of the cyber domain. CY105 meets LO#5. A total of 80.1% of the cadets met the standard.

f. The percentage of cadets meeting the standards for the learning outcomes is summarized in the table below. There are relatively large differences between academic years. We surmise that the AY20 reflects the initial difficulties stemming from the COVID pandemic. In contrast, AY21 reflects compassionate grading in response to the COVID pandemic. Finally, AY22 shows the return to fully in-person classes with standard grading practices. Of note, between AY21 and AY22, the TEE was moved from open book/notes to closed book and limited notes.

% of Cadets Meeting Standard			
LO	AY20	AY21	AY22
1	72.5%	91.9%	80.9%
2	79.0%	90.7%	80.9%
3	71.3%	88.0%	81.7%
4	74.2%	91.9%	80.2%
5	72.5%	90.4%	80.8%

4. Summary of results of APG-WGCD responsibility evaluations conducted during the year.

WGCDs	Supporting Course Outcomes
5.1 Apply mathematics, science, and computing to model devices, systems, processes, or behaviors.	2, 4
5.4 Apply an engineering design process to create effective and adaptable solutions.	2, 3
5.5 Explain and apply computing and information technology concepts and practices in the context of the cyber domain.	1, 2, 3, 4, 5

5. We saw adequate cadet attainment in APG-WGCD 5.1, 5.4, and 5.5 based on an assessment of all five Course Learning Outcomes. Our two micro-processor labs and summative presentation gave cadets opportunity to exercise and develop competencies with elements 5.1 and 5.4. The entire course supports element 5.5; our assessments indicate that cadets are developing the skills to be competent users in the cyber domain.

6. Significant planned or continuing-into-the-next-academic-year curricular changes.

a. For AY23, CY105 will add a lesson titled Cyber and the DoD. This lesson will provide a brief overview of national and international cyberspace laws. With this framework, the class will then discuss the impact of cyberspace on the modern operating environment.

b. For AY23 and beyond, CY105 will fully implement the transition to using the Circuit Playground Bluefruit for the hardware lab.

7. Significant planned or implemented assessment process changes.

a. AY222 and beyond: The IDE discussed in paragraph 1c is designed to be collaborative. As a result, LO#3 for the TEE in AY222 and for quizzes, WPRs, and TEEs in AY23 and beyond, was/will be assessed via code tracing questions and a single programming question in Canvas. The cadets will not have access to the IDE during the exams.

b. AY23 and beyond: Since programming and understanding of computer components are fundamental aspects of CY105, Course Learning Outcomes #3 and #4 are assessed every year. In addition, one other Course Learning Outcome is assessed on a rotating basis.

c. AY23 and beyond: Instead of relying solely on the TEE for the assessment, we intend to also include the assessments from most of the assignments, daily quizzes, and the WPRs. To do this, each question on the quizzes/WPRs or each assignment will be aligned to a specific learning outcome.

d. The assessment for Learning Outcomes 1, 2, 4, and 5 will come from Canvas test questions. Learning Outcome 3's assessment will come from Canvas test questions, weekly programming problems, and their final project. We continue to update and refine our Canvas test banks to accurately reflect what we expect students to learn and to stay current with the dynamic nature of the world of computing.

8. Updated current assessment timeline.

AY23	AY24	AY25	AY26
All Course Learning Outcomes due to course modifications	Course Learning Outcomes 3, 4 Course Learning Outcome 1	Course Learning Outcomes 3, 4 Course Learning Outcome 2	All Course Learning Outcomes

9. Point of contact is the undersigned at jeremie.regnier@westpoint.edu or 845-938-2407.

JEREMIE M. REGNIER
MAJ, SC
CY105 Course Director

Appendix A: CY105 Course Learning Outcomes

1. Examine legal, ethical, professional, and security issues associated with cyber technology.
2. Function as an ethical, secure, and competent user in West Point's immersive technology and cyber environment.
3. Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process.
4. Describe digital data representation, components and functionality of computer hardware, and computer networks.
5. Describe the attributes, opportunities, and challenges of the cyber domain.

Appendix B: Assessment Strategy for CY105

1. Course Learning Outcome #1: Function as an ethical, secure, and competent user in West Point's immersive technology and cyber environment and Course Learning. These metrics come solely from a 40-question multiple choice TEE.

Category	Description	Standard	Result
III	Meets or Exceeds Standard	.751 – 1.00	80.2%
II	Marginally Meets Standard	.651 – .75	0.7%
I	Fails Standard	0.00 – 0.65	19.1%
Course Standard: 70% in Category II & III			

2. Outcome #2: Examine legal, ethical, professional, and security issues associated with cyber technology. These metrics come solely from a 40-question multiple choice TEE.

Category	Description	Standard	Result
III	Meets or Exceeds Standard	.751 – 1.00	79.4%
II	Marginally Meets Standard	.651 – .75	1.5%
I	Fails Standard	0.00 – 0.65	19.0%
Course Standard: 70% in Category II & III			

3. Course Learning Outcome #3: Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process. For AY221, these metrics come from a four-problem programming TEE. For AY222, these metrics come from fifteen code tracing questions and a single code writing question on the TEE.

Category	Description	Standard	Result
III	Meets or Exceeds Standard	.751 – 1.00	78.4%
II	Marginally Meets Standard	.651 – .75	3.3%
I	Fails Standard	0.00 – 0.65	18.3%
Course Standard: 70% in Category II & III			

4. Course Learning Outcome #4: Describe digital data representation, components and functionality of computer hardware, and computer networks. These metrics come solely from a 40-question multiple choice TEE.

Category	Description	Standard	Result
III	Meets or Exceeds Standard	.751 – 1.00	77.6%
II	Marginally Meets Standard	.651 – .75	2.6%
I	Fails Standard	0.00 – 0.65	19.7%
Course Standard: 70% in Category II & III			

MADN-EECS

SUBJECT: Annual Executive Summary for CY105, AY22

5. Course Learning Outcome #5: Describe the attributes, opportunities, and challenges of the cyber domain. These metrics come solely from a 40-question multiple choice TEE.

Category	Description	Standard	Result
III	Meets or Exceeds Standard	.751 – 1.00	80.1%
II	Marginally Meets Standard	.651 – .75	0.7%
I	Fails Standard	0.00 – 0.65	19.2%
Course Standard: 70% in Category II & III			



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB F

AY2021 COURSE ASSESSMENT (EXSUM)



**DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
ELECTRICAL ENGINEERING AND COMPUTER SCIENCE
WEST POINT, NY 10996**

MADN-EC

15 September 2021

MEMORANDUM THRU

LTC Malcolm Haynes, Core Program Director, Department of Electrical Engineering and Computer Science, USMA, West Point, NY 10996

COL James Raftery Jr., Professor and Head, Department of Electrical Engineering and Computer Science, USMA, West Point, NY 10996

FOR Dean of the Academic Board, USMA, West Point, NY 10996

SUBJECT: Annual Executive Summary for CY105, AY21

1. Status of curricular change initiatives completed during the current or previous academic year and results of follow-up assessment.

a. The COVID-19 pandemic affected our ability to have full, in-person, classes. Our maximum in-person classroom constraint was nine cadets (50% of normal class size). All CY105 sections implemented a hybrid format, with half the section attending in-person while the other half attended remote via MS Teams. Although not an ideal academic environment, we made effective use of MS Teams to engage with cadets not able to attend in-person. We maintained the flexibility to administer WPRs and the TEEs in-person or remotely, greatly benefitting those in a hold or quarantine status. Our CY105 instructors mastered the art of the hybrid classroom and effectively delivered course content to students, regardless of their location.

b. In AY212, CY105 piloted five sections using the NUCLEO-F411RE micro-controller in lieu of the Arduino UNO. The genesis for the pilot was to assess the benefits of using a micro-controller that supports the Python programming language, the same language we are teaching in class. Historically, students struggled to understand the C-like code required for the Arduino. The NUCLEO provides cadets the ability to better understand code and write code on a micro-controller. The pilot sections improved cadet comprehension and involvement during our two hardware labs. We will utilize the NUCLEO micro-controller in all future sections.

2. Summary of results of Student Learning Outcome. An overview of each Learning Outcome (LO) is discussed in this section. More details are provided in Appendix B. CY105 met the standard for all Learning Outcomes:

Learning Outcomes	CY105	
	Results	Evidence
1	Meets Standard	Strong
2	Meets Standard	Strong
3	Meets Standard	Strong
4	Meets Standard	Strong
5	Meets Standard	Strong

a. Evidence: Evidence is based upon data collected from graded events. For LO#3, Python code was utilized. All cadets are required to understand a problem description, determine programming steps to solve the problem, and then implement a solution by writing Python code. For all other LOs, multiple choice, true/false, matching, and selection type problems, implemented in Canvas, were utilized for evidence.

b. **Course Learning Outcome #1:** Examine legal, ethical, professional, and security issues associated with cyber technology; and **Course Learning Outcome #2:** Function as an ethical, secure, and competent user in West Point's immersive technology and cyber environment. The test questions used to evaluate both Learning Outcomes have a large overlap and were consolidated. A total of 91.9% of the cadets met the standard.

c. **Course Learning Outcome #3:** Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process. CY105 meets LO#2. A total of 90.7% of the cadets met the standard.

d. **Course Learning Outcome #4:** Describe digital data representation, components, and functionality of computer hardware, and computer networks. A total of 88.0% of the cadets met the standard.

e. **Course Learning Outcome #5:** Describe the attributes, opportunities, and challenges of the cyber domain. A total of 90.4% of the cadets met the standard.

3. Summary of results of APG-WGCD responsibility evaluations conducted during the year.

WGCDs	Supporting Course Outcomes
5.1 Apply mathematics, science, and computing to model devices, systems, processes, or behaviors.	2, 4
5.4 Apply an engineering design process to create effective and adaptable solutions.	2, 3
5.5 Explain and apply computing and information technology concepts and practices in the context of the cyber domain.	1, 2, 3, 4, 5

We saw adequate cadet attainment in APG-WGCD 5.1, 5.4, and 5.5 based on an assessment of all five Course Learning Outcomes. Our two micro-processor labs and summative presentation gave cadets opportunity to exercise and develop competencies with elements 5.1 and 5.4. The entire course supports element 5.5; our assessments indicate that cadets are developing the skills to be competent users in the cyber domain.

4. Significant planned or continuing-into-the-next-academic-year curricular changes.

a. For AY22, we are adding a second lesson covering the Microsoft 365 software suite specifically designed to enhance cadet's ability to use Excel spreadsheets. Based on the usage of Excel in academia and throughout the Army, we want to provide cadets a deeper foundation with spreadsheets so they can utilize effectively in the future.

b. AY22 and beyond: We are implementing a new web-based Integrated Development Environment (IDE) for all programming problems. Our current IDE (Gadget) does not support current versions of Python, is not maintained, and has several issues affecting cadets (data not saving properly). Our new IDE provides an up-to-date Python interpreter, robust software support, and a collaborative coding environment that allows instructors to monitor and assist students in real-time as they are working on assignments.

c. For AY22, we are adding a new drag-and-drop programming tool based on Google's Blockly. This will provide an additional tool to assist cadets in understanding programming constructs (sequence, iteration, and selection). This in no way replaces the requirement to write code and understand programming syntax.

5. Proposed changes to the Student Learning Outcomes. We made minor administrative changes to the Student Learning Outcomes for AY21 which better reflect the transition from Information Technology to Cyber. These changes did not alter course content. Our five Student Learning Outcomes can be found in Appendix A.

6. Significant planned or implemented assessment process changes.

a. AY22 and beyond: Since programming and understanding of computer components are fundamental aspects of CY105, Course Learning Outcomes #3 and #4 are assessed every year. In addition, one other Course Learning Outcome is assessed on a rotating basis.

b. AY22 and beyond: In AY21 we assessed Learning Outcomes #1 and #2 using the same questions. There is enough differentiation in our test questions that we can align those questions with either Learning Outcome #1 or Learning Outcome #4 (with some overlap). Therefore, starting in AY22, we will assess these outcomes from separate questions.

MADN-EC

SUBJECT: Annual Executive Summary for CY105, AY21

c. The assessment for Learning Outcomes 1, 2, 4, and 5 will come from Canvas test questions. We continue to update and refine our Canvas test banks to accurately reflect what we expect students to learn and to stay current with the dynamic nature of the world of computing.

d. The assessment for Learning Outcome 2 will come from graded programming problems.

7. Updated current assessment timeline.

AY22	AY23	AY24	AY25
- All Course Learning Outcomes to assess curriculum changes.	- Course Learning Outcomes 3, 4 - Course Learning Outcome 5	- Course Learning Outcomes 3, 4 - Course Learning Outcome 1	- All Course Learning Outcomes

8. Point of contact is the undersigned at scott.gale@westpoint.edu or 845-938-3642.

SCOTT D. GALE
LTC, AV
CY105 Course Director

MADN-EC

SUBJECT: Annual Executive Summary for CY105, AY21

Appendix A: CY105 Course Learning Outcomes

1. Examine legal, ethical, professional, and security issues associated with cyber technology.
2. Function as an ethical, secure, and competent user in West Point's immersive technology and cyber environment.
3. Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process.
4. Describe digital data representation, components and functionality of computer hardware, and computer networks.
5. Describe the attributes, opportunities, and challenges of the cyber domain.

Appendix B: Assessment Strategy for CY105

1. Course Learning Outcome #1: Function as an ethical, secure, and competent user in West Point's immersive technology and cyber environment and Course Learning Outcome #4: Examine legal, ethical, professional, and security issues associated with cyber technology. The same multiple-choice questions are used to support both Learning Outcome #1 and #4. These metrics come solely from a 40-question multiple choice TEE.

Category	Description	Standard	Result
III	Meets or Exceeds Standard	.751 – 1.00	77.2%
II	Marginally Meets Standard	.651 – .75	14.7%
I	Fails Standard	0.00 – 0.65	8.1%
COURSE STANDARD: 70% in Category II & III			

2. Course Learning Outcome #2: Apply sequential logic, control flow statements, and code libraries using a modern programming language, algorithmic thinking, and a problem-solving process. These metrics come from a four-problem programming TEE.

Category	Description	Standard	Result
III	Meets or Exceeds Standard	0.751 – 1.00	84.7%
II	Marginally Meets Standard	0.651 - 0.75	6.0%
I	Fails Standard	0.00 – 0.650	9.3%
COURSE STANDARD: 70% in Category II & III			

3. Course Learning Outcome #3: Describe digital data representation, components and functionality of computer hardware, and computer networks. These metrics come solely from a 40-question multiple choice TEE.

Category	Description	Standard	Result
III	Meets or Exceeds Standard	.751 – 1.00	73.4%
II	Marginally Meets Standard	.651 – .75	14.6%
I	Fails Standard	0.00 – 0.65	12.0%
COURSE STANDARD: 70% in Category II & III			

4. Course Learning Outcome #5: Describe the attributes, opportunities, and challenges of the cyber domain. These metrics come solely from a 40-question multiple choice TEE.

Category	Description	Standard	Result
III	Meets or Exceeds Standard	.751 – 1.00	73.5%
II	Marginally Meets Standard	.651 – .75	16.9%
I	Fails Standard	0.00 – 0.65	9.6%
COURSE STANDARD: 70% in Category II & III			



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION II

EXAMINATIONS



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB G

WPR 1

CY105 AY26-1 WPR 1

ACCESS CODES:

	<u>Day 1</u>	<u>Day 2</u>	<u>Alternate</u>
Part A	crimson	silent	velvet
Part B	lantern	echo	thunder
H.A.S.	juniper		

GUIDANCE:

8-10x knowledge check questions (35 points total)

- Every option must be 1 point each
- True False, multiple choice, multi-answer, matching all valid question types

8x Code Tracing Questions (25 points total)

- Every option must be 1 point each

1x Open-Ended Question (15 points)

- Textbox, holistic rubric

1x Code Writing Question

- 25 points, holistic rubric
- Should cover iteration (for), basic types, built-in functions

LEGEND: Updates to be made. Question Answers. Feedback Section. Configured (XO-Only)

KC/CT Question Template:

Question Title: Lesson Title X

Question Type: update

Question Points: update

Question Stem: update

****insert question stem****

Feedback:

Knowledge Check Questions:

Use tables for Matching, True/False, DropDown

Feedback: Enter problem feedback that students will see here

Commented [HT1]: Insert your KC questions here and comment with your initials (e.g. HT) to start the comment thread for your question.

CY105 AY26-1 WPR 1

MS365 (7 PTS)

Question Title: Microsoft 365

Question Type: Matching

Question Points: 4

Question Stem: Select the correct Microsoft 365 tool for the given application.

Send an encrypted and digitally signed message	Outlook
Store files and notes for personal reference	OneDrive
Collaborate real-time with several people on a project	Teams
Maintain a repository for often-referenced company data	SharePoint

Feedback:

Send an encrypted and digitally signed message	Only Outlook will send encrypted and digitally signed emails.
Store files and notes for personal reference	OneDrive is cloud storage, intended primarily for the assigned user.
Collaborate real-time with several people on a project	Teams allows real-time chat, voice, and video communication with team members.
Maintain a repository for often-referenced company data	SharePoint will host folders and files that can be viewed and edited by other authorized users.

Question Title: Microsoft 365 II

Question Type: Dropdown

Question Points: 3

Question Stem: Determine whether each of the following statements is True or False about OneDrive and similar cloud services:

They allow you to backup your data to the cloud `True`

They have features that enable you to permanently delete your data securely from the cloud as needed. `False`

They can provide you with access to your data even if you do not have your personal computer with you. `True`

Feedback:

Commented [CJWL2]: JC5

Commented [MH3]: MSH

Commented [MS4]: Let's split this up into two different options. "They allow you to quickly backup your data to the cloud"

and

"The allow you to quickly and securely delete your data from the cloud"

Commented [LP5]: I could not find in readings. Readings most talk about how you can't permanently delete

Commented [MS5R2]: Agreed. Let's just replace this option with one of the two I list above.

CY105 AY26-1 WPR 1

- The correct answer is **True**. Backing up data to the cloud is easy to do.
- The correct answer is **False**. In general, it is very difficult to guarantee that any data uploaded to the cloud is permanently deleted.
- The correct answer is **True**. In general, you can access your cloud-stored data from any computer.

Computers and Data

Question Title: Computers and Data I (4 PTS)

Question Type: Matching

Question Points: 4

Question Stem: Rank these storage solutions in ascending order of speed (1 – slow, 4 – Extremely Fast)

Permanent Storage (HDD, SSD)	1
CPU Registers	4
Main Memory (DRAM)	2
L1/L2/L3 Cache (SRAM)	3

Feedback:

Permanent Storage (HDD, SSD)	Huge capacity, cheap, slow
CPU Registers	Tiny capacity, highly expensive, extremely fast
Main Memory (DRAM)	Medium capacity, medium price, fast
L1/L2/L3 Cache (SRAM)	Low capacity, expensive, faster

AI and Data (7 PTS)

Question Title: AI and Data 1

Question Type: Fill in the Blank (Dropdown)

Question Points: 3

Question Stem: ChatGPT is an example of a(n) ____.

True	ChatGPT is an example of a(n) <u>Large Language Model</u> .
False	ChatGPT is an example of a(n) <u>Turing Test</u> .
False	ChatGPT is an example of a(n) <u>Artificial General Intelligence</u> .

Commented [JH6]: JEH

Commented [MH7]: MSH

Commented [JH8]: JEH

Commented [MS9]: While I like the theme of this question, I don't like the implementation. In general, I am against 1-point questions. Is there a way to transform this question so that way it is worth more points, but have many options?

Maybe a matching question "select the term that matches its definition"? Two of the options could be Large Language Model, Turing Test, maybe ANI could be a third, and Deep Learning for a fourth?

Commented [JH9R2]: I'm not sure I understand the suggestion. I could turn this into a multiple select, but I'm not sure that's any better.

Commented [MS9R3]: That will make it a 4 point question, since it is matching, instead of a 1 point question.

Commented [MS9R4]: Let's discuss how to make this into a 2-pointer

CY105 AY26-1 WPR 1

Feedback:

Large Language Model	ChatGPT is a tool for general-purpose language understanding and generation
Turing Test	A Turing Test is an abstract tool to determine how close a computer's response is to a human's
Artificial General Intelligence	ChatGPT cannot perform *any* task that a human can. It is still an example of Artificial Narrow Intelligence.

Question Title: AI and Data II

Question Type: DropDown

Question Points: 4

Question Stem: Indicate if the following are factors that led to the recent "AI renaissance".

Answer choices: 'Not a contributing factor', 'Contributing factor'.

Commented [AK10]: ALK

Emergence of quantum computing	Not a contributing factor
Low-cost, powerful, specialized hardware (GPUs)	Contributing factor
Blockchain smart contracts	Not a contributing factor
Vast quantities of freely available data	Contributing factor

Feedback:

Emergence of quantum computing	Quantum computing does not contribute to the effective implementation of AI.
Low-cost, powerful, specialized hardware (GPUs)	Use of powerful, specialized hardware at affordable pricing has enabled the necessary computational requirements of modern AI.
Blockchain smart contracts	The decentralized contracts enabled by blockchain are not utilized in AI.
Vast quantities of freely available data	The quantity (and phenomenal growth) of data available on the Internet is required for the effective training of modern AI.

Microsoft 365 GenAI Lab (4 pts)

Question Type: Matching

Question points: 4

Question Stem: Match the term with its best definition

Commented [MS11]: SJM

CY105 AY26-1 WPR 1

An artificial intelligence designed to help users accomplish a business-focused task	agent
A technique for adding context to prompts by using information from uploaded files and websites	grounding
The smallest unit of language that a large language model uses	token
Controls the creativity and variability of a large language model's output	temperature

Problem Solving and Algorithms (9 PTS)

Question Title: PS&A 1

Question Type: Dropdown

Question Points: 4

Question Stem: Which of the following are part of the process of decomposing a problem and creating an algorithm to solve it? Note that there could be multiple true options.

True	Determining the input, output, and assumptions
True	Breaking a complex task into smaller, manageable subtasks
True	Determining the order and flow of subtasks
False	Implementing subtasks by correctly calling built-in functions

Feedback:

Determining the input, output, and assumptions	Yes, do this at the very beginning of problem solving.
Breaking a complex task into smaller, manageable subtasks	Yes, this is problem decomposition.
Determining the order and flow of subtasks	Yes, the steps of an algorithm must be in an ordered sequence.
Implementing subtasks by correctly calling built-in functions	No, implementing the steps of an algorithm goes beyond problem decomposition and writing an algorithm. Built-in functions are programming language-specific, and an algorithm must be generic enough that we could accurately program it in any programming language.

Question Title: PS&A 2

Question Type: drop-down

Commented [TW12]: TW

Commented [HT13]: @Kedrowitsch, Alexander L LTC

Commented [AK14]: ALK

Commented [MS15]: change to drop down.

CY105 AY26-1 WPR 1

Question Points: 5

Question Stem: Which characteristics are true of pseudocode?

It is more precise than natural language	True
Only readable by programmers	False
Not executable by a computer	True
Uses structural elements like sequencing and loops	True
Written in a specific programming language	False

Commented [HT16]: We went over on points; ended up 77/75. cutting these two options

Feedback:

It is more precise than natural language	Pseudocode uses structured elements and language to be less ambiguous than natural language.
Only readable by programmers	While not a natural language, pseudocode is still readable by the general population.
Not executable by a computer	Pseudocode is not a direct implementation of a program and not executable on any computer.
Uses structural elements like sequencing and loops	The structured elements of sequencing, loops, and conditionals are key attributes enabling the precision of pseudocode.
Written in a specific programming language	Pseudocode retains sufficient generality to apply to many different systems and is not written in an individual programming language.

Code Tracing Questions:

Python Basics (7 PTS)

Python Basics

Question Title: Python Basics I

Question Type: Drop-down

Question Points: 4

Question Stem: Which of the following variable assignments are valid?

<code>class = 'AI105'</code>	Invalid
<code>str(14.5) = x_string</code>	Invalid

Commented [HT17]: Insert your CT questions here and comment with your initials (e.g. HT) to start the comment thread for your question.

Commented [HT17R2]: @Kedrowitsch, Alexander L LTC reminder to input assigned questions for WPR1

Commented [MS18]: Need to eliminate 4 points

Commented [WL19]: TW

Commented [TW20]: TW

CY105 AY26-1 WPR 1

<code>_mY_vAr1aBle_X = 14.5</code>	Valid
<code>a_replicated = 'a' * 10</code>	Valid

Feedback:

<code>class = 'AI105'</code>	This is invalid because class is a Python keyword.
<code>str(14.5) = x_string</code>	This is invalid because the variable, x_string, needs to be on the left side of the assignment operator.
<code>_mY_vAr1aBle_X = 14.5</code>	This is valid even though it's an ugly variable name.
<code>a_replicated = 'a' * 10</code>	This is valid even though the * operator seems unnatural between a string and an integer.

Question Title: Python Basics II

Question Type: dropdown

Question Points: 3

Question Stem: You type in the following program. When you run it, you see no output. Why?

```
1 def main():
2     print("Hello, world!")
3
4     main()
```

False	You misspelled "world".
False	The Python interpreter displays an error
True	Line 4 is indented and should not be

Commented [HT21]: MF

Built-in Functions (7 PTS)

Question Title: Built in Functions 1

Question Type: Matching/ Multiple Choice?

Question Points: 4

Question Stem: Match the output of the expression on the left with the output on the right

Expression	Output
<code>len(str(abs(-42)))</code>	2

Commented [MF22]: MF

CY105 AY26-1 WPR 1

<code>max([3,7,2]) - min([3,7,2])</code>	5
<code>int("10")//3</code>	3
<code>round(15/4)</code>	4

Distractors: Error, 1

Feedback:

<code>len(str(abs(-42)))</code>	Evaluates to 2 because the absolute value is 42, which becomes the string '42' with two characters
<code>max([3,7,2]) - min([3,7,2])</code>	Evaluates to 5 because the maximum is 7 and the minimum is 2, 7-2=5
<code>int("10")//3</code>	Evaluates to 3 because the string is converted to the integer 10 and the integer division truncates the result to 3
<code>round(15/4)</code>	Evaluates to 4 because 15 divided by 4 is 3.75 which rounds up to 4

Question Title: Built in Functions II

Question Type: Multiple selection

Question Points: 3

Question Stem: Consider the following program (assume the user properly types numbers):

```
1 def main():
2     grade_try1 = float(input("Enter score for 1st attempt: "))
3     grade_try2 = float(input("Enter score for 2nd attempt: "))
4     final_grade = max(grade_try1, grade_try2)
5     print("Your final score is:", final_grade)
6     main()
```

Indicate if the following statements are True or False

The input() function on line 2 is fruitful.	True
After line 4, final_grade will be the larger of the two arguments provided to the max() function.	True
The print() function on line 5 is fruitful.	False

Feedback:

True	The input() function is always fruitful.
True	The max() function returns the largest value within the sequence it is given.
False	The built-in print() function is not fruitful.

Commented [HL23]: MH

CY105 AY26-1 WPR 1

Iteration – Part I (For Loops) (6 PTS)

Question Title: For Loops

Question Type: Dropdown

Question Points: 4

Question Stem: Select whether each statement about the following snippet of code is true or false.

```
count = 0
for item in [0, 0, 0, 0]:
    count = count + 1
print(count)
```

We could change the values of the items in the list without causing any effect on the output.	True
It prints 4 one time.	True
It prints 0 four times.	False
The variable named <code>item</code> is an example of an accumulator.	False

Feedback:

We could change the values of the items in the list without causing any effect on the output.	This is true. The body of the loop does not reference the variable named <code>item</code> at all.
It prints 4 one time.	This is true. The loop causes the count variable to be incremented four times, so the snippet prints 4 at the end.
It prints 0 four times.	This is false. The loop causes the count variable to be incremented four times, so the snippet prints 4 at the end.
The variable named <code>item</code> is an example of an accumulator.	This is false. We have an example of the accumulator pattern in this snippet, but the accumulator variable is named <code>count</code> .

Question Title: For Loops-II

Question Type: Multiple choice (Dropdown?, 1 pt ea)

Question Points: 4

Question Stem: Which numbers are printed as part of this code's output?

```
for item in range(2, 11, 3):
    print(item)
```

Commented [HT24]: @Catudal, Joseph W LTC missing iteration problem

Commented [CL24R2]: complete, below

Commented [TW25]: TW

Commented [TW25R2]: @Matthews, Suzanne Dr I have changed this question according to our discussion during the sync meeting.

Commented [CL26]: JCS

Commented [MS27]: Again, this violates my rule of having so many points being associated with a single question. You can rewrite this one to be:

"Indicate if the following numbers appear as part of the output of the following code snippet"

And then have 2, 3,4,5 as options (or something similar).

CY105 AY26-1 WPR 1

3	False
5	True
8	True
11	False

Feedback:

3	Incorrect. range(2,11,3) would start at 2, but skip 3. 3 is the step value.
5	Correct. 5 would be the second value produced (2+3).
8	Correct. 8 would be the third and final value produced (2+3+3).
11	Incorrect. This is the stop value, which is not included in the range(2,11,3) output.

List and Strings (4 PTS)

Commented [CL28]: JCS

Question Title: Lists and Strings

Question Type: Drop-Down

Question Points: 4

Question Stem: Indicate whether the characteristic is true of strings, lists, or both lists and strings.

Can use positive and negative numbers for indexing	Lists and Strings
Elements may be of multiple data types	Lists
Cannot modify elements after assignment (NOT mutable)	Strings
Can use both concatenation (+) and replication (*) operators	Lists and Strings

Feedback:

Can use positive and negative numbers for indexing	Lists and Strings. Both lists and strings can reference their elements by positive and negative index.
Elements may be of multiple data types	Lists. While lists may contain mixed data types between elements, strings only consist of character elements.

CY105 AY26-1 WPR 1

Cannot modify elements after assignment (NOT mutable)	Strings. Although variables containing strings may be reassigned, individual elements cannot be changed in strings. List elements can be changed in place.
Can use both concatenation '+' and replication '*' operators	Lists and Strings. The use of concatenation and replication operators are available for both lists and strings and result in a new list or string that can be used or assigned.

Distractors: Neither lists nor Strings

List and Strings (5 PTS)

Question Title: Lists and Strings II

Question Type: Dropdown

Question Points: 5

Question Stem: Determine what the output to the display will be for each of the `print()` statements in the following code:

```
01 def main():
02     # [ Last, First, Company, GPA, HGHT, WGHT, AFT ]
03     last_name = "Johnson"
04     first_name = "Mortimer"
05     CDT_Exemplar = [last_name, first_name, "A1", 4, 67, 130, 280]
06     print(last_name[0])
07     print(len(CDT_Exemplar[-6]))
08     print(type(CDT_Exemplar[3]))
09     print(CDT_Exemplar[7])
10
11
12
13 main()
```

- A) What prints out when line 8 is executed? **J**, Mortimer, M, Johnson, Error
- B) What prints out when line 9 is executed? **8**, 7, None, Error
- C) What prints out when line 10 is executed? **<class 'int'>**, <class 'float'>, <class 'bool'>, <class 'str'>, <class 'list'>
- D) What prints out when line 11 is executed? **Error**, Johnson, Null, A1
- E) How many unique data types are stored in `CDT_Exemplar`? 1, **2**, 3, 7, 8

Feedback:

Commented [HT29]: @Matthews, Suzanne Dr I added this distractor during configuration—easy to remove if you want

Commented [MS29R2]: Good distractor.

Is it "neither or" or "neither nor"? I always thought it was the latter.

Commented [HT29R3]: I think you're right!

Commented [HL30]: MH

CY105 AY26-1 WPR 1

- A) `last_name[0]` - the answer is 'J' since the `last_name[0]` represents the first character in that string ("J")
- B) `len(CDT_Exemplar[-6])` - the answer is 8 since `[-6]` selects the sixth item in the `CDT_Exemplar` list from the right ("Mortimer") and the `len()` command gives the length of that string (8).
- C) `type(CDT_Exemplar[3])` - the answer is `<class 'int'>` because `[3]` selects the fourth item in the `CDT_Exemplar` list (4) and the type of 4 is integer. Note: The variable `gpa` is not in the list.
- D) `CDT_Exemplar[7]` - the answer is `IndexError`. While there are seven items in the `CDT_Exemplar` list, the `[7]` selects the eighth item (addressing items in a list starts from 0) and there is no eighth item.
- E) `CDT_Exemplar` only contains two unique data types: strings and integers

Open-Ended Problems (15 points)

(Day 1)

Prompt: You are interested in using computer science to improve your platoon's physical fitness. Your fellow platoon leader created a solution that surveyed many different people in their network for existing physical training plans, extracted features (i.e. aerobic vs anaerobic, hours of activity, outdoors vs indoors) and classified those plans (effective, neutral, non-effective) to build a predictive model that can create new, personalized, but effective physical fitness plans. **Which technique did your colleague most likely use?** (choose one): Traditional Programming, Machine Learning, or Deep Learning and **defend your choice**. Limit your answer to no more than 6 sentences.

Feedback:

Machine Learning: Given that there are clearly labeled inputs and a desired output, this is not Deep Learning, where inputs are plentiful but not labeled or classified; the model is essentially a black box. Additionally, it is not Traditional Programming because we are not coding an exact algorithm to calculate the ideal fitness plan, and we are not using statistics to create these plans. Machine Learning uses feature extraction and classification to label categories that a Machine Learning system can use to learn and produce the desired output using a Machine Learning algorithm. It is not a black box such as Deep Learning and is also not based purely on statistics.

(Day 2)

Prompt: Your battalion commander asked his favorite cyber lieutenant, most trusted staff officer to identify patterns in current active offensive cyber operations. While the lieutenant has a variety of unstructured data on-hand such as network logs, camera footage, and operator notes, it is difficult to identify specific features to extract from the data. Without feature extraction and classification, he proceeds with feeding a model with this data and is surprised to see that the model can self-identify features and highlight a variety of patterns within the operations with

Commented [HT31]: @Hutton, John E LTC and @Yan, Jinny Y CPT

Commented [YC31R2]: Apologies, missed the paste earlier

Commented [JY31R3]: Primary feedback from group and plan of action going forward:

- * Review book material and ensure key definition of Programming, Machine Learning, and Deep Learning makes it into the prompt.

- * Consider rewording the prompt (and following along with the grading rubric) to emphasize that other choices could be possible if *defense* of that choice is strong (for partial credit).

- * (if above), provide more than one sample response for the alternative (but less ideal) answers.

Commented [MS32]: Can you please suggest some alternates? What would day2 and the alternate version look like?

Commented [LP33]: The book uses the term Traditional Programming

Commented [MS34]: Can Deep Learning also be a valid answer?

Commented [TW34R2]: The inputs and desired output aren't actually clear to me from the question prompt, but I would argue that "deep learning" is indeed valid if "machine learning" is valid. Anything that can be done with a logistic regression / support vector machine / random forest / k-nearest neighbor classifier, etc. can also be done with a neural network.

Commented [MS34R3]: I agree. We will need to work on this a bit more as a team.

Commented [TW35]: This is not any definition of deep learning. It seems to be talking about supervised vs. unsupervised learning which isn't even within the scope of the question.

Commented [MS36]: generalize the branch. It is likely they will have to do this regardless of what branch they choose.

CY105 AY26-1 WPR 1

seemingly minimal human effort. **Identify the technique that the lieutenant most likely used** (choose one): Programming, Machine Learning, or Deep Learning and **defend your choice**. Limit your answer to no more than 6 sentences.

Feedback:

Deep learning. The data you start with is unstructured, unlabeled with no identified features. You feed this data to a preexisting model that produces its own set of features and patterns. It is not Programming because there is no exact algorithm coded to solve your task. It is not Machine Learning, since the data is unstructured. Machine Learning uses feature extraction and classification to label categories that a Machine Learning system can use to learn and produce the desired output using a Machine Learning algorithm.

(Option 2) Machine Learning;

(Alternate)

Prompt: You are a company commander looking to automate some of your Maintenance Monday vehicle care tasks. Your fellow platoon leader tapped into data feeds that are exposed from the equipment for average temperature, vehicle mileage, and terrain impact. She also separately receive vehicle performance ratings from your soldiers. Due her subject matter expertise, she was able to identify an algorithm that ties these four variables in a predictable alerting system for which vehicles are at the highest risk of failure and require the most attention. **Identify the technique your colleague most likely used** (choose one): Programming, Machine Learning, or Deep Learning and **defend your choice**. Limit your answer to no more than 6 sentences.

Feedback:

Programming; You have identified a specific algorithm with which you attack the problem. This algorithm was implemented in a predictable manner, so a traditional program can be use. It is not Machine Learning because you did not extract features and feed them to a model for the model to classify. It is not Deep Learning because you did not feed the data into a black box model. Deep Learning is also not based on simply statistics as implied in the problem.

19SEP25: Your Company Commander is looking to automate some “motor pool Monday” vehicle maintenance tasks. Your Executive Officer (XO) has extensive operational knowledge of key facts related to the vehicle fleet: typical age of vehicles, known common faults, and which vehicles soldiers complained most about breaking down. Using these three variables, the XO develops a digital tool that generates alerts about which vehicles are at the highest risk of failure and require the most attention. **Identify the technique your XO most likely used** (choose one): Programming, Machine Learning, or Deep Learning and **defend your choice**. Limit your answer to no more than 6 sentences.

(Option 2) Machine Learning;

Commented [AK37]: Recommend changing to “human effort”. Deep learning takes considerable computational effort.

Commented [MS38]: I think this one needs to be a bit more obviously traditional programming. Does an algorithm definitely mean that they are going to use traditional programming?

Recommend adding some language like, “she writes some code that deterministically evaluates different scenarios and...”

Commented [HT38R2]: Tipp version (that is in Canvas now): Your Company Commander is looking to automate some “motor pool Monday” vehicle maintenance tasks. Your Executive Officer (XO) has access to several data points from the vehicle fleet: average temperature, vehicle mileage, terrain impact, and vehicle performance ratings from Soldiers. Using these four variables, the XO develops a system that can deterministically generate alerts about which vehicles are at the highest risk of failure and require the most attention. **Identify the technique your XO most likely used** (choose one): Programming, Machine Learning, or Deep Learning and **defend your choice**. Limit your answer to no more than 6 sentences.

Commented [TW38R3]: @Matthews, Suzanne Dr @Tippett, Hayden J MAJ I just tackled this question in my run-through with fresh eyes (I didn’t see it or this discussion until now), and I thought the answer was **machine learning**. I saw that we have some labeled data (if we treat the vehicle performance rating as the label for the tuple of other data points), but not enough for deep learning (the company, battalion, brigade, etc. fleet would not yield the many thousands or millions of data points needed for deep learning), and it appears that the XO did some feature engineering to hand-select the features that would be most predictive of the vehicle performance rating. Once they are trained, most machine learning classifiers (even neural networks, unless we’re talking about Bayesian neural networks) are deterministic (i.e. they will make the same ... [1])

Commented [TW38R4]: Thinking about this more, my suggestion is to say that “the XO has no historical data points from the fleet, but has an idea of what factors are likely contribute to maintenance issues going forward.” Then it would be clear that there’s nothing from which he could train a machine learning model.

Commented [MS39]: @Walsh, Timothy C LTC Thanks for your suggestions. Here is an updated version. Let us know what you think!

Commented [TW39R2]: I do like this more. It would be much more difficult to argue (credibly, without introducing a lot of assumptions about the nature of the XO’s “knowledge”) that ML or DL is possible and appropriate here.

CY105 AY26-1 WPR 1

Code Writing (25 points)

DAY 1 - CODING PROBLEM – PACKING LIST STATISTICS

Write a program that will produce statistics about a packing list provided by the user. Prompt the user for how many items there are, then iteratively prompt the user for the mass of each item in kilograms (kg) and add each to a list. Upon completion, show the total mass and the mass of the heaviest item. Round outputs to one decimal point. Assume that the user will always enter a positive integer for the number of items, and a positive numerical input for each item's mass.

Sample Input/Output:

How many items are on the packing list? 2

Enter weight of item in kg: 10.5

Enter weight of item in kg: 15

The total mass is 25.5 kg

The heaviest item is 15.0 kg

Provided to Students:

```
def main():  
    # enter code here  
  
main()
```

Sample Solution:

```
"""  
Problem: packing list statistics  
Input: number of items, mass of each item  
Output: total mass, heaviest item  
"""  
  
def main():  
  
    # 1. initialize an empty list  
    item_list = []  
  
    # 2. prompt for how many items are on the packing list  
    num_items = int(input("How many items are on the packing list? "))  
  
    # 3. for each item, prompt for the mass and add it to the list while  
    # accumulating the total  
    total_mass = 0
```

Commented [HT40]: @Kedrowitsch, Alexander I. MAJ and @Walsh, Timothy C. LTC

Commented [TW40R2]: We've updated this based on the discussion last week, and we've added the Day 2 and Alternate now.

Commented [LP41]: I would include that after you prompt for how many items that for each item add it to a list.. You could have them display the list before the stats

Commented [TW41R2]: I think I understand what you're saying, and I made a change to the second sentence. Does that look good now?

Commented [LP42]: I would not have them include the item number on input and keep it simple. How many kg is item

Commented [TW42R2]: I'm just trying to think of a way to make the prompt seem reasonably natural while omitting the number...

Commented [MS42R3]: Maybe we can change it to

Enter weight of item 1 in kg

This will remove the necessity of using string concatenation

Commented [TW42R4]: That sounds like a good compromise. It will just take me a few minutes to change later today.

CY105 AY26-1 WPR 1

```
item_list = []
for i in range(num_items):
    item = float(input("Enter weight of item in kg: "))
    total_mass += item
    item_list += [item]

# 4. round the sum of the list to one decimal place
total_mass = round(total_mass, 1)

# 5. get the max of the list and round it to one decimal place
heaviest_item = round(max(item_list), 1)

# 6. display the final statistics
print("The total mass is", total_mass, "kg")
print("The heaviest item is", heaviest_item, "kg")

# invoke the procedure called main
main()
```

Test Cases:

- 1) Test Case Description 1
a) Inputs: 2, 10, 15
- 2) Test Case Description 2
a) Inputs: 1, .1
- 3) Test Case Description 3
a) Inputs: 3, 12.25, 0.24, 22.2

Tested Concepts:

- Python Basics
- Built-In Functions
- Iteration I (For Loops)
- Lists and Strings

DAY 2 - CODING PROBLEM – TEMPERATURE READING STATISTICS

Write a program that will produce statistics about a list of temperature readings provided by the user. Prompt the user for how many temperature readings there are, then iteratively prompt the user for each temperature in degrees Celsius and add each to a list. Upon completion, show the average and highest temperature. Round outputs to one decimal point. Assume that the user will always enter a positive integer for the number of temperatures, and a numerical input for each temperature.

Sample Input/Output:

Commented [HT43]: alt

Commented [JD44]: in degrees Celsius.

Commented [TW44R2]: Done.

CY105 AY26-1 WPR 1

How many temperatures are in the list? **2**

Enter a temperature in C: **10.5**

Enter a temperature in C: **15**

The average temperature is 12.8 C

The **highest** temperature is 15.0 C

The range of heights on the roster is (70 inches to 84 inches).

Commented [HT45]: Range float

Provided to Students:

```
def main():  
    # enter code here  
  
main()
```

Sample Solution:

```
"""  
Problem: Compute temperature statistics  
Input: Number of temperature readings and each temperature  
Output: Average temperature and the highest temperature  
"""  
  
def main():  
  
    # 1. initialize an empty list  
    temperature_list = []  
  
    # 2. prompt for how many temperatures are in the list  
    num_temperatures = int(input("How many temperatures are in the list? "))  
  
    # 3. for each one, prompt for the temperature, add it to the list, and  
    # accumulate the sum of temperatures so that we can find the mean later  
    total_sum = 0  
    for i in range(num_temperatures):  
        temp = float(input("Enter a temperature in C: "))  
        total_sum += temp  
        temperature_list += [temp]  
  
    # 4. divide by the length to get the mean, and round to one decimal place
```

CY105 AY26-1 WPR 1

```
mean_temp = round(total_sum / num_temperatures, 1)

# 5. get the max and round to one decimal place
max_temp = round(max(temperature_list), 1)

# 6. display the final statistics
print("The average temperature is", mean_temp, "C")
print("The highest temperature is", max_temp, "C")

main()
```

- 4) Test Case Description 1
 - a) Inputs: 2, 10, 15
- 5) Test Case Description 2
 - a) Inputs: 1, .1
- 6) Test Case Description 3
 - a) Inputs: 3, 12.25, 0.24, 22.2

Tested Concepts:

- Python Basics
- Built-In Functions
- Iteration I (For Loops)
- Lists and Strings

ALT - CODING PROBLEM – PRICE STATISTICS

Commented [HT46]: Day 2

Write a program that will produce statistics about a list of prices provided by the user. Prompt the user for how many prices there are, then iteratively prompt the user for each price and add each to a list. Upon completion, show the average price and range of prices (in other words, the difference between the highest and the lowest price). Round each output to the nearest dollar. Assume that the user will always enter a positive integer for the number of prices, and a positive numerical input for each price.

Sample Input/Output:

How many prices are in the list? **2**

Enter a price: **10.5**

Enter a price: **15**

The average price is \$13

The range is \$4

Commented [HT47]: cheapest

CY105 AY26-1 WPR 1

Provided to Students:

```
def main():  
    # enter code here  
  
main()
```

Sample Solution:

```
"""  
Problem: Compute price statistics  
Input: Number of prices and each price  
Output: Average price and range of prices  
"""  
def main():  
  
    # 1. initialize an empty list  
    price_list = []  
  
    # 2. prompt for how many prices are in the list  
    num_prices = int(input("How many prices are in the list? "))  
  
    # 3. for each one, prompt for the price, add it to the list, and accumulate  
    # the total of the prices so we can calculate the average later  
    total_sum = 0  
    for i in range(num_prices):  
        price = float(input("Enter a price: "))  
        price_list += [price]  
        total_sum += price  
  
    # 4. divide by the accumulator by the length to get the mean and  
    # round to nearest dollar  
    mean = round(total_sum / num_prices)  
  
    # 5. calculate the range and round to nearest dollar  
    price_range = round(max(price_list) - min(price_list))  
  
    # 6. display the final statistics  
    print("The average price is $" + str(mean))  
    print("The range is $" + str(price_range))  
  
main()
```

Test Cases:

7) Test Case Description 1

CY105 AY26-1 WPR 1

- | |
|--|
| a) Inputs: 2, 10.5, 15
8) Test Case Description 2
a) Inputs: 1, .1
9) Test Case Description 3
a) Inputs: 3, 12.25, 0.24, 22.2 |
| Tested Concepts: <ul style="list-style-type: none">• Python Basics• Built-In Functions• Iteration I (For Loops)• Lists and Strings |

Unused

Question Stem: What does the following code print as its output

```
x = 7
y = 3
# Swap the values of x and y
x = y
y = x
print(str(x) + ', ' + str(y))
```

7,3	False
3,3	True
3,7	False
7,7	False

Feedback:

7,3	The code to swap the values is written incorrectly, but this is not the output.
3,3	Correct, this probably wasn't the intent of the programmer, given the comment, but the Python interpreter only executes the code as written, regardless of what the comment says.
3,7	You might have been confused by the comment that does not match the code. The Python interpreter only executes the code as written, regardless of what the comment says.
7,7	The code to swap the values is written incorrectly, but this is not the output.

Question Title: Built-in Functions

Question Type: Matching

Question Points: 2

Question Stem: Please match the built-in function with a sample input parameter(s).

chr() - 35

Commented [JD48]: Are we trying to "trick" them by putting in an incorrect comment?

Commented [JD48R2]: Also seems like a lot of points since it's only one question.

Commented [MS48R3]: Agreed. Let's cut this and replace this with variable names.

Commented [WL48R4]: Essentially yes, the idea was to test their understanding of the fact that the interpreter does what the code says to do, not what the comments say to do. But I'm happy to cut it and write something different about valid variable naming.

Commented [WL48R5]: Now, above this I've inserted a question to test valid variable names and assignments.

Commented [YC49]: JY

Commented [YC49R2]: Note: I will be OOO week of Aug 1

Commented [HL49R3]: Is it a problem that len() works with everything but 35? So if they flipped the list with, say, the string, they should lose points on input() but not on len().

Commented [MS50]: my vote is cutting this question entirely.

Commented [MS51]: maybe replace this one with int() and 3.14

CY105 AY26-1 WPR 1

ord() - 'A'
input() - "What would you like to eat today? "
len() - ["cat", "dog", "turtle"]

Commented [MS52]: Let's cut this one. Technically, input() can get any string input, so I struggle to see why 'A' is not a valid parameter for input()

Feedback:

Using the reference card, you will be able to understand the possible accepted values of certain built-in functions. It does additionally require the cadet to understand if a certain value is an accepted input to that built-in function.

- o chr(): Accepts an integer number, in this case 35 is a Python int due to it being a number without a decimal point.
- o ord(): Accepts a Unicode character, in this case 'A' is a Python char due to the single quotes and one single character.
- o input(): Accepts a string prompt, in this case "What would you like to eat today? " is a Python str due to the double quotes.
- o len(): Accepts a sequence type, which includes lists. In this case, [...] indicates that ["cat", "dog", "turtle"] is indeed a list.

Iteration — Part I (For Loops)

Question Title: For Loops

Question Type: True/False

Question Points: 1

Question Stem: True or False: Without using loops, a program cannot repeat the same behavior multiple times.

Feedback:

True	Incorrect, a programmer can write any number of identical lines of code that perform the same behavior. Using a feature such as a "for" loop just produces cleaner, more flexible, and often more efficient code.
False	Correct, a programmer can write any number of identical lines of code that perform the same behavior. Using a feature such as a "for" loop just produces cleaner, more flexible, and often more efficient code.

Commented [WL53]: TW

Commented [MS54]: I strongly dislike one point questions. I also think that since they are going to get tested on for loops on the code tracing component, there is no need for a knowledge check question here.

Commented [WL54R2]: So it's appropriate to just cut the question? That's fine with me. I thought another knowledge check question on this topic was assigned to me.

Commented [MS54R3]: I think the goal was to have a code tracing question for for loops, not a knowledge check one. Sorry that wasn't clear.

Question Title: Computers and Data

Question Type: Multiple Choice

Question Points: 1

Question Stem: In nature, most data is:

Commented [MS55]: I strongly dislike 1 point questions. They increase the question count for not much benefit.

Commented [HL55R2]: I'm not sure what you mean by "benefit." We could easily make the question worth two points. Or we could attempt to hide multiple questions in a single question to justify a higher point count, but I'm not sure how that would be better than more, lower point questions.

Commented [MS55R3]: It's just another context that students have to cover during a 55 minute WPR. Having fewer questions with more options helps them keep time better.

CY105 AY26-1 WPR 1

Analog	True
Digital	False

Feedback:

Computers require digital data, but most things in nature are analog, because they have continuous values.

Question Title: Python Basics

Question Type: Drop-down

Question Points: 4

Question Stem: Which of the following Python statements will result in an error?

x = 5 + 3.2	False
y = "5" + "3"	False
z = "5" - "3"	True
a = 10 * 2	False

Commented [MS57]: change to "Error" or "No error"

Feedback:

x = 5 + 3.2	Adding an integer to a float is OK.
y = "5" + "3"	The addition operator performs concatenation for two strings, so this is OK.
z = "5" - "3"	The subtraction operator is not defined for two strings, so this raises an error.
a = 10 * 2	Multiplying two integers is OK.

Commented [MS58]: "results in an error"

Commented [WL58R2]: Done.

Question Title: Lists and Strings

Question Type: True/False

Question Points: 5

Question Stem: Look at the provided code snippet and evaluate the following statements as true or false

```
def main():
    names = ['Al', 'Bob', 'Clara']
    result = "" #1
    for i in range(len(names)): #2
        result += names[i] #3
    names = names * 2 #4
    print(names) #5
```

Commented [FM59]: MF

Commented [HL59R2]: The line below #5 will also produce output, which would make E. False.

Commented [FM59R3]: Hmm I guess I should've made this a bit clearer I meant the only numbered line, I only put in the print so the variable got used, because I think it's bad form to not use a variable

Commented [HL59R4]: Got it. So maybe just saying "of the five numbered lines, the only one to produce output is..."

CY105 AY26-1 WPR 1

```
main()
```

- A. The line labelled #1 will produce an error **False**
- B. The line labelled #2 iterates directly over the values in the list names **False**
- C. The line labelled #3 adds each item in the list names to the variable result **True**
- D. The line labelled #4 produces ['AlAl', 'BobBob', 'ClaraClara'] **False**
- E. The line labelled #5 is the only numbered line which produces output **True**

Feedback:

- A. This is simply an empty string declaration
- B. We are iterating over the indices not the list directly
- C. We are concatenating to the result string
- D. This replicates the list ergo ['Al', 'Bob', 'Clara', 'Al', 'Bob', 'Clara']
- E. All other lines produce no output

Commented [KL60]: Recommend wording changed to "...the only numbered line which..." to reduce potential confusion/misreading

[@Matthews, Suzanne Dr](#) [@Tippett, Hayden J MAJ](#) I just tackled this question in my run-through with fresh eyes (I didn't see it or this discussion until now), and I thought the answer was **machine learning**. I saw that we have some labeled data (if we treat the vehicle performance rating as the label for the tuple of other data points), but not enough for deep learning (the company, battalion, brigade, etc. fleet would not yield the many thousands or millions of data points needed for deep learning), and it appears that the XO did some feature engineering to hand-select the features that would be most predictive of the vehicle performance rating. Once they are trained, most machine learning classifiers (even neural networks, unless we're talking about Bayesian neural networks) are deterministic (i.e. they will make the same prediction for the same input every time), so saying "deterministic" doesn't point me towards traditional programming.



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB H

WPR2

AI105 AY26-1 WPR 2

ACCESS CODES:

	<u>Day 1</u>	<u>Day 2</u>	<u>Alternate</u>
Part A	trellis	journey	sunlight
Part B	captain	velocity	fracture
H.A.S.	wildfire		

GUIDANCE:

6x knowledge check questions (25 points total)

- 3 questions from Block 1
- Every option must be 1 point each
- True/False, multi-answer(dropdown), word bank, matching, and multiple choice (w/ shared stem) all valid question types

8x Code Tracing Questions (35 points total)

- Every option must be 1 point each
- 1 code tracing question from last WPR

1x Free-Response Question (15 points)

- Textbox, holistic rubric

1x Code Writing Problem

- 25 points, holistic rubric
- Should cover iteration (for), basic types, built-in functions

LEGEND: Updates to be made. Question Answers. Feedback Section. Configured (XO-Only)

KC/CT Question Template:

1. Lesson {KC/CT} [x pts]

Question Type: update

Question Stem:

****insert question stem****

Answer

Feedback:

- ..

AI105 AY26-1 WPR 2

PART A: Knowledge Check & Code Tracing

1. Review 1 {KC} [4 pts]

Rank these storage solutions in ascending order of *capacity* on a typical computer (1 – largest, 4 – smallest):

Hard Disk Drive – 1
Random Access Memory – 2
Cache – 3
CPU Registers – 4

Feedback:

In terms of capacity:

- Hard disk drives are the largest (hundreds of GB to TB)
- Random Access Memory is the next largest (tens of GBs)
- Cache is the next (tens to hundreds of MBs)
- Registers are the smallest (Bytes)

The memory hierarchy is depicted as a triangle. The x-axis indicates the capacity.

2. Review 2 {KC} [4 pts]

Question Type: Matching

Question Stem: Match the term with its best definition:

An early benchmark for a computer's intelligence	Turing Test
Determines the "working memory" of an LLM	Context Window
The smallest unit of language that a large language model uses	Token
Controls the creativity and variability of a large language model's output	Temperature

Distractors:
Hallucination
Agent
Word

Feedback:

AI105 AY26-1 WPR 2

- The **Turing Test** was used as an early benchmark for how “intelligent” a computer is
- The **context window** is the number of tokens that an LLM is expected to “remember” during a conversation.
- A **token** is the smallest unit of language that a LLM uses.
- The **temperature** is a setting that determines how much the output of an LLM will typically vary.

3. Review 2 {CT} [4 pts]

Question Type: Fill in the blank

Question Stem: Which of the following variable assignments are valid?

ABC = len('ABC')	valid
my_list = [2] * 2	valid
3rdVar = 14.5	invalid
10 + 2 = add	invalid

Feedback:

- The first assignment is **valid**. The value 3 will be assigned to the variable ABC.
- The second assignment is **valid**. When applied to lists, the * operator performs replication. In this case, the list [2,2] is assigned to the variable my_list
- The third assignment is **invalid**. Variables cannot start with a number.
- The fourth assignment is **invalid**. Variables must always be on the left-hand side (LHS) of the assignment operator.

4. Modules 1 {CT} [3 pts]

Question Type: Multi-Answer

Question Stem: Consider the following code.

```
01 import math
02 import statistics
03
04 def main():
05     numbers = [1.8, 5.0, 5.0, 10.0]
06
07     a = math.floor(numbers[0])
08     b = math.ceil(numbers[0])
09     c = round(numbers[0])
10     d = statistics.mode(numbers)
11     is_a_larger = a > b
12     is_b_larger = b > c
13 main()
```

Commented [HT1]: @Hicks, Matthew S LTC

Commented [HT1R2]: @Hicks, Matthew S LTC should be Code Tracing question as opposed to Knowledge check

Commented [MH1R3]: I added a Code Trace below.

AI105 AY26-1 WPR 2

What is the value of is_a_larger on line 11?	False
What is the value of is_b_larger on line 12?	False
What is the value of d on line 10?	5; 5.0; 5.45; There is no middle value.

Feedback:

- `math.floor(1.8)` returns **1**, and `math.ceil(1.8)` returns **2**.
Since `1 > 2` is **False**, the variable `is_a_larger` is **False**.
- `b = math.ceil(1.8) → 2`, and `c = round(1.8) → 2`
Since `2 > 2` is **False**, the variable `is_b_larger` is **False**.
The `round()` function rounds to the *nearest integer*, and values ending in `.5` round to the *nearest even integer*.
- `statistics.mode(numbers)` looks for the *most frequently occurring value* in the list `[1.8, 5.0, 5.0, 10.0]`. Because `5.0` appears twice (more than any other number), the mode is **5.0**.

5. Modules 2 {CT} [4 pts]

Question Type: Multi-Answer

Question Stem: Consider the following code. For each question, assume execution reaches that line of code.

```
01 import statistics
02
03 def main():
04     my_list = []
05     for i in range(5):
06         my_list += [random.randint(1, 10)]
07     x = stdev(my_list)
08     print(x)
09
10 main()
```

Line 01 will cause an error	False
Line 06 will cause an error	True
Line 07 will cause an error	True
You are limited to importing only 2 modules in a single Python program	False

Feedback:

- Line 01 will cause an error – False; this simply imports the statistics module
- Line 06 will cause an error – True; The random module has not been imported

Commented [HT2]: @Hutton, John E LTC

Commented [JH2R2]: Should I add to the question stem the guidance that each question should assume that execution reaches the line in question?

Commented [HT2R3]: ^ I think that is an important distinction to include

Commented [JH3]: Changed to my_list to minimize incorrect interpretation.

AI105 AY26-1 WPR 2

- Line 07 will cause an error – True; The stdev() function is in the statistics module, so you need statistics.stdev()
- You are limited to importing only 2 modules in a single Python program – False; You may import as many modules as you like.

6. Unary/Binary Conditionals 1 [CT] [3 pts]

Question Type: Multi-answer

Question Stem: Consider the following code snippet.

```
01 def main():
02     x = 5
02     y = 9
03     z = (x == y)
04     if z:
05         print("Equality")
06     print("All done")
07
08     main()
```

After line 03, the value of z will be 14	False
The program will print "Equality"	False
The program will print "All Done"	True

Feedback:

- After line three, the value of z will be – False; This is a valid Boolean assignment.
- The program will print "Equality" – False; Since z will be false, line 05 will not execute.
- The program will print "All Done" – True; This line executes no matter what z evaluates to.

7. Unary/Binary Conditionals 2 [CT] [4 pts]

Question Type: Dropdown

Question Stem: Consider the following code snippet:

```
villain = input("Who did Fred unmask? ")
if villain == "Old Man Wickles":
    print("Blast it!")
if villain == "Farmer Bascomb":
    print("You meddling kids!")
else:
    print("It was Captain Cutler!")
```

Match the correct program output to each of the specified inputs:

Input (given):	Output (the matching text aligned with the given input):
----------------	--

Commented [HT4]: @Hutton, John E LTC

Commented [HT5]: Is there a specific reason we are not using main?

Commented [JH5R2]: Not really. This snippet could come from a main or any other function.

Commented [MS5R3]: Suppose that students say error for the first question. That will cause them to select False for everything else. Are we ok with them earning 1/3 on this question?

Commented [JH5R4]: Added the caveat as in the earlier problem to assume execution always reaches the line in question. This may have the effect of encouraging the selection of error on line 03, though. Any thoughts? This reduces the knock on effect of getting line 03 wrong by at least one other question, though if you choose error, line 04 still becomes difficult. Conversely, a student might reason that since we ask about line 05, line 03 cannot error.

Commented [DJ6]: I'd remove the parentheses here

Commented [HT7]: @Kedrowitsch, Alexander LTC

Commented [AK7R2]: AK

AI105 AY26-1 WPR 2

"Old Man Wickles"	"Blast it!" and "It was Captain Cutler!" are both printed
"Farmer Bascomb"	"You meddling kids!"
"Stuart Wetherby"	"It was Captain Cutler"
"Shaggy"	"It was Captain Cutler"

Include all the above statements as possible options

"You meddling kids!"

"It was Captain Cutler"

"Blast it!"

"Blast it!" and "It was Captain Cutler!" are both printed

Feedback:

- "Old Man Wickles" outputs "Blast it! It was Captain Cutler!" as the first conditions check is True, which prints "Blast it!" and the second conditions check is False, defaulting to the 'else' code block, which prints "It was Captain Cutler!"
- "Farmer Bascomb" outputs "You meddling kids!" as the first conditions check is False, the second conditions check is True, which prints "You meddling kids!", and the 'else' code block is skipped.
- "Stuart Wetherby" outputs "It was Captain Cutler!" as the first and second conditions checks are False, defaulting to the 'else' code block, which prints "It was Captain Cutler!"
- "Shaggy" outputs "It was Captain Cutler!" as the first and second conditions checks are False, defaulting to the 'else' code block, which prints "It was Captain Cutler!"

8. Chained/Nested Conditionals 1 [CT] [3 pts]

Question Type: Dropdown

Question Stem: Based on this code snippet, match the function calls with the print output.

```
def what_to_wear(temperature, raining):
    if temperature < 60:
        if raining:
            print("Wear a raincoat.")
        else:
            print("Wear a jacket.")
    else:
        if raining:
            print("Take an umbrella.")
        else:
            print("Just a t-shirt is fine.")
```

Commented [HT8]: @Kedrowitsch, Alexander L LTC this line break will not configure for a matching or dropdown. One option is to adjust the stem to be something like "identify which statement will print" and then there can be an option for "more than one print statement will execute." Alternatively, I can use the code as a stimulus, and we can ask separate multi-answer or multiple choice questions off of the same stem. MA/MC answers have rich text features

Commented [MS8R2]: we can also change the first one to be:

"Blast it!" and "It was Captain Cutler!" will both print

Commented [AK8R3]: I updated the solution output to: "Blast!" and "It was Captain Cutler!" are both printed'

Commented [HT9]: @Walsh, Timothy C LTC I know you mentioned having a draft of these, sir. If you wrote a KC already, please feel free to trade with LTC Kedrowitsch.

Commented [TW9R2]: TW

AI105 AY26-1 WPR 2

<code>what_to_wear(45, True)</code>	Wear a raincoat.
<code>what_to_wear(75, False)</code>	Just a t-shirt is fine.
<code>what_to_wear(60, True)</code>	Take an umbrella.

Distractor:

Wear a jacket.

Feedback:

- **what_to_wear(45, True):** Remember that 45 is less than 60, and raining is True, so the inner condition for a raincoat should trigger.
- **what_to_wear(75, False):** Since 75 is not less than 60 and it's not raining, only the final else (t-shirt) should execute.
- **what_to_wear(60, True):** 60 is *not* less than 60, so the first if fails — make sure you understand that equality goes to the else branch, leading to “Take an umbrella.”

9. Unary/Binary/Chained/Nested Conditionals 2 [KC] [3 pts]

Question Type: Multi-answer

Question Stem:

Indicate whether the following statements are True or False regarding conditional constructs in Python:

The following a valid chained conditional structure: <code>if ... elif ... elif ...</code>	'True'
Exactly one code block of a chained conditional with an <code>else</code> statement will execute.	'True'
Nested conditionals are not suitable for when a condition depends on the outcome of a previous condition.	'False'

Feedback:

- True — `if ... elif ... elif ...` is a valid structure for chained conditional statements.
- True — chained conditional statements with an `else` statement are guaranteed to execute one code block.
- False — Nested conditional statements are specifically suited for complex decision making when a condition depends on the outcome of conditions before it.

10. While Loops 1 [CT] [4 pts]

Question Type: Multi-answer

Question Stem: For each of the following replacements for LINE X, identify whether the loop will execute a definite, indefinite, or infinite number of times based on the replacement code.

Commented [HT10]: @Kedrowitsch, Alexander I LTC sir, please note that this is a KNOWLEDGE CHECK question about conditionals

Commented [AK10R2]: AK

Commented [HT11]: To make this a bit more difficult, we could drop this else off as well: `if...elif...elif`

Commented [AK11R2]: Concur - updated per recommendation

Commented [HT12]: Recommend “code block” to maintain consistent with reading/slides

Commented [AK12R2]: Updated per recommendation

Commented [HT13]: @Catudal, Joseph W LTC

Commented [CJWL13R2]: jc5

Commented [HT14]: I love this question!! For the stem, I think it's important that we've already entered the loop when we hit LINE X. so perhaps the wording here needs to be something like, “...identify whether the loop will continue to execute a definite, indefinite, or infinite number of times based on the replacement instructions below.”

Commented [J14R2]: adjusted as per rec

AI105 AY26-1 WPR 2

```

item = "gum"
cost = 75
total = 0
coin = 5

while total < cost and item != "quit":
    # LINE X – fill in with code from questions below
    if coin in [5, 10, 25, 50]:
        total += coin
    else:
        print("Invalid coin type!")
print("Please enjoy your", item + "!")

```

coin = 25	Definite
coin = 1	Infinite
coin = int(input("Insert coin: "))	Indefinite
item = "quit"	Definite

Feedback:

coin = 25	Definite. Loop runs exactly 3 times (0->25->50->75).
coin = 1	Infinite. coin = 1 is an invalid coin type, so total never changes; loop never ends.
coin = int(input("Insert coin: "))	Indefinite. Number of iterations depends on user input.
item = "quit"	Definite. Loop terminates immediately after 1 iteration since item becomes "quit".

11. Writing Functions 1 [CT] [5 pts]

Question Type: Fill in the blank (Dropdown)

Question Stem: Consider the following code.

```

01 full_mag = 30
02 def reload():
03     ammo = 0
04     while ammo < 5:
05         ammo += 1
06
07 def main():
08     ammo = 10
09     while ammo > 0:
10         print("FIRE!")

```

Commented [HT15]: @Hicks, Matthew S.LTC

Commented [HT15R2]: If we end up re-distributing some points above, I could see 2-3 additional points being added to this question.

Commented [MS16]: can we rename this to something else? It never gets used thanks to variable shadowing. Right now, it's in there as a distractor. given the complexity of the other problems, I think this will primarily be a distractor.

Maybe change it to magazines = 20

or something else. They are going to have enough trouble keeping track of the ammo declarations there are.

Commented [DJ16R2]: If you change that then the third question also needs to be changed.

Commented [MH16R3]: Changed it to full_mag in the code, answer, and feedback.

AI105 AY26-1 WPR 2

```

11     ammo -= 1
12     reload()
13     print(ammo)
14 main()

```

How many times will the <code>while</code> loop on line 04 execute?	0; 5; 10; 20
What value will line 13 print to the user?	0; 5; 10; 20; print never executes
There is a variable, <code>full_mag</code> , declared at line 01 – what type of variable is this?	Global Variable; Local Variable; Instance Variable
The <code>reload()</code> function is called twice in this program	True, False
The <code>reload</code> function is an example of a fruitful function	True, False

Commented [HT17]: Recommended distractor

Feedback:

- In `reload()`, `ammo` is set to 0 locally (line 03). The loop runs while `ammo < 5`, and each pass does `ammo += 1`. That's 5 iterations (when `ammo` goes 0→1→2→3→4; it stops once `ammo` becomes 5).
- In `main()`, `ammo` is a **local** variable starting at 10 (line 08). The loop on lines 09–11 fires 10 times and decrements `ammo` to 0. Calling `reload()` doesn't change `main`'s `ammo` because `reload()` uses its **own local** `ammo` and returns nothing. So `print(ammo)` prints 0.
- Line 01 assigns `full_mag` the integer value of 30. As this is defined at the top-level of a program, and not inside the scope of any function, it is a global variable.
- `reload()` is only called once, on line 12.
- `reload()` is not a fruitful function as it has no return (e.g. fruit).

12. Writing Functions 2 [CT] [5 pts]

Question Type: Multi-answer

Question Stem: Consider the following code.

```

01 import math
02
03 def insideCircle(x, y, r):
04     d = math.sqrt(x**2 + y**2)
05     return d <= r
06
07 def main():
08     answer = insideCircle(2, 4, 9)
09
10 main()

```

Commented [HT18]: @Hutton, John E LTC

Commented [MS19]: wrap in def main()?

AI105 AY26-1 WPR 2

How many parameters does the <code>insideCircle</code> function take?	0; 1; 2; 3 ; 4
Considering only lines 01 through 05, how many different return values are possible from the <code>insideCircle</code> function	1; 2 ; 3; unknown – it could be any number of possibilities; it does not return any value
When line 07 executes, it will assign x the value 4 within the <code>insideCircle</code> function.	False
Line 05 will cause an error.	False
the <code>insideCircle</code> function is an example of a fruitful function	True

Commented [HT20]: Recommend rewording, "...it will temporarily assign x the value of 4 within the `insideCircle` namespace."

Commented [JH20R2]: Reworded.

Feedback:

- The `insideCircle` function receives three parameters, x, y, and r.
- Since the function returns the results of a comparison, the only possible values are True or False.
- Line 07 will cause the 4 to get loaded into y.
- It is perfectly valid to return the results of an expression.
- the `insideCircle` function is an example of a fruitful function because it returns a Boolean value on line 05

13. Networks & Data 1 (KC) [3 pts]

Question Type: Multi-Answer

Question Stem:

Answer True or False for each statement:

A bandwidth of 100 Mbps means a connection can transfer up to 100 megabits of data per second.	True
Latency measures the maximum number of bits per second that can travel across a network connection.	False
Poor signal strength in wireless networks makes data more likely to be lost during transmission.	True

Feedback:

A bandwidth of 100 Mbps means a connection can transfer up to 100 megabits of data per second.	True.
Latency measures the maximum number of bits per second that can travel across a network connection.	False. Latency measures the <i>time delay</i> between sending and receiving, not the bit rate.

Commented [HT21]: @Catudal, Joseph W LTC

Commented [CJWL21R2]: jc5

AI105 AY26-1 WPR 2

Poor signal strength in wireless networks makes data more likely to be lost during transmission.	True.
--	-------

14. Networks & Data 2 {KC} [4 pts]

Question Type: Matching

Question Stem: Match each network layer with its appropriate description:

Network Access Layer	Layer incorporating the physical connections between devices and transmission mediums
Internet Layer	Layer assigning addresses to connected devices and determines how packets are routed from one network to another
Transport Layer	Layer managing data delivery between two devices and ensuring reliability of data, if needed
Application Layer	Layer that standardizes data formatting between originating programs/apps and servers across the Internet

Feedback:

- The Network Access Layer incorporates protocols, transmission mediums, and physical connections between network devices.
- The Internet Layer assigns IP addresses, performs data packetization, and conducts packet navigation/routing between a source and destination.
- The Transport Layer manages packet delivery and reliability between two end-points, utilizing protocols that provide assured delivery like TCP, or 'best effort' delivery like UDP.
- The Application Layer standardizes the formatting of data sent across the Internet originating from applications, such as web browsers and apps.

15. Internet & Web 1 {KC} [3 pts]

Question Type: Dropdown (Drag and drop)

Question Stem: Match each protocol to the most analogous thing in the physical world.

DNS	A book that allows you to look up someone's contact information given their name.
TCP	A courier that will repeatedly try to deliver a package until the recipient is able to take delivery and sign for the package.
IP	A system of street names and building numbers that enables you to navigate to your destination.

Distractors:

UDP
HTTP
IMAP

16. Internet & Web 2 {KC} [4 pts]

Question Type: Dropdown

Question Stem:

Match each term to its correct definition.

Commented [HT22]: @Kedrowitsch, Alexander L LTC

Commented [AK22R2]: AK

Commented [MS22R3]: @Kedrowitsch, Alexander L LTC
Please rewrite. This is too hard and does not meet the objectives of the lesson.

Instead, please make this about layers (Application, Transport, Internet and Network access) and have a definition associated with each.

Commented [AK22R4]: Updated per request

Commented [MS22R5]: Thanks @Kedrowitsch, Alexander L LTC !! I added some remarks on how to improve it further.

Commented [MS23]: ensuring reliability of delivery, if needed?

Commented [AK23R2]: Sounds more better than my original wording - updated per request!

Commented [MS24]: there needs to be something about communicating across the Internet here.

Commented [AK24R2]: Updated. Is this in line with your intent?

Commented [HT25]: @Walsh, Timothy C LTC

Commented [TW25R2]: TW

Commented [DJ26]: navigate to your destination

Commented [HT27]: Recommend distractors on this question: UDP, HTTP, TLS

Commented [HT28]: @Catudal, Joseph W LTC

Commented [CJWL28R2]: jc5

AI105 AY26-1 WPR 2

JavaScript	A programming language that adds interactivity and actions to a webpage.
URL	The unique address of a webpage on the web, which includes protocol, domain name, path, and file.
HTML	Defines the content and structure of a webpage using tags such as <head>, <title>, <body>, and .
CSS	Specifies the style of a webpage, such as colors, fonts, and layout.

Free Response Questions [15 points]

(Day 1)

Imagine you are a journalist who needs to communicate securely and anonymously with a sensitive contact who could be in danger if their identity is revealed. Would you recommend that your contact use an application on the surface web, deep web or dark web to communicate with you? Defend your choice of layer in 3-6 sentences and state any assumptions. In your defense, you MUST discuss why the layer you chose is preferable to the other layers.

Sample Answer: The surface web includes public websites that anyone can find through a search engine, such as social media or news sites, which would not be secure for confidential communication. The deep web includes private or password-protected spaces like email accounts or cloud storage, which offer some privacy but can still be tracked or monitored. The dark web, on the other hand, requires special software such as Tor to access and is designed to keep users anonymous by encrypting their data and hiding their locations. For a journalist working with a sensitive source, the dark web would be the most appropriate choice because it provides the strongest protection for identity and communication privacy.

(Day 2)

Prompt: Imagine you are watching an animated film on Netflix using your laptop. Which layer of the web (surface web, deep web, or dark web) are you most likely using to accomplish this, and why? In your answer, explain why watching a Netflix movie does not primarily take place on the other two layers. State any assumptions and keep your answer to 3–6 sentences.

Sample Answer: Netflix content is part of the deep web because the videos and account pages are not indexed by search engines and require a user login or subscription to access. The surface web consists of publicly available pages like news articles or a studio's marketing site, which anyone can find and view without signing in. The dark web requires special tools like Tor and is used for anonymous activity; it is unrelated to legitimate, subscription-based streaming. Because Netflix restricts access behind authenticated accounts and does not expose its full media library to web crawlers, the deep web is the correct category.

(ALT)

Prompt: You are tasked with establishing communications between two remote bases. They are separated by a large hill and approximately 20 kilometers of distance in contested territory. The bases

Commented [HT29]: @Catudal, Joseph W LTC @Hicks, Matthew S LTC

Commented [CJWL29R2]: jc5

AI105 AY26-1 WPR 2

require a suitable connection for data transfer and secure communication. Explain which of the following technologies—WiFi, Fiber-optic Cable, or Satellite—would be the best choice to connect the two bases. In 3-6 sentences, describe: Why the selected technology is appropriate for the distance and terrain involved, AND why each of the other technologies would be less effective or impractical in this scenario. State any assumptions you made.

Sample Answer: The best choice to connect the two bases would be satellite. Wi-Fi would be ineffective because its range is limited to a few hundred feet and cannot reliably pass through terrain. Fiber optics allows for communication across long distance, high reliability, and bandwidth for continuous secure communication. However, in contested territory, it is unclear if the cable could be laid. Coaxial cable can carry signals only short distances before losing strength, making it impractical for 20 km. In contrast, satellite communication would enable a suitable Internet connection over a larger geographic area without needing to lay out any additional equipment. Low-earth orbital satellite services like Starlink allow for encrypted communication.

PART B: Code Writing Problem [25 points]

DAY 1 - CODING PROBLEM – WEAPONS QUALIFICATION

Your unit is conducting weapons qualification. Write a program that repeatedly prompts the user to enter a weapons score until the user enters -1. Each time, classify the score as Expert, Sharpshooter, Marksman, or Unqualified, according to the table below.

Score	Result
36-40	Expert
30-35	Sharpshooter
23-29	Marksman
0-23	Unqualified

When the user enters -1, the program should terminate and print the percentage of scores that were expert, rounded up to the nearest whole number percentage. See the sample I/O below.

Input/Output:

```
Enter shooter score, or enter -1 to quit: 40
Expert
Enter shooter score, or enter -1 to quit: 35
Sharpshooter
Enter shooter score, or enter -1 to quit: 20
Unqualified
```

Commented [HT30]: @Walsh, Timothy C LTC @Hutton, John E LTC

Commented [JH31]: @Matthews, Suzanne Dr Day 1 and Alt made similar to Day 2

Commented [AK32]: Unqual and Marksman share value of '23'

AI105 AY26-1 WPR 2

Enter shooter score, or enter -1 to quit: **35**
Sharpshooter
Enter shooter score, or enter -1 to quit: **36**
Expert
Enter shooter score, or enter -1 to quit: **-1**
40% of the shooters qualified Expert.

Provided to Students:

Sample Solution:

```
'''
Problem: Classify shooters as Expert, Sharpshooter, Marksman, or Unqualified,
and print the percentage that qualified Expert.
Input: One score at a time until the user enters -1
Output: The percentage of shooters qualifying Expert
'''

import math

def main():
    #1 initialize accumulators for the total number of
    # shooters and the scores that are Expert
    total = 0
    experts = 0

    #2 get an initial score from the user
    score = int(input('Enter shooter score, or enter -1 to quit: '))

    #3 repeat the following until score is -1
    while score != -1:
        #4 increment the total
        total += 1

        #5 classify it as Expert, Sharpshooter, Marksman, or
        # Unqualified, and increment the number of experts if appropriate
        if score >= 36:
            print('Expert')
            experts += 1
        elif score >= 30:
            print('Sharpshooter')
        elif score >= 23:
            print('Marksman')
        else:
```

AI105 AY26-1 WPR 2

```
print('Unqualified')

#6 get the next score from the user
score = int(input('Enter shooter score, or enter -1 to quit: '))

#7 calculate the percentage of experts and print out the ceiling of that
print(str(math.ceil(100 * experts / total)) + '% of the shooters qualified
Expert.')
```

main()

Test Cases:

1) Test Case 1

a) Inputs:

- i) 40
- ii) 35
- iii) 20
- iv) 35
- v) 36
- vi) -1

b) Outputs:

- i) Expert
- ii) Sharpshooter
- iii) Unqualified
- iv) Sharpshooter
- v) Expert
- vi) 40% of the shooters qualified Expert.

2) Test Case 2

a) Inputs:

- i) 23
- ii) 22
- iii) 29
- iv) -1

b) Outputs:

- i) Marksman
- ii) Unqualified
- iii) Marksman
- iv) 0% of the shooters qualified Expert.

3) Test Case 3

a) Inputs:

- i) 12
- ii) -1

AI105 AY26-1 WPR 2

b) Outputs:

- i) Unqualified
- ii) 0% of the shooters qualified Expert.

Tested Concepts:

- Conditionals I and II
- Using Modules
- Iteration II

AI105 AY26-1 WPR 2

DAY 2 - CODING PROBLEM – HOT, MILD, OR COLD

Problem Specifications:

Write a program that repeatedly prompts the user to enter a daily temperature until the user enters -99. Each time, classify the temperature as being hot, mild, or cold using the table below:

Classification	Temperature
Hot	Above 75 degrees
Mild	Above 55 and up to 75 degrees
Cold	Above 32 to 55 degrees
Freezing	32 and below

When the user enters -99, the program should terminate and print the percentage of days that were mild, rounded up to the nearest whole number percentage. You may assume that the user will enter at least one temperature and no temperature will be lower than -99.

Sample Input/Output

Enter a temperature (or -99 to quit): **55**

Mild

Enter a temperature (or -99 to quit): **5.5**

Cold

Enter a temperature (or -99 to quit): **105**

Hot

Enter a temperature (or -99 to quit): **-99**

34% of the temperatures were mild.

Commented [JH33]: @Matthews, Suzanne Dr , here's what we were talking about in the sync. If this fits the bill, Tim and I will rework day 1 and alt to be similar.

Commented [MS33R2]: I've made some more changes. Please follow this format for the other code writing problems.

Commented [MS33R3]: Thanks for your work on this! @Hutton, John E LTC and @Walsh, Timothy C LTC

Commented [AK34]: Should be Cold (according to current temperature classification)

Commented [AK35]: Should be freezing

Commented [AK36]: Should be 0% (since no temperatures were 'Mild')

Provided to Students:

Sample Solution:

```
'''
Problem: Classify temperatures as being hot, mild, or cold, and find the
percentage of mild temperatures
Input: One temperature at a time until the user enters 0
Output: The percentage of mild temperatures
'''

import math
```

AI105 AY26-1 WPR 2

```
def main():
    #1 initialize accumulators for the total number of
    # temperatures and the number that are mild
    total_days = 0
    mild_days = 0

    #2 get initial input
    temperature = float(input('Enter a temperature above 0, or enter 0 to quit:
'))

    #3 repeat the following lines until temperature is 0
    while temperature != 0:

        #4 increment the total, classify it as hot, mild,
        # or cold, and increment the number for mild if appropriate
        total_days += 1
        if temperature > 75:
            print('Hot')
        elif temperature > 50:
            print('Mild')
            mild_days += 1
        else:
            print('Cold')

        #5 get the next input
        temperature = float(input('Enter a temperature above 0, or enter 0 to
quit: '))

    #6 compute the percentage of days and the ceiling (outside the loop)
    print(str(math.ceil(100 * mild_days / total_days)) + '% of the temperatures
were mild.')

main()
```

Test Cases:

2) Test Case 1

c) Inputs:

- i) 55
- ii) 5.5
- iii) 105
- iv) 0

d) Outputs:

- i) Mild
- ii) Cold

AI105 AY26-1 WPR 2

iii) Hot
iv) 34% of the temperatures were mild.
4) Test Case 2
a) Inputs:
i) 75
ii) 76
iii) 50
iv) 51
v) 0
b) Outputs:
i) Mild
ii) Hot
iii) Cold
iv) Mild
v) 50% of the temperatures were mild.
5) Test Case 3
a) Inputs:
i) 12
ii) 0
b) Outputs:
i) Cold
ii) 0% of the temperatures were mild.
Tested Concepts:
- Conditionals I and II - Using Modules - Iteration II

ALT - CODING PROBLEM - URINALYSIS LINEUP

At the company urinalysis, there are three stations. The left station is for Soldiers with last names that begin with A through H, the middle station is for Soldiers with last names beginning with I through Q, and the right station is for Soldiers with last names beginning with R through Z.

Name begins with	Station
A – H	Left
I – Q	Middle
R – Z	Right

Write a program that continually asks the user for Soldier last names until the user enters "quit". After every name is entered print whether that name goes with the left, middle, or right station. Count the number of Soldiers who go with the left station, and at the end of the program, print the percentage of the total that were at the left station, rounded to the next highest integer. Remember that percentage is 100 times the fraction.

Commented [MS37]: I feel like this is a fantastic WPR 3 question, and is rather difficult for a WPR 2 question.

I know you all created this question based on WPR 2 last semester. I think one of the big takeaways from last semester is that they didn't have enough practice writing functions yet to succeed. I remember one of the comments being that the WPR 3 code writing question would have better served as the WPR 2 problem, and the WPR 2 problem as the WPR 3 problem.

For this year, my preference is that there is just a simple loop question.

So, write a function that performs X (where X requires an indefinite while loop). Then, main() simply calls that function.

OR

Just a while loop in main.

Commented [HT38]: @Matthews, Suzanne Dr I'm sorry I did not catch this before now, but are you comfortable making this the ALT and pulling the current ALT into Day 1? I think the string comparison makes this version significantly more difficult?

Commented [MS38R2]: good catch! Thanks @Tippett, Hayden J MAJ . Yes, let's switch the alt and the day 1.

Commented [MS38R3]: Should we add a few more temperature categories to day 2 to make it more equivalent? Right now, day 1 will have 5 categories, while alt/day2 has only three.

Commented [MS39]: went to?

Commented [DJ39R2]: Or reported to?

Commented [AK40]: Necessary? Day 1/2 questions don't include this verbiage.

AI105 AY26-1 WPR 2

You may assume all names will begin with a capital letter.

Sample Input/Output:

Enter name, or enter "quit" to quit: **Alexander**

Left

Enter name, or enter "quit" to quit: **Michaels**

Middle

Enter name, or enter "quit" to quit: **Smith**

Right

Enter name, or enter "quit" to quit: **quit**

34% of the Soldiers went left.

Provided to Students:

```
'''
Problem: Send names to the left, middle, or right station, and find the
percentage going to the left.
Input: One name at a time until the user enters 'quit'
Output: The percentage of people at the left station
'''
```

```
import math
```

```
def main():
```

```
    #1 initialize accumulators for the total number of
    # names and the number that have gone left
```

```
    total = 0
```

```
    lefts = 0
```

```
    #2 initial prompt
```

```
    name = input('Enter name, or enter "quit" to quit: ')
```

```
    #3 repeat the following lines until the user enters "quit"
```

```
    while name != "quit":
```

```
        #4 increment the total, classify it as left, middle, or
        # right, and increment the number of lefts if appropriate
```

```
        total += 1
```

```
        if name >= "R":
```

```
            print('Right')
```

```
        elif name >= "I":
```

```
            print('Middle')
```

```
        else:
```

AI105 AY26-1 WPR 2

```
print('Left')
lefts += 1

#5 prompt the user for a name
name = input('Enter name, or enter "quit" to quit: ')

#6 calculate and print the ceiling of the percentage of Lefts
print(str(math.ceil(100 * lefts / total)) + '% of the Soldiers went left.')

main()
```

Test Cases:

1) Test Case 1

a) Inputs:

- i) Alexander
- ii) Michaels
- iii) Smith
- iv) quit

b) Outputs:

- i) Left
- ii) Middle
- iii) Right
- iv) 34% of the Soldiers went left.

2) Test Case 2

a) Inputs:

- Adams
- Harrison
- Ianelli
- Roman
- quit

b) Outputs:

- i) Left
- ii) Left
- iii) Middle
- iv) Right
- v) 50% of the Soldiers went left.

3) Test Case 3

a) Inputs:

- i) Jones
- ii) quit

b) Outputs:

- i) Middle
- ii) 0% of the Soldiers went left.

AI105 AY26-1 WPR 2

Tested Concepts:

- Conditionals I and II
- Using Modules
- Iteration II

UNUSED

17. Using Modules 1 {KC} [3 pts]

Question Type: Matching

Question Stem: Match each Python module with its primary intent and the types of functions or variables it provides:

math	Provides functions and constants for performing mathematical operations like square roots, trigonometry, and powers.
statistics	Provides functions for working with averages, medians, and standard deviations using numerical data.
random	Provides functions to generate random numbers, shuffle sequences, and select random items.

Distractors:

Provides tools for creating and manipulating text strings, including searching, splitting, and formatting text.

Provides functions for working with dates and times, such as calculating time differences or formatting timestamps.

Feedback:

- The `math` module provides mathematical constants (like `pi`) and functions such as `sqrt()`, `sin()`, and `pow()` for performing advanced math operations.
- The `statistics` module focuses on analyzing numerical data, offering functions like `mean()`, `median()`, and `stdev()` for statistical calculations.
- The `random` module is used for generating random numbers or selecting random elements, such as with `randint()`, `choice()`, or `shuffle()`.

18. While Loops 1 {CT} [3 pts]

Question Type: Multi-answer

Commented [HT41]: @Hicks, Matthew S LTC

Commented [HT41R2]: @Hicks, Matthew S LTC should be Code Tracing question as opposed to Knowledge check

Commented [MH41R3]: I added a Code Trace below.

Commented [HT42]: @Walsh, Timothy C LTC

Commented [TW42R2]: TW

AI105 AY26-1 WPR 2

Question Stem: Based on this code snippet, indicate whether the following statements are true or false:

```
secret_word = "eagle"
guess = ""
while guess not in [secret_word, "quit"]:
    guess = input("Guess the secret word: ")
    if guess == "quit":
        print("Giving up already?")
    elif guess != secret_word:
        print("Nope, try again!")
if guess == secret_word: # line 9
    print("You guessed it!")
```

The loop will not execute if the user types "eagle" on the first try.	False
Typing "quit" will cause the loop to end even though it's not the secret word.	True
If secret_word was assigned the value "quit", the program would eventually print "You guessed it!"	True

Feedback:

- False because the loop needs to execute at least once to get and check the user input.
- True because this will cause the conditional after the while keyword to become False.
- True because the final if statement on line 9 will execute regardless of how the while loop terminated.

Commented [MS43]: possible for a student to misinterpret this. If a user types "quit" the first iteration, it will still run the loop once.

Typing "quit" will cause the loop to end.

Commented [TW43R2]: I'm still not sure what the misinterpretation could be, but I think I've changed it now to what you're suggesting.

Commented [MS44]: the code snippet is not numbered.

Commented [TW44R2]: I've added a comment in the code to mark line 9.



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB I

WPR 3

AI105 AY26-1 WPR 3

ACCESS CODES:

	<u>Day 1</u>	<u>Day 2</u>	<u>Alternate</u>
Part A	papyrus	mariner	fragment
Part B	scroll	helm	shard
H.A.S.	relic		

GUIDANCE:

10x knowledge check questions (41 points total)

- 2-3 questions from Block 1/2
- Every option must be 1 point each
- True/False, multi-answer(dropdown), word bank, matching, and multiple choice (w/ shared stem) all valid question types

5x Code Tracing Questions (19 points total)

- Every option must be 1 point each
- 3 code tracing question from last WPR
- 2 bluefruit tracing questions

1x Free-Response Question (15 points)

- Textbox, holistic rubric

1x Code Writing Problem

- 25 points, holistic rubric
- Should cover iteration (for), basic types, built-in functions

LEGEND: Updates to be made. Question Answers. Feedback Section. Configured (XO-Only)

Outcome Alignment

KC/CT Question Template:

1. Lesson {KC/CT} [x pts]

Question Type: update

Question Stem:

Outcome Alignment: SLO1/SLO2/SLO3/SLO4/SLO5/SLO6

insert question stem

Answer

Feedback:

AI105 AY26-1 WPR 3

PART A: Knowledge Check & Code Tracing

1. [Review 1 {KC} [4 > 3 pts]

Question Type: Drag and Drop

Question Stem: Match each protocol to the most analogous thing in the physical world.

Outcome Alignment: SLO1/SLO2/SLO3/**SLO4**/SLO5/SLO6

DNS	A dictionary that gives you the definition of unknown words
UDP	A delivery driver that chucks packages at your door while driving by your house.
IP	A system of street names and building numbers that enables you to navigate to your destination.

Distractors:

TCP
HTTP
IMAP

Feedback:

DNS is analogous to a dictionary. UDP is “best effort” packet delivery (there is no real check that it reached its destination). IP defines routing, which is similar to how you navigate in the real world.

Commented [HT1]: @Matthews, Suzanne Dr

Commented [MS1R2]: SM

Commented [HT2]: @Tippett, Hayden J MAJ NEED TO ADD ONE MORE OPTION

2. [Review 2 {KC} [4 pts]

Question Type: drag and drop/matching

Question Stem: Identify a valid communication protocol and match it to its associated layer

Outcome Alignment: SLO1/SLO2/SLO3/**SLO4**/SLO5/SLO6

Ethernet	Network Access Layer
IP	Internet Layer
UDP	Transport Layer
HTTP	Application Layer

Distractors:

URL, Router, HTML, Packet

Feedback:

- Protocols on the Network Access layer include Ethernet and WiFi
- IP is the main protocol on the Internet Layer that we discuss in AI05
- UDP and TCP are examples of protocols that belong to the Transport Layer
- HTTP, DNS and TLS are examples of protocols on the application layer

Commented [HT3]: @Matthews, Suzanne Dr

Commented [MS3R2]: SM

AI105 AY26-1 WPR 3

- URLs, Routers, HTML, and Packets are not examples of communication protocols.

3. Cyberspace & Threat Actors 1 {KC} [4 pts]

Question Type: Multiple answer

Question Stem: Match the activity with the threat goal.

Outcome Alignment: SLO1/**SLO2**/SLO3/SLO4/SLO5/SLO6

An adversary uses a botnet to flood your internet connection with a large amount of data	Deny , Alter, Disclose,
A social media app gathers data without a user's consent to sell it to third-party brokers	Disclose , Alter, Deny
A deep fake video shows your favorite athlete telling you that you have won a prize	Alter , Deny, Disclose
Ransomware	Deny , Alter, Disclose

Feedback:

The botnet is trying to crowd out valid traffic so you cannot reach your own resources. This is a DENY goal.
The social media is collecting information to give to a third party. This is a DISCLOSE goal.
Deep fakes ALTER the original subject.
Ransomware encrypts your hard drive so you do not have access to your data nor applications. This is a DENY goal.

Commented [HT4]: @Hutton, John E LTC

4. Cyberspace & Threat Actors 2 {KC} [4 pts]

Question Type: Multi-answer

Question Stem: State whether the following statements regarding the layers of cyberspace are True or False

Outcome Alignment: SLO1/SLO2/SLO3/**SLO4**/SLO5/SLO6

The logical layer dictates hardware placement in the physical layer.

False

A user's location geographically located always limits which cyber-persona they can use.

False

The logical layer operates independently of physical location.

True

An individual may have more than one cyber-persona.

True

Feedback:

- The logical organization of a network does not place strict requirements on where physical network components are located.
- Cyber-personas are largely independent from the location of users.

Commented [HT5]: @Kedrowitsch, Alexander L LTC

Commented [AK5R2]: ALK

Commented [MS6]: delete?

I want to cut down the number of words.

Commented [MS7]: A user's location limits which cyber-person they can use.

AI105 AY26-1 WPR 3

- The physical location of network hardware is abstracted from how a network is logically organized.
- Individuals often adopt more than one cyber-persona when interacting with networks.

5. Cybersecurity 1 {KC} [6 pts]

Question Type: Matching

Question Stem: Match two malware-type labels to each malware description: (0.66 1pts for each correct label)

Outcome Alignment: SLO1/SLO2/SLO3/SLO4/SLO5/SLO6

A harmless looking program was downloaded from a website and installed, now the computer constantly displays numerous advertisements

Trojan, Adware

Malware that independently spreads from computer to computer and encrypts data, demanding money for users to regain access to files

Worm, Ransomware

Malware that has infected a valid spreadsheet file and steals passwords by recording keystrokes

Virus, Spyware

(Available malware-type labels (non-reusable): Trojan, Adware, Worm, Ransomware, Virus, Spyware)

Feedback:

- Trojan malware masquerades as innocent software; Spyware generates advertisements to display to users.
- Worms self-replicate and self-propagate between computers without user interaction; Ransomware denies user access to their data through encryption and refuses to disclose the decryption key without some form of payment.
- Viruses attach themselves to legitimate user files and are triggered by user interaction; Spyware monitors user actions on the computer, such as with a keylogger, and can steal user credentials.

6. Cybersecurity 2 {KC} [4 pts]

Question Type: Multi-answer

Question Stem: Identify whether each statement about the goals of cybersecurity is true or false.

Outcome Alignment: SLO1/SLO2/SLO3/SLO4/SLO5/SLO6

non-repudiation

Confidentiality can be ensured with encryption that allows only a sender and recipient to read the plaintext version of a message.	True
Encryption ensures the confidentiality of a message sent across a network.	
Digital signatures are commonly used to achieve authentication, integrity, and non-repudiation.	True
Digitally signing a message helps authenticate the user sending the message	

Commented [HT8]: @Kedrowitsch, Alexander L LTC

Commented [AK8R2]: ALK

Commented [MS9]: @Tippett, Hayden J MAJ can we match two like this? I didn't think this was possible in Canvas.

Commented [TM9R2]: this is doable. I don't see any reason why we couldn't do a word bank without re-use and just have a table column with two blanks each.

Commented [HT10]: @Walsh, Timothy C LTC

Commented [TW10R2]: TW

Commented [MS11]: @Walsh, Timothy C LTC We don't spend too much time on non-repudiation! I think it would be better to focus on one of the big three: confidentiality, authentication and integrity.

Commented [TW11R2]: How about this? I modified it to touch on most of the goals of cybersecurity instead of focusing on non-repudiation.

Commented [MS11R3]: I think this is harder. We need to cut down the number of words.

For example, for the first one. What if it is changed to:

Encryption ensures the confidentiality of a message sent across a network.

Digitally signing a message helps authenticate the user sending the message

A message with integrity guarantees means the sender cannot lie about who they are.

The primary purpose of non-repudiation is to verify the identity of a sender

Commented [TW11R4]: OK, I cleaned up a couple things here.

AI105 AY26-1 WPR 3

Integrity means that a user cannot lie about sending a message that they did not really send. The cybersecurity goal of integrity guarantees that a message's sender cannot lie about who they are.	False
Non-repudiation is primarily concerned with verifying the identity of the sender to prevent impersonation. The primary purpose of non-repudiation is to ensure data is not disclosed in transit.	False

Feedback:

- True.
- True.
- False. Integrity is about ensuring that a message can't be altered, or that a recipient of an altered message can detect that it was altered.
- False. This describes the goal of confidentiality. Non-repudiation is about ensuring that a sender cannot later deny sending some data.

7. Online Tracking and Ethics {KC} [4 pts]

Question Type: Multi-answer

Question Stem: Identify whether each statement about anonymity is true or false.

Outcome Alignment: SLO1/SLO2/SLO3/SLO4/SLO5/SLO6

Your device, operating system, and browser configuration can be used by websites to uniquely identify you.	True
Data that has been stripped of personal identifiers (like names and Social Security numbers) prevents re-identification and is therefore considered to be anonymous. Proximity-based apps such as Jodel guarantee the anonymity of its users	False
The effectiveness of anonymization techniques depends partly on what other datasets are publicly available. It is often possible to reconstruct an individual's identity by cross-referencing anonymized data to other databases.	True
Grouping or averaging data from multiple users can reduce, but not eliminate, the risk of identifying individuals.	True

Feedback:

- True. These details are routinely transmitted during web browsing and a set of them can serve as a unique identifier.
- False. While Jodel may guarantee anonymity to other users on the app, it does not hide the identity of the users to the service provider
- True. Cross-referencing with other sources of data can compromise anonymity.
- True. Aggregation helps to obscure individual data, but uniqueness in behavior or small group sizes can still lead to re-identification.

Commented [HT12]: @Dougherty, Ryan E CIV

Commented [TW12R2]: I already have something prepared for this topic. I'll put it here and own it, or you can overwrite it.

Commented [MS13]: @Walsh, Timothy C LTC I like this question! I would challenge you to see if you can make the writing a bit more succinct, to cut down the amount the cadets have to read to get through this question. This is especially an issue for our foreign exchange cadets.

Commented [TW13R2]: Sure. I cut down the number of words in that one and the fourth one.

Commented [RD14]: Not entirely sure what this means - @Walsh, Timothy C LTC can you explain? This might be reasonable but the wording seems somewhat unclear to me.

Commented [TW14R2]: It's trying to allude to aggregation. For example, in chapter 26.4, there's a link to the story about Massachusetts Governor William Weld's medical records, saying "...it is often possible to reconstruct an individual's identity by cross-referencing anonymized data to other databases that contain identifiers but no sensitive information... She used Weld's readily available zip code and birth date to scan the Massachusetts Group Insurance Commission (GIC) database for his records and confirmed the identity using voter-registration records from Cambridge, Massachusetts." So that's the idea, but maybe this isn't obvious enough from the reading and lesson.

Commented [TW14R3]: I realized that I could just insert the quoted statement above as the question stem, so I've done that.

AI105 AY26-1 WPR 3

8. GenAI Wireshark Lab {KC} [4 pts]

Question Type: Multi-answer

Question Stem: You are walking through Grant Hall with a friend and notice someone running Wireshark. Your know-it-all friend makes four statements. Determine whether each of the statements below are True or False:

Outcome Alignment: **SLO1**/SLO2/SLO3/**SLO4**/SLO5/SLO6

"Using a packet sniffer on someone else's network to capture packets is a legitimate exercise of 1st Amendment rights."	False
"A packet sniffer is unable to capture any packets from the WREN because the WREN Wi-Fi is encrypted."	False
"A packet sniffer's sole purpose is to assist hackers in the reconnaissance and scanning phases of an attack."	False
"Wireshark allows a computer to capture all packets it can see on the local area network, not just those intended for it."	True

Feedback:

- False. Capturing packets on someone else's network without permission will often be illegal, not protected speech.
- False. Wi-Fi encryption does not prevent all packet capture; encrypted frames can still be captured, though contents may be unreadable.
- False. Packet sniffers like Wireshark are *neutral tools* – used by both network administrators and attackers. Their purpose depends on authorization and intent.
- True. Wireshark can be configured so that a computer captures *all* packets visible on the local network segment, not just those addressed to its own MAC address. This is useful for network monitoring and analysis but can also be misused for eavesdropping if done without authorization.

9. Robotics 1 {KC} [4 pts]

Question Type: Matching

Question Stem: Recall that all robots fulfill the sense-think-act paradigm of robotics. Evaluate whether each given device is a robot using this framework.

Outcome Alignment: SLO1/SLO2/**SLO3**/SLO4/SLO5/SLO6

A programmable toy car that drives in a figure 8 pattern.	Missing sense
A weather station that records temperature and humidity and predicts the weather for the next 24 hours.	Missing act
An autonomous vacuum that maps a room and cleans it.	Robot
A security camera that rotates 360 degrees to capture its surroundings.	Missing think

Answer bank (Bank with Reuse): Missing sense, Missing act, Missing think, Robot

Commented [HT15]: @Hicks, Matthew S LTC

Commented [MS15R2]: @Hicks, Matthew S LTC This question is also due by COB today. Please get this done!

Commented [HL16]: @Matthews, Suzanne Dr This is the changed one.

Commented [MS16R2]: @Hicks, Matthew S LTC Can we please rephrase?

"Wireshark allows a computer to capture all packets it can see on the local area network, not just those intended for it"

My concern is that there will be some ambiguity on what "network segment" means and that some will be confused with what "those addressed to it" means.

Commented [HL16R3]: @Matthews, Suzanne Dr Fixed.

Commented [HT17]: @Hicks, Matthew S LTC

Commented [MS17R2]: @Hicks, Matthew S LTC this question is due COB today. Please get this done!

Commented [MS18]: recommend saying something like "drives in a figure 8 pattern". The ter "pre-set route" suggests that some sensing is needed.

AI105 AY26-1 WPR 3

- Feedback:

A programmable toy car that drives a figure-8 pattern – Missing sense

The car follows coded instructions but cannot detect obstacles or changes in its environment, so it lacks the ability to *sense* anything around it.

- A weather station that records temperature and humidity – Missing act**

The station *senses* environmental data and *thinks* by processing or storing it, but it doesn't *act* or respond to the readings in any way.

- An autonomous vacuum that maps a room and cleans on its own – Robot**

It *senses* obstacles and dirt, *thinks* by planning an efficient cleaning path, and *acts* through movement and cleaning – fulfilling all three parts of the sense-think-act framework.

- A security camera that rotates 360 degrees to capture its surroundings – Missing think**

The camera *senses* visually and *acts* by rotating, but it doesn't *analyze* or *decide* what to focus on—it just follows a programmed pattern without real reasoning.

10. Robotics 2 {KC} [4 pts]

Question Type: Multiple Choice

Question Stem: For each of the following statements about the Robotics lab, answer whether it is True or False.

Outcome Alignment: SLO1/SLO2/SLO3/SLO4/SLO5/SLO6

The robot uses its servo to sense light levels.	False
The robot uses the Bluefruit microcontroller to make decisions.	True
The robot can move its actuator without requiring the Bluefruit.	False
The robot's accelerometer is directly responsible for making the robot move.	False

Feedback:

The robot uses its servo to sense light levels.	This is false – a sensor can sense light levels, not a servo.
The robot uses the Bluefruit microcontroller to make decisions.	This is true – this is precisely the “Think” part of the Sense-Think-Act paradigm.
The robot can move its actuator without requiring the Bluefruit.	This is false – there always is a “thinking” step required.
The robot's accelerometer is directly responsible for making the robot move.	This is false as the accelerometer is not directly responsible for making the robot move.

11. Review 3 {CT} [4 => 3 pts]

Question Type: update

Commented [HT19]: @Dougherty, Ryan E CIV

Commented [MS19R2]: @Dougherty, Ryan E CIV The question is due today by COB. Please get this done!

Commented [MS20]: @Dougherty, Ryan E CIV Overall, I like this question! However, I challenge you to think about how you can cut down the writing. For a 55-minute exam, it is in our students' interest to not have a lot to read (especially our foreign exchange students).

For example, this first statement could be adjusted to be: "The robot used its servo to sense light levels"

Commented [DJ21]: I think the feedback needs to be updated to match the revised wording of the question.

Commented [HT22]: 3 options >> 4 pts?

AI105 AY26-1 WPR 3

Question Stem:

Outcome Alignment: SLO1/SLO2/**SLO3**/SLO4/SLO5/SLO6

```
01 def main():
02     x = 5
03     y = 4
04     z = (x >= y)
05     if z:
06         print("Hooray")
07         print("All Done")
08
09     main()
```

After line 04, the value of z will be 9	False
The program will print "Hooray"	True
The program will print "All Done"	True

Feedback:

- After line 4, the value of z will be – False; This is a valid Boolean assignment.
- The program will print "Hooray" – True; Since z is True, line 06 will execute.
- The program will print "All Done" – True; This line executes no matter what z evaluates to.

12. Bluefruit 1 {CT} [4 pts]

Question Type: Multiple answer (fill in the blank)

Question Stem: Consider the code snippet below from the Robotics Lab and select the best answer.

```
bitmap = displayio.OnDiskBitmap("/EECS.bmp")

while True:
    display_image(display, bitmap)
    time.sleep(3)
```

The loop shown is an example of a(n) ____ loop.	Definite, Indefinite, Infinite
It will take a minimum of ____ seconds to execute one loop.	3 , 6, 9, 12
From where is an image retrieved just before it is displayed?	On the laptop hard drive, In your laptop's RAM, On Canvas, On the Bluefruit device

Commented [HT23]: @Hutton, John E LTC

Commented [MS24]: @Hutton, John E LTC I like this question! My only concern about this is that this is one of the two activities that cadets had to do. Thes ones who chose this option will have a much greater advantage (from a time and understanding perspective) than those who chose a song. I think that is an unfair advantage. Would it be possible to remove the three images and just do one image? That's one they all were supposed to do

AI105 AY26-1 WPR 3

There are two possible images that can display in this code.	True, False
--	-------------

Outcome Alignment: SLO1/SLO2/**SLO3**/SLO4/SLO5/SLO6

Feedback:

The loop has no way to change the condition, and there is no code to break out of the loop, so it is infinite.
The sleep accounts for 9 seconds (i.e. 3x 3 second sleeps), so the loop must take at least that long. It will take some, but very little time to display the images, so it will be done in noticeably less than 12 seconds.
We had to copy the image on the Bluefruit, and the image is loaded from the file on the Bluefruit.
There is only one image that will appear – EECS.bmponly

- The loop has no way to change the condition, and there is no code to break out of the loop, so it is infinite.
- The sleep accounts for 9 seconds (i.e. 3x 3 second sleeps), so the loop must take at least that long. It will take some, but very little time to display the images, so it will be done in noticeably less than 12 seconds.
- We had to copy the image on the Bluefruit, and the image is loaded from the file on the Bluefruit.
- There is only one image that will appear – EECS.bmponly

13. Bluefruit 2 {CT} [4 pts]

Question Type: Multi-answer

Question Stem: Consider this code, which you used during the Robotics Lab, and answer the following questions about it.

```
import time
from adafruit_circuitplayground import cp

def button_A_or_B():
    while True:
        # checks to see if button A is pressed
        if cp.button_a:
            print("Button A pressed")

        # checks to see if button B is pressed
        if cp.button_b:
            print("Button B pressed")
        time.sleep(0.1)

def main():
    button_A_or_B()
```

Commented [HT25]: @Walsh, Timothy C.LTC

Commented [WL25R2]: TW

AI105 AY26-1 WPR 3

main()

It's possible to get both "Button A pressed" and "Button B pressed" to print during the same iteration of the loop.	True
A certain sequence of button A and B presses will cause the program to terminate.	False
The loop waits for the user to press a button before continuing execution.	False
Holding down button A for about one second will cause "Button A pressed" to print about ten times. Holding down button A for a second, without letting go, will cause "Button A pressed" to print multiple times.	True

Outcome Alignment: SLO1/SLO2/**SLO3**/SLO4/SLO5/SLO6

Feedback:

- True. On each iteration, the loop checks the status of both buttons independently, and both print statements can execute.
- False. This is an infinite loop. The condition is simply True and no sequence of button presses will trigger a break or return statement.
- False. The program checks the status of the buttons every tenth of a second and continues looping regardless of whether the buttons are pressed.
- True. The program checks the status of the buttons every tenth of a second and continues looping, so holding the button down for a second will cause the print statement to appear about ten times.

14. Review 4 {CT} [4 pts]

Question Type: Dropdown

Question Stem: Consider the following code.

```
01 x = 10
02 def add2(x):
03     return x + 2
04
05 def main():
06     num = 3
07     total = 0
08     while num > 0:
09         total += add2(num)
10         num -= 1
```

Commented [MS26]: @Walsh, Timothy C LTC Overall, I really like this question! I just worry this may be a little too hard. Instead, can we just say it will cause "Button A pressed" to print multiple times?

Commented [TW26R2]: That's fair. Done. I also added a few other words to make it more clear.

Commented [TM27]: @Walsh, Timothy C LTC please confirm feedback based on changes

Commented [TW27R2]: @Tippett, Hayden J MAJ Confirmed.

Commented [MS28]: SM

AI105 AY26-1 WPR 3

```
11     print(total)
12     print(x)
13 main()
```

How many times will the <code>while</code> loop on line 08 execute?	0; 3; 2; 1
What value will line 11 print to the user?	0; 12; 36; 48
Executing line 12 results in an error	True, False
The <code>add2</code> function is an example of a fruitful function	True, False

Outcome Alignment: SLO1/SLO2/**SLO3**/SLO4/SLO5/SLO6

Feedback:

- In the `main()` function, `num` is set to 3 (line 06). The loop runs while `num > 0`, and each iteration `num` is subtracted by 1. Therefore, the loop will run for three iterations (when `num` is 3, 2, and 1).
- Line 11 prints 12 to the user, since `total` is incremented by `num+2` each iteration of the loop. Therefore, `total` is `5+4+3`, or 12.
- Incorrect. Since `x` is defined in the global scope, line 12 will print out 10.
- `add2()` is an example of a fruitful function since it does return a value (e.g. fruit).

15. Review 5 {CT} [4pts]

Question Type: True/False

Question Stem: The code below is designed to determine the number of cadets that have the letter "A", "B" or "C" in their names. However, it does not work as intended. Indicate if the statements about the code are True or False.

Consider the following code written by a fellow cadet to determine how many cadets in his/her company are assigned each possible detail, which are "A", "B", and "C". Each cadet is represented as a string of the detail(s) the cadet is assigned to. For example, if a cadet is represented by "BA", then they are assigned to both "A" and "B" detail, once each. Note that each cadet may be assigned a detail more than once; e.g., "AAB" means participating in detail "A" twice, and "B" once.

For example, if the cadet list has two cadets represented as "AAB" and "CA", then the code should return the list [3, 1, 1] as there were three occurrences of "A" across both cadets, and 1 of the other details.

For each of the following questions, determine if the answer is True or False.

```
01 def count_cadets(L): # L is a list of cadets
```

Commented [MS29]: I like this problem, but not the story behind it. Can we simplify it to just be the cadets with "A" "B" or "C" in their names? That's easier to explain, and will require less text for the WPR. The code then stays the same.

AI105 AY26-1 WPR 3

```

02 total_count = [0, 0, 0] # counters for A, B, and Cs
03 for cadet in L:
04     if "A" in cadet:
05         total_count[0] += 1
06     elif "B" in cadet:
07         total_count[1] += 1
08     elif "C" in cadet:
09         total_count[2] += 1
10 return total_count # returns final counters as a list

```

Commented [MS30]: I like this problem, but not the story behind it. Can we simplify it to just be the cadets with "A" "B" or "C" in their names? That's easier to explain, and will require less text for the WPR. The code then stays the same.

The if/elif chain makes it impossible to determine counts when a cadet has a combination of As, Bs, and Cs in their names.	True
The count_cadets function can take in more than 1 argument (e.g., when there are many cadets to count).	False
The list indices used on lines 05, 07, and 09 are invalid, and thus cause the program to crash.	False
The code correctly handles any cadets who don't have As, Bs, or Cs in their names.	True

Commented [DJ31]: I find this wording a little confusing.

Commented [RD31R2]: Made an update

Outcome Alignment: SLO1/SLO2/**SLO3**/SLO4/SLO5/SLO6

Feedback:

The if/elif chain makes it impossible for determining detail counts when a cadet has more than 1 detail assignment.	This is true – the if/elif chain makes it impossible for the code to execute in more than 1 of the if or elif's.
The countEachDetail function can take in more than 1 argument (e.g., when there are many cadets to count).	This is false – the function always takes in exactly one argument, the list L. This list can itself have many cadets within it, but the countEachDetail function itself only takes L as a parameter.
The list indices used on lines 05, 07, and 09 are invalid, and thus cause the program to crash.	This is false – on line 02, total_count is initialized with three entries (all set to zero). Thus the valid indices of total_count are 0, 1, and 2, which are what are used on lines 05, 07, and 09. With that said, this code is “fragile”. For instance, if one adds another detail, then the code needs to be updated in several places to keep everything in-sync. Think about ways that one can update the code to be less fragile.
The code correctly handles any cadets who don't have As, Bs, or Cs in their names.	This is true – when this happens, every if and elif check returns False (as none of the details are in

Commented [DJ32]: I find this wording a little confusing.

Commented [RD32R2]: Made an update

AI105 AY26-1 WPR 3

	a string that's empty), and thus total_count is not updated at all, as expected.
--	--

Free Response Questions [15 points]

Focus: AI and Ethics

Outcome Alignment: **SLO1**/SLO2/SLO3/SLO4/**SLO5**/SLO6

(Day 1)

Imagine that the Army issues your company a weapon system that does automatic target recognition using AI. It consists of a machine learning model trained to detect targets in camera images, a gun that automatically aims at whatever gets marked as a target, and several cameras. How do these components map to the sense-think-act paradigm of robotics? Discuss one benefit and one ethical concern that can arise from using this weapon system.

V1: Imagine that the Army issues your company a machine learning-enabled turret system with an enhanced scope that does automatic target recognition, orientation (the system will turn and pivot towards the identified target), and engagement. How do the components of this weapon system map to the sense-think-act paradigm of robotics? State any assumptions you make. Discuss one benefit and one ethical concern that can arise from using this weapon system.

Does this comply—and why or why not—with the ethical principle stating that AI systems must be governable, i.e. able to be disengaged or deactivated if they demonstrate unintended behavior?

You can validate and tune the system settings towards either higher precision or higher recall. Demonstrate your understanding of these metrics and the ethical considerations by presenting your best argument for tuning the system towards precision instead of towards recall.

Sample Answer:

The scope corresponds to the “sense” part of the robotics paradigm, the machine learning model corresponds to “think” part, and the mechanical actuators that aim the gun correspond to the “act” part.

List of benefits that a student could give: provides an advantage over the enemy by speeding up our operations; enables us to keep up with an enemy who might already be using this kind of technology; helps a human operator to make more accurate decisions and avoid civilian casualties or friendly fire.

List of ethical concerns that a student could give: an unbalanced training dataset could introduce an unacceptable bias; training data and testing conditions that aren’t representative of the true deployment environment could cause it to be very inaccurate when deployed; a stressed out or tired human operator could be misled into engaging civilians or friendly forces; it could desensitize operators to killing and erode their sense of responsibility.

Commented [HT33]: @Kedrowitsch, Alexander L LTC @Walsh, Timothy C LTC Free Response Day 1, 2, ALT with AI & Ethics focus

Commented [HT33R2]: FRAGO, updated to LTCs Walsh and Kedrowitsch.

Commented [TW33R3]: TW

Commented [MS34]: I think this part is too hard. Can't we simplify it to ask them to discuss a benefit and an ethical concern?

Commented [MS35]: We don't spend enough time on these topics. There is no way the students are going to be able to answer this.

Instead, why not ask them to map the different components of the weapon to sense, think, act and the legal and ethical considerations of using such a weapon?

Commented [TW35R2]: Ok, I rewrote this one along those lines. I think that asking generally about the legal and ethical considerations would make it very hard for us to grade - like that's a long list of potentially correct answers, and I don't think any of us are philosophy or legal scholars - so I'm trying to force them to address and show understanding of one specific consideration from the reading.

AI105 AY26-1 WPR 3

As described, the system complies with the “governable” principle because it only aims automatically; it does not pull the trigger, so a human operator in the loop is able to make the final decision about whether an object is really a target, and whether to engage or not.

Tuning the system towards precision means trying to reduce the number of false positives, i.e. the likelihood of the system falsely classifying an object as a target when the object is really a non-target. Because we’re talking about lethal effects, this is desirable because we need to minimize civilian casualties and friendly fire to comply with the law of war and preserve our own forces. Tuning the system towards recall would mean the opposite, i.e. reducing the number of false negatives, but likely increasing the number of false positives. That would be ethically unacceptable unless other measures to prevent civilian casualties and friendly fire are in place.

(Day 2)

Outcome Alignment: **SLO1**/SLO2/SLO3/SLO4/**SLO5**/SLO6

An autonomous drone or missile can fly to its intended target without a pilot’s intervention and precisely crash munitions into the target with lethal effects. How do the components of the drone or missile map to the sense-think-act paradigm of robotics? State any assumptions you make. Discuss one benefit and one ethical concern that can arise from using this weapon system.

How can the developers of this system ensure compliance with the ethical principle stating that AI systems must be reliable?

Assume that we can validate and tune the system settings towards either higher precision or higher recall. Demonstrate your understanding of these metrics and the ethical considerations by presenting your best argument for tuning the system towards recall instead of towards precision.

Sample Answer:

The instruments on the drone or missile correspond to the “sense” part to get data about its position, orientation, heading, airspeed, altitude, etc. relative to the earth and its intended target. The software that reads the instrument data and computes what adjustments are needed corresponds to the “think” part. The actuators that move the control surfaces or adjust the engine power level up and down correspond to the “act” part.

List of benefits that a student could give: provides an advantage over the enemy by speeding up our operations or being a force multiplier; enables us to keep up with an enemy who might already be using this kind of technology; possibly makes fewer mistakes than a drone or missile that’s remotely controlled by a human operator.

List of ethical concerns that a student could give: poorly designed algorithms or testing conditions that aren’t representative of the true deployment environment could cause it to be very inaccurate or unreliable when deployed; it could desensitize operators to killing and erode their sense of responsibility.

Tuning the system towards recall means trying to reduce the number of false negatives, i.e. the likelihood of the system falsely classifying an object as a non-target when the object is really a target. Assuming that targets are hostile and will use lethal effects on our forces and possibly civilians, tuning towards recall is desirable because we need to recognize as many of them as possible. Failing to detect even one hostile target could get us killed. Tuning the system towards precision would mean the opposite, i.e. reducing the number of false positives, but missing more targets. False positives are problematic too, but we can take other measures to prevent civilian casualties and friendly fire, like having the human operator approve each target before engaging.

Commented [MS36]: same comment as the previous

Commented [MS37]: same comment as the previous

Commented [TW37R2]: I’ve tried to do something similar to the above here.

Commented [TW37R3]: I’ve rewritten this whole thing again to mirror the Day 1 prompt that we settled on.

AI105 AY26-1 WPR 3

(ALT)

Outcome Alignment: **SLO1**/SLO2/SLO3/SLO4/**SLO5**/SLO6

Imagine that you are developing an AI weapon system that does automatic target recognition; i.e. sensing and classifying objects in the environment as either targets or non-targets for lethal effects. What is the problem with training it on a dataset that is heavily imbalanced, containing far more examples of targets than non-targets, or vice versa?

What metrics can you use to evaluate whether there is a problem with your system?

Sample Answer:

A system trained on a heavily imbalanced dataset may become biased towards the majority class, because erring on the side of the majority class when uncertain allows it to be correct more often. Our reading gave the example of a dataset containing 950 examples of croissants and just 50 examples of cats, on which a classifier could score 95% accuracy by simply labeling every example as a croissant. For a weapon system, that kind of behavior would raise serious ethical concerns because erring towards false positives, i.e. mistakenly labeling civilians or friendly forces as targets, could result in killing the wrong people.

We should instead look at precision, recall, and balanced accuracy to get a more complete picture of how the system performs, especially on examples of the minority class.

PART B: Code Writing Problem [25 points]

NOTE TO FACULTY: This should be of similar difficulty to the last WPR!

The difference is that this time, they are **required** to write a function and refer to it in main()!

Please look at WPS 8 for some inspiration.

Outcome Alignment: SLO1/SLO2/SLO3/SLO4/SLO5/SLO6

DAY 1 - CODING PROBLEM – LATENESS TRACKER

Problem Specifications: You are placed in the role of tracking arrival times of cadets in your squad and determining who is very late for formation for that day. Your program should first ask for the time of formation (an integer). Afterwards, your program should continually prompt the user for the arrival time of the cadet, until the value -1 is entered. Your program should output the number of cadets that are late (in other words, if they arrive after the formation time).

Note: you are **required** to write a function that takes a cadet's arrival time and the formation time as its two parameters. The function should print "On time" if the arrival time is before or at the formation time or "Late" if the arrival time is after the formation time. The functions should also **return True** if the cadet is very late and **return False** otherwise. Your main program must use the function that you wrote.

Commented [TW38]: @Matthews, Suzanne Dr Are you OK with this one, though?

Commented [MS38R2]: I am honestly a little uncomfortable, since we do not explicitly cover precision and recall in class.

Commented [TW38R3]: Adjusted per our discussion.

Commented [HT39]: @Hicks, Matthew S LTC @Dougherty, Ryan E CIV

Commented [HT39R2]: FRAGO, updated to LTC Hicks and Dr. Dougherty for CWP

Commented [MS39R3]: @Hicks, Matthew S LTC @Dougherty, Ryan E CIV The code writing problems for the WPR are still missing. They are due by COB today!

Commented [RD39R4]: Done.

AI105 AY26-1 WPR 3

Sample Input/Output:

```
Enter the start of formation: 0700
Enter a cadet's arrival time: 0659
On time
Enter a cadet's arrival time: 0703
Late
Enter a cadet's arrival time: 0650
On time
Enter a cadet's arrival time: 0706
Late
Enter a cadet's arrival time: 0710
Late
Enter a cadet's arrival time: 0658
On time
Enter a cadet's arrival time: -1
The number of late cadets is: 3
```

Provided to Students:

No code provided

Test Cases:

Test Case 1

Inputs:

```
0700
0659
0703
0650
0706
0710
0658
-1
```

Outputs:

```
On time
Late
On time
Late
Late
On time
The number of late cadets is: 3
```

Test Case 2

Inputs:

```
0900
0858
0857
```

Commented [RD40]: There's a potential rare issue: if the formation time is 0010 or something really early, and the cadet arrives at 2359, then they should be marked as on time but are really marked as late as the solution below only considers the int representation.

Commented [RD40R2]: One easy solution is to assert up front that all entered times are on the same day. None of the test cases I wrote have a problem with this, and I doubt any submissions will account for it. Just wanted to point it out just in case.

Commented [MS40R3]: I've added a clause in the top that says "for that day".

AI105 AY26-1 WPR 3

0856
0855
-1

Outputs:

On time
On time
On time
On time
The number of late cadets is: 0

Test Case 3

Inputs:

0620
0621
0622
0619
0618
-1

Outputs:

Late
Late
On time
On time
The number of very late cadets is: 2

Sample Solution:

```
def cadet_is_very_late(cadet_time, formation_time):  
  
    if cadet_time - formation_time <= 0:  
        print("On time")  
        return False # on-time cadets are never late  
    else:  
        print("Late")  
        return True  
  
def main():  
    time = int(input("Enter the start of formation: "))  
    arrival = 0  
    num_late = 0  
  
    while arrival != -1:  
        arrival = int(input("Enter a cadet's arrival time: "))  
        if arrival != -1 and cadet_is_very_late(arrival, time):  
            num_late += 1
```

AI105 AY26-1 WPR 3

```
print("The number of late cadets is:", num_late)

main()
```

Tested Concepts:

- Indefinite iteration
- Writing Functions

DAY 2 - CODING PROBLEM – REASONABLE ROOMS

Problem Specifications:

V1: You are asked to monitor the noise levels in different barracks for study purposes. Your program should first ask for the maximum number of decibels (an integer) allowed. Your program should then continually prompt the user for the room's noise level in decibels, until -1 is entered. Each time a level is inputted, print whether the room is "too loud" or "reasonable". Your program should output the number of rooms that are too loud.

Note: you are **required** to write a function that takes a room's noise level and the maximum allowed decibels as its two parameters. The function should **return True** if the room is reasonable and **return False** otherwise. Your main program must use the function that you wrote.

V2: Write a function `room_is_too_loud` that takes 2 integer parameters. The first parameter is the room's noise level and the second parameter is the maximum allowed noise level in decibels. Your function should display "reasonable" and return the boolean `True` if the room's noise level is below the maximum allowed noise level. Otherwise, your function should display "too loud" and return the boolean `False`.

Outside of your function in main, you should first prompt the user for the maximum allowed noise level in decibels (an integer). Then continually prompt the user for the room's noise level in decibels until they enter -1. Each time a level is inputted call your `room_is_too_loud` function. When the user enters -1 display how many times the room was "too loud".

Sample Input/Output:

```
Enter the maximum allowed decibels: 50
Enter a room's noise level (dB): 48
reasonable
Enter a room's noise level (dB): 57
too loud
Enter a room's noise level (dB): 24
reasonable
Enter a room's noise level (dB): 27
reasonable
Enter a room's noise level (dB): 30
```

Commented [RD41]: Can just make an isomorphic version of this if the "story" doesn't work.

Commented [HT42]: @Tippett, Hayden J MAJ Update all specifications

AI105 AY26-1 WPR 3

reasonable

Enter a room's noise level (dB): -1

The number of rooms that are too loud is: 1

Provided to Students:

No code provided

Test Cases:

1. Test Case 1

a. Inputs:

50
48
57
24
27
30
-1

b. Outputs:

- i. reasonable
- ii. too loud
- iii. reasonable
- iv. reasonable
- v. reasonable
- vi. The number of rooms that are too loud is: 1

2. Test Case 2

a. Inputs:

39
40
40
40
39
39
-1

b. Outputs:

- i. too loud
- ii. too loud
- iii. too loud
- iv. reasonable
- v. reasonable
- vi. The number of rooms that are too loud is: 3

3. Test Case 3

a. Inputs:

49
50

Commented [MS43]: There seems to be something wrong here.. the first number is the max right? why are the rooms shown all reasonable?

Commented [HL43R2]: @Dougherty, Ryan E CIV I think I fixed this now.

AI105 AY26-1 WPR 3

50

51

51

-1

b. Outputs:

i. too loud

ii. too loud

iii. too loud

iv. too loud

v. The number of rooms that are too loud is: 4

4. Test Case 4

a. Inputs:

1

1

1

100

-1

b. Outputs:

i. reasonable

ii. reasonable

iii. too loud

iv. The number of rooms that are too loud is: 1

Sample Solution:

```
def room_is_too_loud(room_db, max_decibels):
    if room_db > max_decibels:
        print("too loud")
        return True
    else:
        print("reasonable")
        return False

def main():

    max_decibels = int(input("Enter the maximum allowed decibels: "))
    level = 0
    num_too_loud_rooms = 0
    while level != -1:
        level = int(input("Enter a room's noise level (dB): "))
        if level != -1 and room_is_too_loud(level, max_decibels):
            num_too_loud_rooms += 1
    print("The number of rooms that are too loud is:", num_too_loud_rooms)
```

AI105 AY26-1 WPR 3

main()

Tested Concepts:

- Lists and Strings
- Writing Functions

ALT - CODING PROBLEM – SUPPLY LOAD REPORT

Problem Specifications: You oversee distributing supply loads to trucks departing the base. Your program should first ask for the max suggested load for these trucks (an integer). Afterwards, your program should continually prompt the user for the load of each truck, until the value -1 is entered. Your program should output the number of trucks that are overloaded (in other words, if they are loaded with more than the max suggested load).

Note: you are **required** to write a function that takes a truck's max suggested load and the truck's current load as its two parameters. The function should **print** whether the truck is "Below maximum load." or "At load limit." or "Overloaded." The function should **return True** if the truck is overloaded and **return False** otherwise. Your main program must use the function that you wrote.

Sample Input/Output:

```
Enter the max suggested load of the trucks: 1500
Enter a truck weight: 1200
Below maximum load.
Enter a truck weight: 1500
At load limit.
Enter a truck weight: 1600
Overloaded.
Enter a truck weight: 1300
Below maximum load.
Enter a truck weight: -1
The number of overloaded truck(s) is: 1
```


AI105 AY26-1 WPR 3

Provided to Students:

No code provided

Test Cases:

Test Case 1

Inputs:

1500
1200
1500
1600
1300
-1

Outputs:

Below maximum load
At load limit
Overloaded
Below maximum load

The number of overloaded truck(s) is: 1

Test Case 2

Inputs:

1000
1500
1200
800
1200
1000
-1

Outputs:

Overloaded
Overloaded
Below maximum load
Overloaded
At load limit

The number of overloaded truck(s) is: 3

Test Case 3

Inputs:

15000
1200
800
1000
1500
1200

Commented [MS44]: This isn't the only valid output. What about the print statements? Please revise [@Hicks](#) [Matthew S LTC](#)

Commented [MS44R2]: Make sure you make the changes for the other test cases too.

AI105 AY26-1 WPR 3

-1

Outputs:

Below maximum load
Below maximum load
Below maximum load
Below maximum load
Below maximum load
The number of overloaded truck(s) is: 0

Sample Solution:

```
def truck_is_overloaded(truck_weight, max_truck_weight):
    if (truck_weight - max_truck_weight) < 0:
        print("Below maximum load")
        return False # on-time cadets are never late
    if (truck_weight - max_truck_weight) == 0:
        print("At load limit")
        return False
    else:
        print("Overloaded")
        return True

def main():

    max_weight = int(input("Enter the max suggested load of the trucks: "))
    truck_weight = 0
    num_overloaded = 0

    while truck_weight != -1:

        truck_weight = int(input("Enter a truck weight: "))
        if truck_weight != -1 and truck_is_overloaded(truck_weight,
max_weight):
            num_overloaded += 1

        print("The number of overloaded truck(s) is:", num_overloaded)

main()
```

Tested Concepts:

- Lists and Strings
- Writing Functions

AI105 AY26-1 WPR 3

AI105 AY26-1 WPR 3

UNUSED

My so-called “friend” does not like when a list has duplicate elements and prefers “unique” ones. An element is “unique” if it does not appear elsewhere in the list. For example, in the list $L = [1, 2, 7, 2]$, elements 1 and 7 are unique (and 2 is not). He claims that he wrote the following Python code snippet to determine the number of unique elements in the list:

```
1. def countUniques(L): # L is a list
2.     count = 0
3.     for item1 in L:
4.         for item2 in L:
5.             if item1 != item2:
6.                 count += 1
7.     return count
```

Commented [MS45]: We do not cover doubly-nested loops in the course. Therefore, we cannot use this question. I am moving it to "unused"

However, you find that the countUniques function does not behave as expected. Which of the following explanations for the function’s behavior are accurate?

- | | |
|---|-------|
| Lines 5 and 6 add to the count when any two elements are different individually, not necessarily unequal to all other elements. | True |
| The function does not return the correct value for any list L. | False |
| The function returns the incorrect value and should return $\text{count} / 2$ instead of count. | False |
| item1 and item2 can be equal, thus influencing the count. | False |

Feedback:

- | | |
|---|---|
| Lines 5 and 6 add to the count when any two elements are different individually, not necessarily unequal to all other elements. | This is true – the “item1 != item2” comparison only involve the individual items item1 and item2; they both can be equal to some other element in the list L. |
| The function does not return the correct value for any list L. | This is false – for example, if L is an empty list, then the for loop does not execute (as there are no elements in L), so the function accurately returns 0. |
| The function returns the incorrect value and should return $\text{count} / 2$ instead of count. | This is false – for example, suppose L is a list with one element. Then the function returns 0 (incorrectly) whereas it should return 1. However, $0 / 2$ is still 0. (Many other list examples work here.) |
| item1 and item2 can be equal, thus influencing the count. | This is false – the if statement explicitly says to only modify the count if they are not equal. |

AI105 AY26-1 WPR 3

However, the observation is useful in “fixing” the code because if items A and B are compared, then they are compared again (both for loops iterate over the same list independently).

16. Bluefruit Tracing 1 {CT} [4 pts]

Question Type: Multiple answer (fill in the blank)

Question Stem: Consider the code snippet below from the Robotics Lab and select the best answer.

Outcome Alignment: SLO1/SLO2/**SLO3**/SLO4/SLO5/SLO6

```
01 bitmap1 = displayio.OnDiskBitmap("/EECS.bmp")
02 bitmap2 = displayio.OnDiskBitmap("/DA_seal.bmp")
03 bitmap3 = displayio.OnDiskBitmap("/USMA_Crest.bmp")
04
05 while True:
06     display_image(display, bitmap1)
07     time.sleep(3)
08     display_image(display, bitmap2)
09     time.sleep(3)
10     display_image(display, bitmap3)
11     time.sleep(3)
```

Commented [HT46]: @Hutton, John E LTC

Commented [MS47]: @Hutton, John E LTC I like this question! My only concern about this is that this is one of the two activities that cadets had to do. Thes ones who chose this option will have a much greater advantage (from a time and understanding perspective) than those who chose a song. I think that is an unfair advantage. Would it be possible to remove the three images and just do one image? That's one they all were supposed to do

The loop shown is an example of a(n) _____ loop.	Definite, Indefinite, infinite
It will take a minimum of _____ seconds to execute one loop.	3, 6, 9 , 12
From where is an images retrieved just before it is displayed?	On the laptop hard drive, In your laptop's RAM, On Canvas, On the Bluefruit device
Which line(s) could you change to ensure USMA_Crest.bmp is the first image displayed, but all three images were still used?	01, 01 and 03 , 01 and 06, 10

Feedback:

The loop has no way to change the condition, and there is no code to break out of the loop, so it is infinite.
The sleep accounts for 9 seconds (i.e. 3x 3 second sleeps), so the loop must take at least that long. It will take some, but very little time to display the images, so it will be done in noticeably less than 12 seconds.
We had to copy the image on the Bluefruit, and the image is loaded from the file on the Bluefruit.
You could change either line 01 or line 06 to get the USMA crest up first, but you also have to change line 03 with line 01 or line 10 with line 06 to ensure all three images are still displayed.

AI105 AY26-1 WPR 3

DAY 1 - CODING PROBLEM – LATENESS TRACKER

Problem Specifications: You are placed in the role of tracking arrival times of cadets in your company and determine who is very late. Write a program that asks, 5 times, for cadet arrival times in minutes after 0700 hours (an integer) and stops when the user enters 999; a negative value indicates the cadet arrived before 0700, and a positive value is after 0700.

Any cadet is "very late" if they are both late (i.e., their arrival time is larger than 0) and if their arrival time is more than 5 minutes later than the median of the entire group. Determine the number of very late cadets and print that number.

Note: you are **required** to write a function that takes (at minimum) a cadet's time and overall median arrival time as arguments and returns a Boolean value indicating if that cadet is considered very late.

Sample Input/Output:

```
Enter a cadet's arrival time: -1
Enter a cadet's arrival time: 15
Enter a cadet's arrival time: 2
Enter a cadet's arrival time: 32
Enter a cadet's arrival time: -3
The number of very late cadets is: 2
```

Provided to Students:

No code provided

Test Cases:

1. Test Case 1
 - a. Inputs:
 - i. -1
 - ii. 15
 - iii. 2
 - iv. 32
 - v. -3
 - b. Outputs:
 - i. The number of very late cadets is: 2
2. Test Case 2
 - a. Inputs:
 - i. -1
 - ii. -2
 - iii. -3
 - iv. -4
 - v. -5
 - b. Outputs:

Commented [MS48]: This is so arbitrary.

Commented [MS49]: @Dougherty, Ryan E CIV This is still too hard and too long for a WPR question. Please simplify it to what I originally asked!

Commented [PL50]: Could the function just determine late or not late and anything > 1900 is Late. Make it a while loop so you can keep checking until you enter -1 then in main an accumulator to add up how many were late

AI105 AY26-1 WPR 3

- i. The number of very late cadets is: 0

3. Test Case 3

a. Inputs:

- i. 6
- ii. 7
- iii. 8
- iv. 13
- v. 14

b. Outputs:

- i. The number of very late cadets is: 1

Sample Solution:

```
import statistics

def cadet_is_very_late(cadet_time, median_of_all):
    if cadet_time <= 0:
        return False # on-time cadets are never late
    if cadet_time <= median_of_all + 5:
        return False
    else:
        return True
    # alternatively, can instead do:
    # return cadet_time > 0 and cadet_time > median_of_all+5

def main():
    arrival_times = []
    for i in range(5):
        time = int(input("Enter a cadet's arrival time: "))
        arrival_times += [time]
    median = statistics.median(arrival_times)
    num_late_cadets = 0
    for cadet_time in arrival_times:
        if cadet_is_very_late(cadet_time, median):
            num_late_cadets += 1
    print("There number of very late cadets is:", num_late_cadets)

main()
```

Tested Concepts:

- Lists and Strings
- Using Modules
- Writing Functions



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB J

Term End Examination

	17DEC	20DEC
Part A - LockDown	frosty	sparkle
Part B - LockDown	garland	candle
Part A (TIPPETT ONLY)	jingle	winter
Part B (TIPPETT ONLY)	ribbon	festive
H.A.S.	holiday	

Format: This exam is 250 points, or 25% of your entire grade. You will have **3.5 hours (210 minutes)** to complete the exam. This exam will cover material from the entire course. cop

- 30 multiple choice or similar style questions for a total of 125 points
- 11 Code Tracing questions for a total of 45 points (4-5 points each).
- 2 Open-Ended Responses (30 points)
- 2 Code Writing question (zyBooks) for 50 points.

Key: _____

Questions adapted from the AY24-2 TEE:

New Questions

Answers to questions are shown in yellow

Configured in Canvas

Microsoft 365 (6pts)

[Q1, 4PTS] Which of the following statements are true about OneDrive? (True/False, 1 pt each)

- It creates a virtual drive that backs up data in the cloud
- If a user gets a new computer and uses O365, they can download data stored on their OneDrive onto their new computer.
- Only files that are created by the Microsoft O365 suite of tools (such as PowerPoint, Excel, etc.) can be stored in OneDrive
- You cannot access OneDrive from the web; using Windows Explorer (your folder view) is the only way to access it.

[Q2, 2PTS] (Multiple Choice) Your TAC tasked you to maintain a spreadsheet that tracks the ACFT scores of everyone in the company. The spreadsheet needs to be a long-term record of company ACFT performance. The spreadsheet needs to be accessible by all squad leaders,

Commented [TW1]: Change to AFT?

platoon leaders, and the TAC and TAC NCO, and potentially members of the regiment and brigade leadership. Which O365 service would be the most appropriate place to store the spreadsheet so that it meets all the goals?

- a. OneDrive (0.75pts)
- b. Microsoft Teams (1.5pts)
- c. SharePoint (2pts)
- d. PowerPoint (0pts)

Problem Solving and Algorithms (6pts)

[Q3, 3PTS] What is an algorithm?

- A. A step-by-step process written in a human language describing how to solve a problem.
- B. A special Python function that combines mathematical operations, often used to solve complex problems involving sequencing, conditionals and loops.
- C. A way to document your code with statements that get ignored by the computer. These statements give the code organizational structure and assist others in understanding the intent of code.
- D. A special type of memory contained within the CPU.

Commented [MS2]: @Filcik, Daniel MAJ please update the wording in this question in the TEE. We are now using the terms sequencing, conditionals and loops instead of "sequence, selection and iteration"

[Q4, 3PTS] When an algorithm uses conditionals, it:

- A. Chooses which steps to execute
- B. Executes a group of steps multiple times
- C. Only goes from one step to the next step directly
- D. Stops executing

Commented [MS3]: @Filcik, Daniel MAJ Please update the wording on this one too

Python Basics (7pts)

[Q5, 4PTS] [Fill in the blank] Indicate what the following Python expressions evaluate to:

5 / 2 is 2.5

5 // 2 is 2

5 % 2 is 1

Commented [HT4]: This question feels like too much of a gimme (all answers are directly stated on reference card). Could instead ask a series of questions like:
7%3 results in ____;
7//3 results in ____;
what is the result of type(9/3);
what is the result of int(2.33)

Commented [MS4R2]: I agree. We should replace this with something else.

5 ** 2 is 25

[Q6, 3PTS] (matching/dropdown) Consider the code below:

```
def my_loop():  
    fruits = ["apples", "cherries", "oranges"]  
    for i in range(len(fruits)):  
        print(fruits[i])
```

Indicate the data type of each of the following expressions:

i integer

fruits[i] string

fruits list

Distractors:

float

boolean

Computers and Data (13 pts)

[Q7, 7PTS] True/False: For each of the computer components listed below, determine if it is an **input/output (I/O) device**. All I/O devices should be marked as True. If it is not an I/O device, select False.

- A. Keyboard
- B. Battery
- C. Microphone
- D. Speakers
- E. Ethernet Cable
- F. Webcam
- G. Touchscreen

[Q8, 6PTS] True/False: Indicate TRUE for the components below that are **primarily** intended to **store data**.

- A. Battery
- B. Motherboard
- C. Random Access Memory
- D. Wireless Router
- E. Hard Drive
- F. GPU

Conditional Statements (6pts)

[Q9, 6PTS] True/False: Suppose the following variables are initialized as follows:

```
x = 3
y = 21
z = 4
```

For each of the following boolean expressions, indicate if they evaluate to True or False

<code>x >= 3 and y < 21</code>	FALSE
<code>x > 3 or z < 5</code>	TRUE
<code>not (z != 4) and y <= 21</code>	TRUE
<code>3 < z < 21 or not (x == 3)</code>	TRUE
<code>not (y < 22)</code>	FALSE
<code>not (x == 3 and y == 20) and not (z == 3)</code>	TRUE

Built-In Functions (4pts)

[Q10 (4PTS) [Dropdown] Suppose a Python program contains the following lines of code:

```
y = "CY105"
x = range(len(y))
```

Commented [M55]: New question on Built-In Functions

Commented [TW6]: AI105

print(len(x)) evaluates to
print(max(x)) evaluates to
print(min(x)) evaluates to
The data type of y is

5
4
0
str

Data Sources and Representation (4pts)

[Q11, 4PTS] True/False: Select the statements that are true about analog and digital systems

- A. An analog system can **only** have discrete values
- B. Digital systems process and store continuous values
- C. Computers primarily operate on digital data
- D. Analog systems excel at representing physical phenomena

Networks and Data (10 pts)

[Q12, 5PTS] True/False: Evaluate if each of the following statements are true about the Internet

- A. It is owned exclusively by the Department of Defense
- B. Information is sent over it in one large chunk
- C. It is a decentralized collection of networks
- D. It is bound by a set of protocols that define how computers communicate on it
- E. It is composed of several networking layers

[Q13, 5PTS] True/False Select TRUE for the items below that are basic components of IP packets.

- A. Data Length
- B. Source IP addresses
- C. Data
- D. Router
- A. Ethernet

Internet and the Web (4 pts)

[Q14, 4PTS] Match the following terms to the definitions that best describe them

- | | |
|------|--|
| HTML | Defines the structure and content of a webpage |
| CSS | Defines the style of various elements of a webpage |

JavaScript Adds interactivity or action elements to a webpage
URL Uniquely identifies a webpage in the world wide web

Cyberspace and Threat Actors (7 pts)

[Q15, 4PTS] Match the malware with the best corresponding description

Ransomware Denies access to a user's data through encryption, demands payment
Trojan Masquerades as a harmless piece of software
Spyware Covertly tracks information about a user without their knowledge
Worm Self-propagates itself from user to user through a network

[Q16, 3PTS] Match the following terms with its corresponding definition as it relates to cyberspace:

Physical Layer Network of ISPs and infrastructure that support cyberspace
Logical Layer Network of data and applications not attached to a single physical device, but can describe the digital assets of an organization
Cyber-persona Layer Network of user accounts, whether human or automated, and their relationship to one another

Cybersecurity (5 pts)

[Q17, 5PTS] True/False: Evaluate if the following statements are true about asymmetric cryptography (also known as public key cryptography):

- A. It involves two parties sharing a common key
- B. A private key warehouse stores everyone's private key
- C. It ensures that the only person who can read a message is the intended recipient
- D. It is a technology underpinning encryption
- E. It can help ensure that a person sending a message is who they say they are

Commented [MS7]: Apparently too hard a vocab term for cadets. Several cadets asking what this means. We should consider changing the wording next semester.

Cyber Hygiene (5 pts)

[Q18, 5PTS] True/False: Evaluate if the following statements are good advice related to cyber hygiene:

- A. Use a password manager whenever possible

- B. When given the option, it is a good idea to accept all cookies
- C. Use multifactor authentication whenever possible
- D. If something appears on an App Store, you can automatically assume that it is trustworthy.
- E. It's ok to reuse passwords, if they are sufficiently long (>20 chars).

Copyright Lecture (6pts)

[Q19, 4PTS] True/False: Select TRUE if the item below is protected by U.S. copyright law:

- A. A great idea for the next book series that you have not written down.
- B. A dance routine recorded in a notebook
- C. Music released online
- D. Software code stored in a public repository

[Q20, 2PTS] Multiple Choice: A photographer takes a picture of the winner of a pie-eating contest during the competition. Who owns the copyright to that photo?

- A. The winner of the pie-eating contest
- B. The photographer
- C. The maker of the pies
- D. Whoever files for copyright first

Robotics Lab (9 pts)

[Q21, 4PTS] True/False: Which characteristics define ALL robots?

- A. Sense the world around them
- B. Make decisions based on input
- C. Perform physical actions
- D. Replicate human behavior

[Q22, 5PTS] True/False: Evaluate if robots are well-suited to perform the following tasks:

- A. Perform dull (repetitive or menial) tasks
- B. Perform jobs that are potentially harmful for humans
- C. Perform tasks that require a great deal of precision
- D. Provide psychological advice to a human suffering emotional distress

- E. Provide compassionate end-of-life care for a loved one

Online Tracking and Ethics (11pts)

[Q23, 3PTS] True/False: Suppose you post a photo of yourself online using a photo-sharing service like Instagram. Evaluate if each of the statements below are True or False.

- A. Whoever has access to the photo can make perfect copies of that photo without your knowledge.
- B. Deleting the photo means that the data is permanently erased from the photo-sharing service.
- C. If you own the copyright of the photo, the photo-sharing service cannot use it for training A.I. tools.

Commented [MS8]: @Filcik, Daniel MAJ update to wording

Commented [AB9]: Legally they might not be able to. To disambiguate, recommend changing to "will not"

Commented [MS9R2]: according to what law?

[Q24, 4PTS] Match the following terms with their associated definitions

Cookie	A special file (or ticket) stored in your browser that gives a web server information about you.
Web-Beacon	Invisible image or website that is embedded in websites or emails
Browser Fingerprint	Uses browser's unique configuration to track activity
Geotag	Using GPS to automatically determine the location that data was created or accessed

Commented [AB10]: web server?

[Q25, 4PTS] Evaluate which of the following statements are true about cookies. Note that multiple statements can be true!

- A. All cookies require consent prior to being stored on a web browser.
- B. Cookies can enable you to login automatically to websites that you visit frequently
- C. Cookies can be used to build accurate ad-profiles about you
- D. Cookies, once created, cannot be removed and exist forever.

AI and Data (11pts)

[Q26, 3 PTS] (Matching):

Match the following terms with the text that BEST describes their purpose.

Artificial Intelligence	Enables computers to perform tasks normally requiring human intelligence
Machine Learning	Helps computers learn how to do a particular task, without explicitly programming the rules to do so
Deep Learning	Uses multilayered neural networks to help computers learn how to do a particular task normally done by humans

[Q27, 3PTS] (Matching)

Match the following terms with the text that BEST describes them:

Artificial Narrow Intelligence	A.I. capable of performing only a specific task
Artificial General Intelligence	A.I. capable of performing a multitude of tasks equally as well as humans.
Artificial Super-Intelligence	A.I. that can outperform humans in all tasks

Commented [MS11]: @Filcik, Daniel MA check spelling in configured exam. noticed there was a spelling error

[Q28, 5PTS] (True/False)

Which of the following are common use-cases of artificial intelligence? Note that there may be multiple true answers.

- A. Crawling the web to create an accurate picture of the Internet
- B. Suggesting video recommendations based on what you watched online.
- C. Helping identify a picture of plant that you took a photo of while on a walk.
- D. Tracking how many times a website was viewed
- E. Helping doctors diagnose a patient's level of cancer risk

AI and Ethics (4pts)

[Q29, 4PTS] You are working on a battalion staff at your first duty station when the Battalion Operations Officer, Major Newtech, demonstrates to the office a machine learning (ML) program he downloaded. He claims he could enter the history of Uniformed Code of Military Justice (UCMJ) actions in the Battalion and the program will be able to advise the Commander on appropriate punishments for Soldiers under disciplinary action in the future. When he asks for feedback from the group, you highlight multiple ethical issues with his proposed plan in a professional manner: (choose all appropriate **ethical concerns**)

Commented [HT12]: On Battalion staff?

- A. ML algorithms are incapable of operating at the speed required for the Commander to make timely judgments.
- B. Using a small set of historical data is likely to produce a very biased model that could disadvantage particular groups of Soldiers.
- C. Even though the Commander has the right to access UCMJ history in the unit, it may violate data privacy regulations and could be insecure to use that information with the application MAJ Newtech found.
- D. Incorporating a sophisticated technological tool will reduce the Commander and the operation group's ability to focus on unit priorities.

Code Tracing

[Q1, 4PTS] Python Basics

True/False: Consider the following program:

```
def main():
    my_num = input("Enter a number: ")
    my_num = my_num * 3
    print("The result is", my_num)

main()
```

Commented [MS13]: Have there been enough examples of this calling convention?

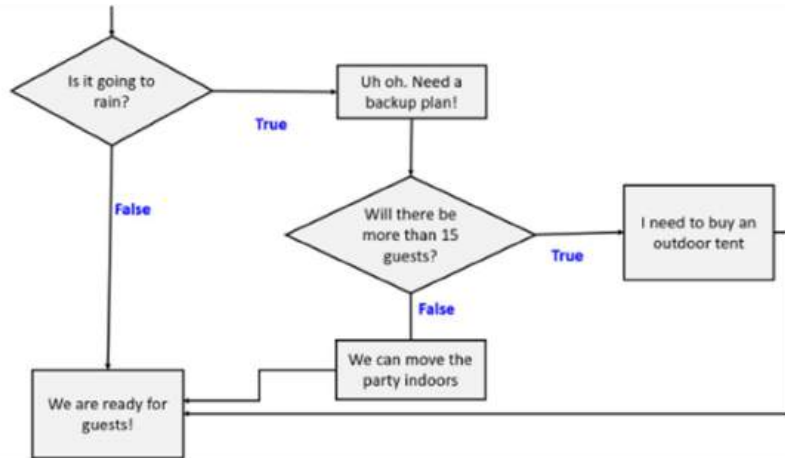
Commented [MS14]: @Filcik, Daniel MAJ Please update this one to main

Evaluate if the following statements are true or false about the above program:

- A. The `main()` procedure prompts the user for input
- B. If the number 1 is entered at the prompt, The `result is 3` will be printed to the screen
- C. The `main()` procedure contains three lines of code as part of its code block
- D. There are no comments in this code.

[Q2, 4PTS] You are planning a party at your house. Based on the forecast, you need to decide if a backup plan is needed and what that would be based on the guest count. Each diamond represents a decision point, while each square represents a statement that should be printed at that point in the process.

Evaluate which statements are true about the represented decision process:



- A. If it isn't going to rain, the statement "We are ready for guests!" prints out
- B. If it is going to rain, and there are at most 15 guests, the statement "We can move the party indoors" is included in the output.
- C. The statement "I need to buy an outdoor tent" prints out only if it's going to rain and there are more than 15 guests.
- D. If it isn't going to rain and there are fewer than 14 guests, the statement "We can move the party indoors" prints out.

Commented [HT15]: Recommend maintaining consistency in the condition and say "If it isn't going to rain" --technically sun showers occur.

[Q3, 4PTS] Read the specifications and select the correct options from the drop-down menus below to complete the blanks in the code.

- Ask the student to enter a number and convert that user input to an integer
- Output the following based on the converted input value:

Converted Value	Output
1	Army
2	Navy
3	Air Force
Any other value	Coast Guard

```

def football_OML():
    ans = [dropdown menu A]
    [dropdown menu B]
    print("Army")

```

```
[dropdown menu C]
    print("Navy")
elif ans == 3:
    print("Air Force")
[dropdown menu D]
    print("Coast Guard")
```

Dropdown menu A:

- int(input("Enter a number: "))
- str(input("Enter a number: "))
- input("Enter a number: ")
- list(input("Enter a number: "))

Dropdown menu B:

- if ans == 1:
- elif ans == 1:
- else if ans == 1:
- else:

Dropdown menu C:

- if ans == 2:
- elif ans == 2:
- else if ans == 2:
- else:

Dropdown menu D:

- if ans == 4:
- elif ans == 4:
- else if ans == 4:
- else:

[Q4, 4PTS] True/False: Consider the following procedure:

```
def pinkyAndTheBrain(option):
    if option == "genius":
        print("Brain")
    elif option == "insane":
        print("Pinky")
    else:
        print("Narf")
```

For each of the following invocations, evaluate if the statements are True or False:

A. If `pinkyAndTheBrain` is invoked with the argument "genius", the string "Brain" will be outputted to the screen.

B. If `pinkyAndTheBrain` is invoked with the argument "genius", the string "Narf" will be outputted to the screen

C. If `pinkyAndTheBrain` is invoked with the argument "insane", the string "Pinky" will be outputted to the screen.

D. If `pinkyAndTheBrain` is invoked with the argument "mice", the string "Narf" will be outputted to the screen.

[Q5, 4PTS] After running the following code:

```
import random
```

```
def main():  
    iterations = random.randint(1,3)  
    p = 0  
    for i in range(iterations):  
        p += 1  
    print(p)
```

```
main()
```

Commented [MS16]: @Flick, Daniel MA here is another one where I changed it to main()

Select all possible outputs

A. 1

B. 2

C. 3

D. 6

[Q6, 4PTS] (True/False) Evaluate if the statements about the following code are true or false. (Note: there may be multiple answers that are true)

```
def sum_nums(limit):  
    total = 0  
    i = 0  
    while i < limit:
```

```

        total += i
        i+=1
    print("The sum from 1 to", limit, "is", total)

def main():
    sum_nums(10)

main()

```

- A. The program outputs The sum from 1 to 10 is 45
- B. When run, the value of limit at the start of the sum_nums procedure is 10
- C. At the end of sum_nums, the value contained in i is the same as limit
- D. The sum_nums procedure contains an infinite loop

[Q7, 5PTS] (Matching) True/False

```

def prepare_goose(weight):
    minutes_per_pound = 15
    cooking_time = weight * minutes_per_pound
    if cooking_time > 300:
        return 300
    return cooking_time

def main():
    weight = int(input("Enter the weight of the goose: "))
    cooking_time = prepare_goose(weight)
    print("The goose will be ready in", cooking_time, "minutes.")

main()

```

For the given code snippet, mark the following statements as True or False:

- A. If weight is set to 40, the function prepare_goose() will return 300 and then 600.
- B. prepare_goose() is a fruitful function that accepts exactly one argument.
- C. The variable minutes_per_pound is accessible in the main() procedure.
- D. Entering a non-numeric value for the goose weight will cause an error in the program.
- E. The prepare_goose function could potentially return a value less than 300 if the goose's weight is less than 20 pounds.

Q8 [7 PTS] Suppose the file `grades.csv` contains the following students and their grades for three exams (exam names are Defense of the Dark Arts, Transfiguration, and Charms, respectively). Note that there's a newline character at the end of each line.

Name,Defense Against the Dark Arts,Transfiguration, Charms
Harry Potter,80,85,87
Hermoine Granger,99,100,100
Ronald Weasley,75,80,82
Gregory Goyle,33,20,10
Luna Lovegood,90,95,100

Consider the following program:

```
import statistics
def main():
    file = open("grades.csv", "r")
    file.readline() #removes header
    OWL = []
    for line in file:
        student_grades = line.strip().split(',')
        OWL.append(int(student_grades[2]))
    result = statistics.mean(OWL)
    print(result)
    file.close()

main()
```

Indicate if the following statements are true or false:

- | | |
|--|-------|
| A. The program opens a file called <code>grades.csv</code> for reading . | True |
| B. The program computes the average grade on the Transfiguration exam. | True |
| C. If the file <code>grades.csv</code> does not exist, the program will create it. | False |
| D. At the end of program execution, the program displays the value 85. | False |
| E. The program did not close the <code>grades.csv</code> file. | False |
| F. The for loop executes a total of 5 times. | True |
| G. The list <code>OWL</code> contains all strings. | False |

[Q9, 4PTS] Consider the following program:

```
import random
```

```
def main():
    incomplete = True
    percent = 0
    count = 0

    while incomplete:
        new_value = random.randint(0,20)
        percent += new_value
        count += 1
        if percent >= 100:
            incomplete = False
        else:
            print("keep going")

    print(percent, "%")
    print(count, "turns")

main()
```

Fill in the blank

The while loop will terminate once percent equals or exceeds the value `100`

The maximum value that new_value is assigned is `20`

Every iteration of the loop, count increments by the value `1`

incomplete is assigned a `boolean` data type

[Q10, 4PTS] Multiple Function calls

Read the specifications below and select the correct options from the dropdown menus below to complete the blanks in the code.

1. Define a function `isPrime` that takes one parameter (an integer) and returns a boolean value. The function should return `True` if the provided integer is a prime number and should return `False` if the provided integer is not a prime number. A number is prime if it is only evenly divisible by itself and 1. ***The code to determine if the number is a prime is provided.**
2. Define a separate `main` function that loops through the numbers 2 to 20 (inclusive) and call the function `isPrime` to determine whether or not each number is prime. Display in the console whether or not the number is prime.

Sample Output

2 is a prime number.
3 is a prime number.
4 is not a prime number.
5 is a prime number.
6 is not a prime number.
7 is a prime number.
8 is not a prime number.
9 is not a prime number.
10 is not a prime number.
11 is a prime number.
12 is not a prime number.
13 is a prime number.
14 is not a prime number.
15 is not a prime number.
16 is not a prime number.
17 is a prime number.
18 is not a prime number.
19 is a prime number.
20 is not a prime number.

[dropdown A]

```
prime = True
for i in range(2,x):
    if x % i == 0:
        prime = False
```

[dropdown B]

[dropdown C]

```
for number in range(2,21):
```

[dropdown D]

```
    print(number, "is a prime number.")
```

```
    else:
```

```
        print(number, "is not a prime number.")
```

Dropdown A:

- def isPrime():
- def main():
- def main(x):

- def isPrime(x):

Dropdown B:

- return prime
- print(prime)
- return x
- print(x)

Dropdown C:

- def main():
- def main(x):
- def isPrime():
- def isPrime(x):

Dropdown D:

- if isPrime(x):
- if not isPrime(x):
- if isPrime(number):
- if not isPrime(number):

[Q11, 4PTS] (True/False) Consider the following program:

```
def doit(x):  
    x = x ** 3  
    print("finished!")  
    return x
```

```
def main():  
    x = 3  
    result = doit(x)  
    print(x)
```

main()

Commented [HT17]: @Tippett, Hayden J MAJ on the canvas item—this is configured RED TEXT inadvertently

Select all the statements that are true

A. The doit function returns the value 9

B. The doit function outputs finished!

C. The main function outputs the value 3

D. The initial value of x in doit is 3

[Q12, 4PTS] Consider the following program (assume lst is of type list):

```
def find_last_num(lst, n):  
    length = len(lst)  
    empty = ''  
    for i in range(1, length+1):  
        indx = i * -1  
        if lst[indx] == n:  
            return n  
    return empty
```

Commented [MS18]: @Filcik, Daniel MA please confirm the indentation in the configured exam

Indicate if each of the following statements are true or false:

A. This function checks to see if the element n exists inside of lst

B. The function checks for elements using forward indexing

C. The function returns the last instance of n in the list, if it exists.

D. If the desired element does not exist in the list, the function returns an empty list



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION III

ASSIGNMENTS



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB K

PROBLEM SET 3

LESSONS COVERED

- Microsoft 365
- Computers and Data
- AI and Data
- Problem Solving and Algorithms

Part A – Canvas Component: 15 pts

- 3-4x Knowledge Check Questions for MS365, C&D, AI&D PS&A [8pts]
- 1x GenAI activity [7pts]

Legend: Updates to be made. Question Answers. Feedback Section. Configured in Canvas/zyBooks.

Canvas Component (8 PTS)

1. Question Title: Microsoft 365

Question Type: Matching

Question Points: 2

Question Stem: Select the best Microsoft 365 tool for a specific scenario

OneDrive	- You are your family's designated historian and you'd like to back-up and share generations worth of family documents and media data.
Excel	- You are the PT officer and would like to keep record and run statistics on all the AFT scores of the members of your company.
PowerPoint	- You are asked to give a presentation on a group project for one of your classes.
Teams	- You are the cyber club cadet-in-charge, you would like to communicate and notify members of your club that the evening meeting has been cancelled.

Feedback:

- OneDrive: tool in Office 365 that is a virtual drive that backs up files remotely and also allows for easy sharing with others.
- Excel: Office 365 tool that is used to manage data, calculation, and basic analytics and visuals. This is the best tool to record data in a tabular manner.
- PowerPoint: the primary tool in Office 365 that is used to conduct presentations in a single delivery.
- Teams: tool in Office 365 used for communication – can send messages and make calls – between individual people or within larger teams. It is the best tool to make announcements at a dynamic rate.

2. Question Title: Computers and Data

Question Type: Fill in the Blank (Dropdown, T/F)

Question Points: 2

Commented [JY1]: JY

Commented [JY1R2]: Note: I will be OOO week of Aug 1

Commented [MS1R3]: I'm happy with this one!

Commented [JY2]: JY

Commented [JY2R2]: Note: I will be OOO week of Aug 1

Commented [MS2R3]: I think this is fine too. I wish there was something in the question stem that somehow unifies the options. Maybe something like: "For the following questions on data, indicate if each statement is True or False"

Commented [TW2R4]: I've changed the question stem as you requested.

Question Stem: For the following statements about data, indicate if each is True or False

- Data from basic physical phenomena is categorized as **analog**. True
Data is stored long-term in **RAM**. False
Lossless compression (such as PNG) allows perfect reconstruction of the original data True
The machines in a **data center** must be identical so that they can share resources like memory and storage. False

Feedback:

1. Data from basic physical phenomena is analog and not digital, but computers need to convert analog data to digital to be useful.
2. Data stored long-term is in disk storage while RAM is not for long-term data storage since it is volatile memory. This means that RAM is frequently changed and lost once a computer is shut off.
3. Lossless compression retains all the information needed to reconstruct the entirety of the original file during decompression.
4. Memory and storage sharing between datacenter servers does not require them to be identical.

Commented [MS3]: This needs to be revised to reflect the question change that catudal is suggesting. @Walsh, Timothy CLTC

Commented [TW3R2]: Got it.

3. Question Title: AI and Data

Question Type: Multiple Answer

Question Points: 3

Question Stem: In machine learning, having a model “overfit” or essentially memorize its training data is problematic because:

It would typically take too long to memorize all the training data.	False
Memorizing all the training data results in a model that requires too much memory and storage space.	False
This can mean that the model has “learned” things that don’t generalize well beyond the training data to new and unseen data.	True

Commented [TW4]: TW

Commented [MS4R2]: This is excellent! Thanks Tim!

Feedback:

It would typically take too long to memorize all the training data.	No, training for too long could result in overfitting, but the time it takes to train is not really the concern here.
Memorizing all the training data results in a model that requires too much memory and storage space.	No, the model size generally doesn’t change based on the amount of training data or how much it has “learned” from its data. A model that has overfit its training data will not take up more space than one that hasn’t.

Commented [RD5]: Very small but I would suggest putting learned in quotes to emphasize again that it’s not “learning” in the way they do or they think the model “learns”.

Commented [TW5R2]: That works for me.

Commented [DJ6]: Possibly confusing as to whether we’re talking about the size of the model or the size of the training data set.

Commented [TW6R2]: I’m open to a suggestion that would make it more clear that we’re talking about the model.

Commented [JD6R3]: Maybe something like this? - Memorizing all the training data results in a model that requires too much memory and storage space.

Commented [JD6R4]: As the prompt, not the feedback.

Commented [TW6R5]: I put that in, thanks.

This can mean that the model has “learned” things that don’t generalize well beyond the training data to new and unseen data.	Yes, overfitting means that the model may have “learned” idiosyncratic features of the training data, to include errors like mislabeled examples, that allow it to score well on its training data but don’t help it to perform well at test time or when it’s deployed in the real world.
---	--

4. [Question Title: Problem Solving and Algorithms](#)

Question Type: **Fill in the Blank (Dropdown, T/F)**

Question Points: 1

Question Stem: True or False: Every step in an algorithm must occur in the order specified.

Feedback:

True: This is a FALSE statement because some steps in an algorithm can be interchangeable in terms of their order.

False: This is a FALSE statement because some steps in an algorithm can be interchangeable in terms of their order.

1. [GenAI Problem](#)

GenAI Problem (7 PTS)

Using what you learned about the CLEAR framework, use a public-facing LLM (such as ChatGPT, Gemini or Claude) to generate a logo of your cadet company, athletic team, or another organization. For full credit you must:

- Upload a copy of your logo
- Provide a link to the full conversation that you had with the AI, or document that contains a full transcript
- Write a paragraph reflection that covers the following elements:
 - What did you decide to create a logo for? Why did you make this choice?
 - What elements of the CLEAR framework did you use in constructing the prompt?
 - What did you learn about prompt engineering by completing this assignment?

Unused

Prompt: Using what you have learned from Chapter 6 (Microsoft 365 GenAI Lab), see if you can get Copilot or another LLM chatbot of your choice to produce a hallucination (without explicitly

Commented [TW7]: TW

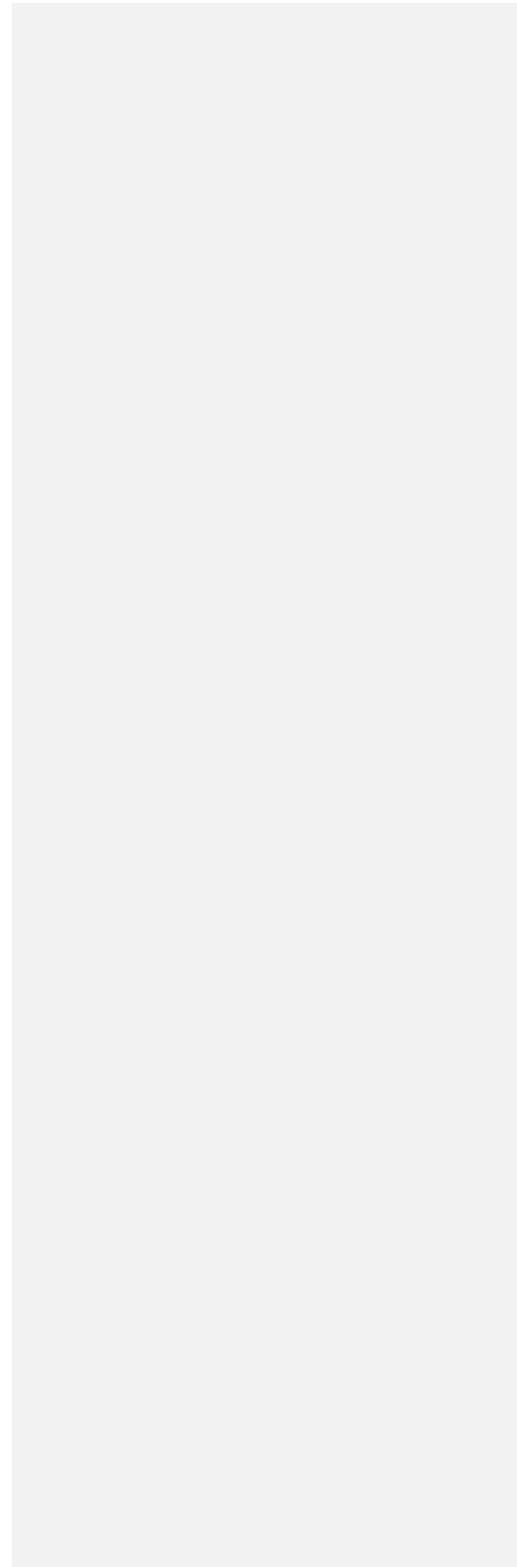
Commented [TW8]: TW

Commented [MS8R2]: I think the last problem you were supposed to create was one for "Problem Solving and Algorithms". Can you please create a problem for that?
[@Walsh, Timothy C LTC](#)

Commented [MS8R3]: We can do a simple multiple choice for this one and have it be just 1 point, since your last problem is three points.

Commented [TW8R4]: Ok. I thought we were responsible for the whole WPS including the GenAI Activity, so this was my suggestion for that. I'm happy to come up with a quick Problem Solving and Algorithms question instead.

telling it to hallucinate), where it invents a reference to a non-existence source or to apparently false information. Describe in a few paragraphs what strategy or techniques you tried, why you thought they might lead to hallucinations, and what the results were. Provide the shareable link to the conversation or a transcript.





UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB L PROBLEM SET 4

CY105 AY26-1 WPS4

LESSONS COVERED

- Python Basics
- Built-in Functions

Part A – Canvas Component: 7 pts

- 1x Knowledge Check (Review, Lessons 2-7)
- 2x Code Tracing Questions

Part B – zyBooks Component: 8 pts

- 1x Code Writing Problem with an *algorithm provided*
- 1x Code Writing Problem

Legend: Updates to be made. Question Answers. Feedback Section. Configured in Canvas/zyBooks

Canvas Component (7 PTS)

1. Knowledge Check: Built-in Function Matching (Lesson 9)

Question Type: Matching

Question Points: 2 points

Question Stem: Please match the application (scenario) of a built-in function to the function itself.

Built-in Function	Application
<code>print()</code>	I calculated the total number of lights in Cullum and want to display it to the user.
<code>input()</code>	I want to find out from the customer how many scoops of ice cream they want.
<code>min()</code>	I have a list of the weights of all the cadets in the company and want to find the lightest.
<code>ord()</code>	I want to create a transposition cipher where each letter is replaced by the letter 13 places further on in the alphabet (ROT13).

Distractors: `int()`, `round()`, `len()`, `lowest()`

Feedback:

<code>print()</code>	<code>print()</code> displays its input to the screen. In this case, we have already calculated a total number of lights from our function and want to display it back to the user.
----------------------	---

Commented [CJWL1]: Confirm the total question points for this section match the section value?

Commented [JY1R2]: Yeah I had trouble with this. If there was 1xKC and 2xCT, that would probably mean the KC only gets 1 point (assuming I knew the CT were of equal difficulty and 3 points each). I guessed in this case and initially allocated only 1 point but changed it afterward because it didn't seem to make sense for a 5-part matching. Should I have assumed what the other instructors were doing and only provided a simple 1-point type of question prematurely?

Commented [CJWL1R3]: Good q, maybe a broader discussion on point assignment with group?

Commented [HT1R4]: The review questions is the lowest priority. At most 2 points, giving 5 pts to share among python basics and built-in functions.

Commented [HT2]: @Yan, Jinny Y CP Review question

Commented [HT3]: Replace with "lights in Cullum" – it's plebe knowledge 😊

Commented [DJ4]: I would prefer to test `int()` instead because we use it so much more often.

Commented [HL5]: A: I calculated the total number of tables in Washington Hall and want to provide it back out to the user.

B: I have a list of the weights of for all of the cadets in my company. I want to iterate over the list and find out our total weight.

C: I want to create a transposition cipher where each letter is replaced by the letter 13 places further on in the alphabet (ROT13).

D: I have a list of the weights of all the cadets in the company and want to find the lightest.

E: I want to find out from the customer how many scoops of ice cream they want.

Commented [JY5R2]: Thank you for these suggestions! I have included it in my answer and added more detailed feedback.

Commented [MS6]: Not sure how comfortable I am with `chr()` as an option here. For any transposition cipher, an additional step is necessary to transform the number to a letter. That requires the use of `chr()`. My recommendation is that you replace the `ord()` one with a description for `round()` instead. That, or remove `chr()` as a distractor.

CY105 AY26-1 WPS4

<code>input()</code>	<code>input()</code> prompts the user for information in order to utilize that data to customize an output. In this case, we are asking the user to provide the number of scoops of ice cream they want. The <code>input()</code> function is the way to get this data at run-time.
<code>min()</code>	<code>min()</code> returns the smallest value in a sequence. In this case, we are trying to find the smallest value within a list of weights representing the entire company.
<code>ord()</code>	<code>ord()</code> returns the integer representation of a particular ASCII/Unicode character. To use the transposition cipher (rotate a single character such as 'A' by 13 places), we first need to convert 'A' to its integer representation, which would be 65. As a second step, we then add 13 (65+13=78) and then take the result (78) and turn it back to a character, 78 is 'N'. A separate function, <code>chr()</code> helps with that conversion.

2. Code Tracing: Lesson 8, Python Basics

Question Type: Dropdown

Question Points: 2

Question Stem:

```
1 def main():
2     x = 0
3     x = x + 3 * 2
4     x = x + 2 * 2
5     power = x + 2 ** 3
6     y = power / 2 + 1
7     print(y)
8
9 main()
```

Which of the following statements are true regarding the procedure `main()`:

a. After line 4 executes, variable `x` equals 10 `True`

Commented [HT7]: @Kedrowitsch, Alexander L MAJ
Python Basics Code Tracing question

Commented [MS8]: Let's add line numbers to these. That will make the wording in the statements below less awkward. @Kedrowitsch, Alexander L MAJ

Commented [HT9]: Write the body of the questions, identifying the dropdown place holders with the following character ` (e.g. `True`)

Commented [KM9R2]: I think I'm correcting this properly - let me know if I still don't have it

CY105 AY26-1 WPS4

- b. In line 5, when assigning the value of `power, x+2` is executed first and the result is then raised to the power of 3 `False`
- c. After line 5 executes, `power` is assigned the value 18.0 `False`
- d. After line 6 executes, `y` is of type float `True`

Feedback:

- a. The correct answer is `True`; The variable `x` is assigned the value of 10. `x` is initially assigned 0, then assigned the value of 6 ($0 + 3 * 2$) on line 3, then assigned the value of 10 ($6 + 2 * 2$) on line 4.
- b. The correct answer is `False`. The order of operations in Python evaluates exponents before addition/subtraction; the proper evaluation would be $x + (2**3)$.
- c. The correct answer is `False`; The variable `power` is assigned the integer value 18.
- d. The correct answer is `True`; `y` is evaluated to be 10.0, as the division operator always produces a floating-point value `True`.

3. Question Title: Lesson 9. Built-in Functions

Question Type: Dropdown

Question Points: 3

Question Stem:

```
1 def main():
2     #Get your name and score
3     cadet_1 = input("What is your name?")
4     cadet_score_1 = input("What is your AFT score?")
5     #Get your battle buddy's name and score
6     cadet_2 = input("What is your battle buddy's name?")
7     cadet_score_2 = int(input("What is your battle buddy's AFT score?"))
8     #Does this work?
9     cadets = cadet_1 + cadet_2
10    #Combine information
11    cadets = [cadet_1, cadet_score_1, cadet_2, cadet_score_2]
12    main()
```

```
1 def main():
2     #Get your name and score
3     cadet_1 = input("What is your name?")
4     cadet_score_1 = input("What is your AFT score?")
5     #Get your battle buddy's name and score
6     cadet_2 = input("What is your battle buddy's name?")
7     cadet_score_2 = int(input("What is your battle buddy's AFT score?"))
8     #Does this work?
9     cadets = cadet_1 + cadet_2
10    #What is the difference?
11    gap = cadet_score_1 - cadet_score_2
12    #display result
```

Commented [DJ10]: Would suggest a slight change in wording. Maybe something like this: In line 5, when assigning the value of `power, x+2` is executed first and the result is then raised to the power of 3. Open to other suggestions that might be better.

Commented [HL11]: It is a little awkward to compare PEMDAS with PEMDAS. It might be better to change the question so that it is not an evaluation of a bad copy/edit but a question about whether a word problem has been properly translated into a math equation.
Suggestion:
`power` is evaluated as $x + 2$ first and the result is then raised to the power of 3

Commented [KM11R2]: The text and Reference Card have the order listed out, which is why the question is so specific, but I do like your suggestion; I'll probably switch to that.

Commented [MS12]: perhaps rephrase as "the final value in `power` is 20.0"

Commented [CJWL12R2]: We already do a code trace in the for loop..perhaps C can state: `power` is a float variable

Commented [KM12R3]: Combined both suggestions

Commented [KM13]: Due to simplifying this portion of the question, I feel these two statements are redundant. Should I eliminate/replace one?

Commented [MS13R2]: yes, get rid of both

Commented [MS14]: Either change this to be `trace_this()` or change the function header to be `trace_it()`. We should also call the `trace_this()` function.

Commented [MS15]: is assigned

Commented [HT16]: @Zitter, Andrew P MAJ Built-in functions code tracing question

Commented [HT16R2]: @Zitter, Andrew P MAJ, make sure to update question according to feedback (for loop ... [3])

Commented [ZM16R3]: Question was re-worked to account for all feedback provided. Variables are now ... [4]

Commented [MS16R4]: @Zitter, Andrew P MAJ This problem still contains concepts that are not covered yet ... [5]

Commented [ZM16R5]: Question stem and feedback updated accordingly

Commented [MS16R6]: My last feedback is this: we are asking cadets to do a lot of complicated code tracing for ... [2]

Commented [ZM16R7]: I had originally wanted students to output the values and types, but I was having trouble ... [1]

Commented [MS16R8]: I think drop down should be able to handle it. We may also be able to do fill in the blank.

CY105 AY26-1 WPS4

```
13 print ("There is a", gap , "point difference between your score and "+
cadet_2 + "'s score.")
14 main()
```

Which of the following statements are true regarding the procedure `main()`:

- a. The variable `cadet_score_1` is a `str` type `True`
- b. The execution of line 9 will cause an error `False`
- c. The assignment to the gap variable on line 11 results in an error `True`

Feedback:

- a. The correct answer is `True`. The `input()` function returns a `str` type by default.
- b. The correct answer is `False`. On lines 3 and 6, the `input()` function returns a `str`. When the `+` operator is applied to two string operands, it performs concatenation.
- c. The correct answer is `True`. On line 4, `cadet_score_1` is assigned the type of `str` (default data type returned from the input statement). On line 7, `cadet_score_2` is assigned the type of `int`. Then, on line 11, the program attempts to assign gap the value of a `str` - `int`, which results in an error (`TypeError: unsupported operand type(s) for -: 'str' and 'int'`).

zyBooks Component (8 PTS)

1. CODING PROBLEM 1 – TRANSPORT ESTIMATOR (4 PTS)

You are tasked to design a program that determines the number of LMTVs required to transport a company for a movement during an FTX. The program should prompt the user for the company name, the number of personnel, and the amount of equipment in cubic feet.

An LMTV can carry up to 20 personnel or up to 500 cubic feet of equipment. Calculate the number of LMTVs required for personnel and for equipment separately, rounding each to a whole number. Then, add one additional LMTV to the final total for overage and backup. Lastly, print out the company name and the total number of LMTVs required. *You may assume the user will always enter a positive integer for the number of personnel and equipment volume.* Make sure the format of your input and output statements EXACTLY MATCH what is shown in the Sample Input/Output section below:

Here is a sample algorithm to help you solve the problem:

Program: Transport Calculator

Input: Company Name, Personnel count, Equipment volume in cubic feet.

Output: Company Name and number of LMTVs required.

Relevant Equations: Per LMTV: 20 personnel OR 500 cubic feet. Round to a whole number for each, then add one for backup.

1. Prompt user for company name
2. Prompt user for personnel count

Commented [JY17]: While using `'i'` as an incrementor can be pretty common in basic Python programming, I would like to suggest using more descriptive names for the variables of `'i'` and `'j'` here. Something like `'i'` as `'item'` and `'j'` as `'int_representation'`

Devil's advocate side of this is that cadets should understand they can name their variables whatever they want (as long as they can keep track and the code is good and clear) and perhaps if we name the variables *too* well, we can sometimes give away the answer to the problem without the student understanding what the code is actually doing.

The main reason for doing better variable naming is to "lead by example" and show them to write good code with descriptive variables; something that all coders should do in an ideal setting.

Commented [HT18]: Write the body of the questions, identifying the dropdown place holders with the following character ``` (e.g. ``True``)

Commented [KM18R2]: I think I'm correcting this properly - let me know if I still don't have it

Commented [HT19]: Write the body of the questions, identifying the dropdown place holders with the following character ``` (e.g. ``True``)

Commented [KM19R2]: I think I'm correcting this properly - let me know if I still don't have it

Commented [HT20]: @Catudal, Joseph W LTC & @Hicks, Matthew S LTC code writing problem with algorithm provided.

Commented [MS20R2]: This is an excellent problem @Catudal, Joseph W LTC and @Hicks, Matthew S LTC Well done! I've made minor edits for the sake of readability.

Commented [DJ21]: rounding each to...

Commented [DJ21R2]: And wouldn't it make more sense to round up? Although we haven't covered `ceil()` yet.

Commented [DJ22]: Is this one additional in total or one additional each for personnel and equipment?

CY105 AY26-1 WPS4

3. Prompt user for equipment volume
4. Calculate number of LMTVs required for personnel, rounded
5. Calculate number of LMTVs required for equipment, rounded
6. Add one LMTV
7. Print company name and required LMTVs

Sample *Input/Output*:

What is the Company name? **F2**

How many personnel? **88**

How much equipment (cubic ft)? **620**

F2 Company requires 6 LMTVs for movement.

Provided to Students:

```
"""
Program: Transport Calculator
Input: Company Name, Personnel count, Equipment volume in cubic feet.
Output: Company Name and number of LMTVs required
Relevant Equations: Per LMTV: 20 personnel OR 500 cubic feet. Round to whole number for each.
Add one LMTV for overage.
"""

def main():
    #Copy and paste algorithm steps from the instructions
    #Select the copied text and press CTRL + / to comment lines
    #Complete the required code for each algorithm step

main()
```

Sample Solution:

```
"""
Program: Transport Calculator
Input: Company Name, Personnel count, Equipment volume in cubic feet.
Output: Company Name and number of LMTVs required
Relevant Equations: Per LMTV: 20 personnel OR 500 cubic feet. Round to whole number for each. Add one LMTV
for overage.
"""

def main():

    # 1.    Prompt user for company name
    co_name = input("What is the company name? ")

    # 2.    Prompt user for personnel count
    pax = int(input("How many personnel? "))

    # 3.    Prompt user for equipment volume
    equip_vol = int(input("How much equipment (cubic ft)? "))

    # 4.    Calculate number of LMTVs required for personnel, rounded
    lmtv = round(pax / 20)

    # 5.    Calculate number of LMTVs required for equipment, rounded
```

CY105 AY26-1 WPS4

```
lmtv += round(equip_vol/500)
# 6.    Add one LMTV
lmtv += 1
# 7.    Print company name and required LMTVs
print(co_name,"Company requires",lmtv,"LMTVs for movement.")

main()
```

Test Cases:

- 1) Test Case Description 1
 - a) Inputs: 'F2',88,620
 - b) Expected Prompt(s):
 - i) "What is the company name? "
 - ii) "How many personnel? "
 - iii) "How much equipment (cubic ft)? "
 - c) Expected Output(s):
 - i) "F2 Company requires 6 LMTVs for movement."
- 2) Test Case Description 2
 - a) Inputs: 'Bandit',19,115
 - i) "What is the company name? "
 - ii) "How many personnel? "
 - iii) "How much equipment (cubic ft)? "
 - b) Expected Output(s):
 - i) "Bandit Company requires 2 LMTVs for movement."
- 3) Test Case Description 3
 - a) Inputs: 'HHC', 21, 1999
 - i) "What is the company name? "
 - ii) "How many personnel? "
 - iii) "How much equipment (cubic ft)? "
 - b) Expected Output(s):
 - i) "HHC Company requires 6 LMTVs for movement."
- 4) Test Case Description 4
 - a) Inputs: 'Whiskey', 1, 20
 - i) "What is the company name? "
 - ii) "How many personnel? "
 - iii) "How much equipment (cubic ft)? "

CY105 AY26-1 WPS4

- b) Expected Output(s):
i) "Whiskey Company requires 1 LMTVs for movement."

5) Test Case Description 5

- a) Inputs: 'Whiskey, 1, 750
i) "What is the company name? "
ii) "How many personnel? "
iii) "How much equipment (cubic ft)? "

- b) Expected Output(s):
i) "Whiskey Company requires 3 LMTVs for movement."

Tested Concepts:

- Python Basics
- String Concatenation
- Built-In Functions

2. CODING PROBLEM 2 – GRADE POINT AVERAGE (4 PTS)

Write a program that prints the maximum, minimum, and average scores for a student across three WPR's.

Prompt the user for the student name and then for that student's grade on each of three WPR's. Assume scores will be a float between 0.0 and 100.0, inclusive. After you have collected all the scores, print the student's maximum score, minimum score, and average score, each rounded to one decimal place. Then calculate how many points the student would need on the TEE (there are 250 points possible) to get an A- (90%). You may assume there are no other grades in play. Your output must match the format in the example below.

Sample Input/Output:

```
Enter student name: Bob
Enter Bob's WPR1 score: 100
Enter Bob's WPR2 score: 25.5
Enter Bob's WPR3 score: 75
Statistics for Bob:
Highest score: 100.0
Lowest score: 25.5
Average score: 66.8
Bob would need a score of 294.5 points on the TEE to make an A-
```

Commented [HT23]: @Zitter, Andrew P MA & @Hutton, John E LTC code writing problem

Commented [HT24]: ITERATION is out of scope for WPS4

Commented [MS24R2]: Maybe modify the problem to take some scores and add them up and compute the average? So it prompts the student for grades on three different WPRs and then computes and displays the average, the min grade and max grade?

Commented [JH24R3]: Problem refactored to remove 'for' and get WPR scores for a single student.

Commented [MS24R4]: @Hutton, John E LTC This problem is still inappropriately scoped. We will not have covered list concatenation. My concern is that students will not understand enough about lists to complete this problem.

Instead, I propose you modify this to have user enter three grades, compute the average and then the minimum score they will need to get on the TEE to earn at least an A-average on all exams. You should tell them the TEE is out of 250 points. Both the score on the TEE and the average score should be rounded to two decimal points. This is then somewhat similar to the previous problem, but they will have to come up with the algorithm itself.

The only downside to this suggestion is that we are using round() quite a bit. But, perhaps that is ok.

Commented [JH24R5]: @Matthews, Suzanne Dr Are built in functions on the table? I see on the 26-1 sched ... [6]

Commented [MS24R6]: Yes! They Built-in Functions are listed at the top of the page.

Commented [JH24R7]: Reverted to old problem and modified to remove the list and add the TEE calculation.

Commented [MS24R8]: Great work @Hutton, John E LTC I really like this version.

Commented [JH24R9]: I left the max and min in because there's not much else exercising built in functions aside from print, input, and round.

Commented [JH25]: Question for the group. Would it be better to give fractional input in the example or to have the grade entries as integers (or to leave it alone)?

Commented [HT25R2]: The first problem uses strictly ints, I think a float for a grade is reasonable here. The ... [7]

Commented [MS26]: recommend making this a percentage (out of 100) to match the format of the rest of them. Alternatively, modify the output to be something ... [8]

Commented [JH26R2]: How about we just call it 295 points in the output? That would make it more consistent with the user input.

Commented [MS26R3]: sounds great to me! Thanks!

CY105 AY26-1 WPS4

Provided to Students:

...

Place problem header here

...

```
def main():
    # Write your program below this line

main()
```

...

```
def main():
    # Prompt the user to enter the student name
    student_name = input("Enter student name: ")

    # Prompt the user to enter the student's score for WPR1 and save it as a
    variable
    wpr1 = float(input("Enter", student_name + "'s WPR1 score: "))

    # Repeat for WPR2
    wpr2 = float(input("Enter", student_name + "'s WPR2 score: "))

    # Repeat for WPR3
    wpr3 = float(input("Enter", student_name + "'s WPR3 score: "))

    # Output the statistics portion, including the highest, lowest, and average
    scores
    print("Statistics for", student_name + ":")
    print("Highest score:", round(max(wpr1, wpr2, wpr3), 1))
    print("Lowest score:", round(min(wpr1, wpr2, wpr3), 1))
    print("Average score:", round((wpr1 + wpr2 + wpr3) / 3, 1))

    # Calculate the needed score for an A-
    needed_score = 0.9 * 550 - (wpr1 + wpr2 + wpr3)

    # output the result
    print(student_name, "would need a score of", round(needed_score, 1),
    "points on the TEE to make an A-")

main()
```

Test Cases:

Commented [MS27]: clever. I like it!

CY105 AY26-1 WPS4

- 6) Test Case Description 1
- a) Inputs: "Bob", 100, 25.5, 75
 - b) Expected Prompt(s):
 - i) "Enter student name: "
 - ii) "Enter Bob's WPR1 score: "
 - iii) "Enter Bob's WPR2 score: "
 - iv) "Enter Bob's WPR3 score: "
 - c) Expected Output(s):
 - i) "Statistics for Bob:"
 - ii) "Highest Score: 100.0"
 - iii) "Lowest Score: 25.5"
 - iv) "Average Score: 66.8"
 - v) "Bob would need a score of 294.5 on the TEE to make an A-"
- 7) Test Case Description 2
- a) Inputs: "Alice", 84.5, 92, 71
 - b) Expected Prompt(s):
 - i) "Enter student name: "
 - ii) "Enter Alice's WPR1 score: "
 - iii) "Enter Alice's WPR2 score: "
 - iv) "Enter Alice's WPR3 score: "
 - c) Expected Output(s):
 - i) "Statistics for Alice:"
 - ii) "Highest Score: 84.5"
 - iii) "Lowest Score: 71.0"
 - iv) "Average Score: 82.5"
 - v) "Alice would need a score of 247.5 on the TEE to make an A-"
- 8) Test Case Description 3
- a) Inputs: "Eve", 75.395, 88.802, 84.555
 - b) Expected Prompt(s):
 - i) "Enter student name: "
 - ii) "Enter Eve's WPR1 score: "
 - iii) "Enter Eve's WPR2 score: "
 - iv) "Enter Eve's WPR3 score: "
 - c) Expected Output(s):
 - i) "Statistics for Eve:"
 - ii) "Highest Score: 88.8"
 - iii) "Lowest Score: 75.4"
 - iv) "Average Score: 82.9"
 - v) "Eve would need a score of 246.2 on the TEE to make an A-"

Tested Concepts:

- [Python Basics](#)

CY105 AY26-1 WPS4

- *String concatenation*
- *Built in Functions*

CY105 AY26-1 WPS4

UNUSED

```
def         ():  
    x = 0  
                                      
    x = x + k *           
    power = x + 2 ** 3  
    y = power / 2 + 1  
    flag =          (y > 10)  
    x = "        "  
    print(flag, x)
```

Commented [HT28]: def main():
custom functions introduced in block 2.

Commented [HT28R2]: Also need to show invocation of
main

Commented [JY29]: I like the -1 step because it tests
something that is conceptually a little out of the box. I did sit
in a STAP class for this module though, and we did not
address what would happen if the step was -1. I worry that
adding this gotcha + the concept of loop + the concept of
the flag using not() to potentially be too many concepts to
test within one coding question.

Commented [TW30]: Loops are not yet introduced in
Python Basics. We could just remove this component and let
it initialize and update a variable once.

Commented [JY31]: Since this code block spans 2 pages,
it looks like the line starting with 'power =' is un-
indented, whereas I think your intention is to have it within the for
loop.

Commented [HT31R2]: This will not be the case once
implemented in Canvas—it's one continuous page for the
WPS/WPRs

Commented [TW32]: The logical operator "not" comes
after Python Basics, so I think we need to take it out of this.

Commented [JY33]: I'm not sure we should get into the
habit of showing cadets they can reassign a variable to a
different data type. I understand why you needed to do this
(no conditionals taught yet besides the logical not) though
so open to having my mind changed.

Commented [KM33R2]: In defense of why I included
this, I wanted to emphasize that variable data types should
not be assumed. True this happens most in Python, but is
common in other languages, such as C pointers.
But I concede that this is probably a 'gotcha' question, so I'll
drop it.

Page 3: [1] Commented [ZM16R7] Zitter, Andrew P MAJ 8/5/2025 10:44:00 AM

I had originally wanted students to output the values and types, but I was having trouble getting that to fit in one of the approved Canvas question types. I would rather the question be set up as you described; what would you recommend for structuring the answer?

Page 3: [2] Commented [MS16R6] Matthews, Suzanne Dr 8/5/2025 10:37:00 AM

My last feedback is this: we are asking cadets to do a lot of complicated code tracing for a question that never asks about output, but only the final types of variables. What are your thoughts about asking cadets to also print out the value, not just the type? Otherwise, all the students need to do is inspect the last four lines to get the types, due to the nature of duck typing.

I would also get rid of line 9.

Page 3: [3] Commented [HT16R2] Tippet, Hayden J MAJ 7/31/2025 10:00:00 AM

@Zitter, Andrew P MAJ, make sure to update question according to feedback (for loop out if scope). Deadline is COB tomorrow. Please shoot me a note once complete.

Page 3: [4] Commented [ZM16R3] Zitter, Andrew P MAJ 7/31/2025 9:37:00 PM

Question was re-worked to account for all feedback provided. Variables are now assigned independent of each other, eliminating the waterfall / cascading error potential. The range() function was eliminated. The variables were named something more interesting. My intent was to force students to keep track of all three variables throughout the program, so they are modified in a random order instead of sequentially (e.g. first foo, then bar, etc.)

Page 3: [5] Commented [MS16R4] Matthews, Suzanne Dr 8/4/2025 9:01:00 AM

@Zitter, Andrew P MAJ This problem still contains concepts that are not covered yet in the course by this lesson. The strip() method is not covered until unit 4. lower() is also a string method, not a built-in function. Again, these concepts have not been covered until Unit 4.

I think you can a question of similar desired complexity for cadets by changing line 1 to just be spam="27", and deleting lines 6, 11

Page 7: [6] Commented [JH24R5] Hutton, John E LTC 7/31/2025 11:50:00 AM

@Matthews, Suzanne Dr Are built in functions on the table? I see on the 26-1 schedule they come before the due date, but I'm guessing we assign the WPS before covering the built-ins.

Page 7: [7] Commented [HT25R2] Tippet, Hayden J MAJ 9/2/2025 7:55:00 AM

The first problem uses strictly ints, I think a float for a grade is reasonable here. The question stem says float, so we should show at least one for sample input.

Page 7: [8] Commented [MS26] Matthews, Suzanne Dr 1/1/1900 12:00:00 AM

recommend making this a percentage (out of 100) to match the format of the rest of them. Alternatively, modify the output to be something like

295.0/250

to make it clear that we are not saying 295%



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB M

PROBLEM SET 5

CY105 AY26-1 WPS5

LESSONS COVERED

- (review) Iteration - For
- (Review) Strings and Lists
- Using Modules

Part A – Canvas Component: 5 pts

- 2-3 KC/CT Questions, focus on Using Modules

Part B – zyBooks Component: 10 pts

- 2x Code Writing Problem (6/4 pt ea)

Legend: Updates to be made. Question Answers. Feedback Section. Configured in Canvas/zyBooks.

Canvas Component

1. Knowledge Check: Using Modules

Question Type: Matching

Question Points: 2

Question Stem: Match replacement for FUNCTION(n) to the resulting output:

```
import math

def main():
    numbers = [2.1, 3.7, 4.6]
    result = []

    for n in numbers:
        result = result + [ FUNCTION(n) ]
    print(result)

main()
```

[2,3,4]	<code>math.floor(n)</code>
[1,1,2]	<code>int(math.sqrt(n))</code>
[2,24,120]	<code>math.factorial(round(n))</code>
[3,4,5]	<code>math.ceil(n)</code>

Distractors: `round(n)`

Feedback:

[2,3,4]	<code>math.floor(x)</code> returns the greatest integer $\leq x$ (rounds down).
---------	---

Commented [PL1]: Instead of adding to the empty string I think it would be more clear for cadets at this point to add to an empty list. I would replace the round with math.floor

Commented [CJWL1R2]: Changed per rec.

Commented [MS2]: Lots of reinforcing the int() function. Does it make sense to change this to round? If we make 4.5 4.6, we will also avoid the bankers rounding issue.

Commented [CJWL2R2]: I have removed the typecast from the question stem and placed in as int(math.sqrt(n)).

Commented [MS2R3]: I like this change. I still think it is an open question on whether it makes sense to change the int() in math.factorial() to round() just to change it up a little.

Commented [CJWL2R4]: Updated per rec

Commented [HT3]: @Matthews, Suzanne Dr instead of a distractor, could I use this as an example in the question stem?

Commented [MS3R2]: I don't think an example needs to be put in the question stem, especially because round is one of the things we are testing them on.

CY105 AY26-1 WPS5

[1,1,2]	int(math.sqrt(x)) returns square root of x, then truncates the decimal portion of the result.
[2,6,24]	math.factorial(round(x)) ensures an int is passed to the math.factorial() function, which multiplies all positive integers from 1 to x.
[3,4,5]	math.ceil(x) returns the smallest integer $\geq x$ (rounds up).

2. Knowledge Check: Strings and Lists

Question Type: **Dropdown**

Question Points: 1

Question Stem: Consider the following code snippet:

```
values = [1, 2]
result = values * 2 + values + [3]
print(result)
```

Which is the correct explanation for how the list operators work and the produced output?

- A. values * 2 repeats [1,2] twice, + values modifies the list to be [2,4,1,2], then [3] is added to the list. Produces: [2, 4, 1, 2, 3]
- B. values * 2 repeats [1,2] twice, + values concatenates one more copy, and finally [3] is concatenated at the end. Produces: [1, 2, 1, 2, 1, 2, 3]
- C. values * 2 makes the list [2,4], + values modifies the list to be [4,8], then [3] is added to the end. Produces: [4, 8, 3]
- D. values * 2 makes the list [2, 4], + values modifies the list to be [3, 6], then [3] is joined to it. Produces: [3, 6, 3]

Feedback:

values * 2 repeats [1,2] twice, + values adds [1,2] to the first two elements, then [3] is added to the list. Produces: [2, 4, 1, 2, 3]	Wrong, + with lists does not add numbers inside, it concatenates lists.
values * 2 repeats [1,2] twice, + values concatenates one more copy, and finally [3] is concatenated at the end. Produces: [1, 2, 1, 2, 1, 2, 3]	Correct, * repeats the entire list, and + concatenates lists. The final [3] is concatenated at the end.
values * 2 makes [2,4], + values makes [4,8], then [3] is added to the end. Produces: [4, 8, 3]	Wrong, * does not multiply elements, it repeats the list.

Commented [MS4]: What's more important? The output or the explanation? I had a really hard time parsing a number of the explanations. D is the easiest to understand. I think reformatting A, B and C to be more like D would make more sense.

Instead of two outputs that are [3,6,3], make one of them

values * 2 makes [2,4], then + values makes [4,8], then [3] is joined to it. Produces: [4,8,3]

I recommend replacing option C with the above.

Commented [CJWL4R2]: Updated per rec

Commented [MS5]: makes the list

Commented [MS6]: makes the list

Commented [MS7]: modifies the list to be

CY105 AY26-1 WPS5

<code>values * 2</code> makes <code>[2, 4]</code> , + <code>values</code> makes <code>[3, 6]</code> , then <code>[3]</code> is joined to it. Produces: <code>[3, 6, 3]</code>	Wrong, list multiplication and addition never do element-wise math.
--	---

3. Code Tracing: Using Modules

Question Type: Fill in the Blank (Dropdown, T/F)

Question Points: 2

Question Stem: Consider the following output:

`[4,4,7]`

For each code snippet below, decide True/False: *Is it possible for this code to produce the exact output `[4, 4, 7]`?* You may assume that the program imported the random module earlier.

<pre>nums = [] for i in range(3): nums += [int(random.random() * 6) + 1] print(nums)</pre>	False
<pre>nums = [] for i in range(3): nums += [random.randint(1, 7)] print(nums)</pre>	True
<pre>nums = random.choices(range(1, 8), 3) print(nums)</pre>	True
<pre>nums = random.sample(range(1, 8), 3) print(nums)</pre>	False

Feedback:

<pre>nums = [] for i in range(3): nums += [int(random.random() * 7)] print(nums)</pre>	False: <code>random.random()</code> produces a float in <code>[0.0, 1.0)</code> . Multiplying the number by 7 and converting to an <code>int</code> type, ultimately results in integers 0-6. It is impossible to produce a 7.
<pre>nums = [] for i in range(3): nums += [random.randint(1, 7)] print(nums)</pre>	True: <code>random.randint(1, 7)</code> returns an integer between 1 and 7 inclusive.
<pre>nums = random.choices(range(1, 8), 3) print(nums)</pre>	True: <code>random.choices()</code> selects items with replacement; duplicates like <code>[4, 4, 7]</code> are possible.
<pre>nums = random.sample(range(1, 8), 3) print(nums)</pre>	False: <code>random.sample()</code> selects items

Commented [PL8]: Last year we did not cover `.append` until the later list lesson. You would have to change to `+= []`

Commented [CJWL8R2]: Changed per rec.

Commented [MS9]: There is a lot of list concatenation tested throughout this problem set. Is that really needed?

I think this problem would be simplified if the for loop just had a `print()` statement inside, and we ask

"For each code snippet below, decide True/False: Is it possible for this code to output the sequence

`4`
`4`
`7`

Commented [CJWL9R2]: `random.choices` and `random.sample` return a list. Placing the other snippets in a list ensure consistent output, so students focus on the call to random functions, not on the output format.

CY105 AY26-1 WPS5

```
print(nums)
```

without replacement; duplicates are not possible.

zyBooks Component

1. CODING PROBLEM 1 – FREE THROW STATISTICS (6 PTS)

As an assistant coach, it is your responsibility to generate statistics on the free throws a player makes during practice. You receive the number of free throw attempts a player makes during practice and the player's free throw success rate for the day (a decimal value between 0 and 1, inclusive) as two separate data points. You must combine these data points to produce a list of successful free throws per day over a specified number of practices and calculate the overall mean, median, and standard deviation (rounded to the nearest single decimal point).

Your program must perform the following:

Objective: Given user input of free throws and success rate per day, produce statistics, rounded.

Input: Number of *days* of practice (once), free *throw* attempts made for a day (multiple), *rate* of successful free throws for a day (multiple).

Output: Print mean, median, and standard deviation values for successful free throws across all provided days, rounded to nearest single decimal point.

Constraints/Assumptions: User will only input valid values: positive integer for *days* and *throws*, and a decimal between 0.0 and 1.0 (inclusive) for *rate*.

Algorithm:

1. Initialize 3 lists to store *throws*, *rate*, and *baskets* made.
2. Prompt and receive *days* from user.
3. For each *day* in *days*:
 - a. Prompt and receive *throws* from user.
 - b. Store throws in *throws_list*
 - c. Prompt and receive *rate* from user.
 - d. Store rate in *rate_list*
4. For each *index* in *throws_list*:
 - a. Multiple entry at *index* in *throws_list* by entry at *index* in *rate_list* to calculate basket count
 - b. Store basket count in *basket_list*
5. Using statistics module, calculate and store mean, median, and standard deviation of *basket_list*.
6. Print mean, median, and standard deviation, all rounded to nearest single decimal point.

For example, given the following record of a player's free throw attempts and success rate over a training period for testing:

Day	1	2	3	4	5	6	7
Free throw Attempts	20	25	22	30	28	24	26
Success Rate	0.75	0.8	0.7	0.85	0.78	0.82	0.76

Commented [PL10]: Change header from Grade Pt Average

Commented [AK10R2]: Good catch. Updated

Commented [HT11]: Round up to the nearest whole number (ceil)

CY105 AY26-1 WPS5

Here is a sample of what the input/output sequence looks like:

```
Enter the number of practice days: 7
Enter the number of free throw attempts for the day: 20
Enter the free throw success rate for the day: 0.75
Enter the number of free throw attempts for the day: 25
Enter the free throw success rate for the day: 0.8
Enter the number of free throw attempts for the day: 22
Enter the free throw success rate for the day: 0.7
Enter the number of free throw attempts for the day: 30
Enter the free throw success rate for the day: 0.85
Enter the number of free throw attempts for the day: 28
Enter the free throw success rate for the day: 0.78
Enter the number of free throw attempts for the day: 24
Enter the free throw success rate for the day: 0.82
Enter the number of free throw attempts for the day: 26
Enter the free throw success rate for the day: 0.76
Free throw mean: 19.6
Free throw median: 19.8
Free throw standard deviation: 3.6
```

Sample *Input/Output*:

Provided to Students:

```
"""
Enter algorithm header here
"""
```

```
def main():
    #enter code here
```

```
main()
```

Sample Solution:

```
"""
Procedure: successful_free_throws
Input: Number of free throw attempts per day for a week; free throw success
rate per day for a week
Output: Mean, median, and standard deviation of successful free throws over
the week
"""
```

```
# Import the statistics module
import statistics
```

```
def main():
    # Specify the number of days being tracked
    num_practice_days = int(input("Enter the number of practice days: "))
```

CY105 AY26-1 WPS5

```
# Initialize arrays of free throw attempts and free throw successes
free_throw_attempt_list = []
free_throw_success_rate_list = []

# Repeat num_practice_days times:
for _ in range(num_practice_days):
    # Request number of free throw attempts for a specific day
    free_throw_attempts = int(input("Enter the number of free throw
attempts for the day: "))
    # Append number of free throw attempts to the free throw attempt array
    free_throw_attempt_list += [free_throw_attempts]
    # Request the free throw success rate for the day
    free_throw_success_rate = float(input("Enter the free throw success
rate for the day: "))
    # Append the day's free throw success rate to the array
    free_throw_success_rate_list += [free_throw_success_rate]

# Initialize the array of successful free throws made
free_throw_success_count_list = []

# Repeat length of free throw attempt list times
for index in range(len(free_throw_attempt_list)):
    # Calculate the free throw success count for the day by multiplying the
value from free throw attempt list by value from free throw success rate list
at the same index
    free_throw_success_count =
free_throw_attempt_list[index]*free_throw_success_rate_list[index]
    # Append the day's free throw success count to the array
    free_throw_success_count_list += [free_throw_success_count]

# Calculate the mean number of successful free throws
free_throw_mean = statistics.mean(free_throw_success_count_list)
# Calculate the median number of successful free throws
free_throw_median = statistics.median(free_throw_success_count_list)
# Calculate the standard deviation of successful free throws
free_throw_std_dev = statistics.stdev(free_throw_success_count_list)

# Print the mean number of successful free throws
print("Free throw mean:", round(free_throw_mean, 1))
# Print the median number of successful free throws
print("Free throw median:", round(free_throw_median, 1))
# Print the standard deviation of successful free throws
print("Free throw standard deviation:", round(free_throw_std_dev, 1))
```

Commented [MS12]: Simplify to "Enter the number of free throw attempts for the day:" just to reduce the amount of code students have to write for this problem.

Commented [MS13]: append is not covered yet.

Commented [MS14]: Same as comment above. Simplify to "Enter the free throw success rate for the day"

CY105 AY26-1 WPS5

main()

Test Cases:

1) Test Case Description 1

a) Inputs:

- i) 7
- ii) 20
- iii) 0.75
- iv) 25
- v) 0.8
- vi) 22
- vii) 0.7
- viii) 30
- ix) 0.85
- x) 28
- xi) 0.78
- xii) 24
- xiii) 0.82
- xiv) 26
- xv) 0.76

b) Expected Prompt(s):

- i) "Enter the number of practice days: "
- ii) "Enter the number of free throw attempts for the day: "
- iii) "Enter the free throw success rate for the day: "
- iv) "Enter the number of free throw attempts for the day: "
- v) "Enter the free throw success rate for the day: "
- vi) "Enter the number of free throw attempts for the day: "
- vii) "Enter the free throw success rate for the day: "
- viii) "Enter the number of free throw attempts for the day: "
- ix) "Enter the free throw success rate for the day: "
- x) "Enter the number of free throw attempts for the day: "
- xi) "Enter the free throw success rate for the day: "
- xii) "Enter the number of free throw attempts for the day: "
- xiii) "Enter the free throw success rate for the day: "
- xiv) "Enter the number of free throw attempts for the day: "
- xv) "Enter the free throw success rate for the day: "

c) Expected Output(s) (using provided input):

- i) "Free throw mean: 19.6"
- ii) "Free throw median: 19.8"
- iii) "Free throw standard deviation: 3.6"

Commented [HT15]: Use math.isclose()

CY105 AY26-1 WPS5

Tested Concepts:

- Iteration - For
- Strings and Lists
- Using Modules

2. CODING PROBLEM 2 – FREE THROW STATISTICS: AI-ASSISTED REMIX (4 PTS)

Problem description - Solve problem 1 in another way using a module not discussed in the course, but recommended by a 3rd party GenAI. Students are required to use a 3rd party GenAI to solve this problem, and demonstrate how they prompted the AI to help them understand how to pick particular functions etc to use; however, in keeping with the course, the cadet must still write the code themselves.

Sample *Input/Output*:

Provided to Students:

```
"""
Enter algorithm header here
"""
```

```
def main():
    #enter code here

main()
```

Sample Solution:

```
"""
procedure: successful_free_throws_redux
Input: Number of free throw attempts per day for a week; free throw success
rate per day for a week
Output: Mean, median, and standard deviation of successful free throws over the
week
```

~~ EXAMPLE AI PROMPTS ~~

Example Prompt 1 (Providing Part 1 Solution Code):

I am solving a Python programming problem for an introduction to technology course. I have already solved it but now need to solve it another way using a Python module other than random, math, statistics. Can you suggest some alternate modules that would work? Do not give me the solution - only suggest some functions and provide generic code on how to use them?

Here is my current solution:

<Part 1 solution code>

Commented [PL16]: Change header from Grade Pt Average

Commented [AK16R2]: Good catch. Updated

Commented [PL17]: Sample solution should show sample prompts to AI

Commented [AK17R2]: Thanks. Added!

Commented [PL18]: I think it will be very difficult for them to implement the suggestions from the prompt but I don't think we want them to use the AI to give them the code to solve it a different way

CY105 AY26-1 WPS5

~~~~~

Example Prompt 2 (Providing Algorithm Description):

I am given the following Python problem. I've already solved it but now need an alternate solution that uses a Python module other than random, math, or statistics. Can you suggest alternates? Don't give me the solution, only suggest some functions and provide generic code descriptions on how to use them.

Problem to solve in Python:

As an assistant coach, it is your responsibility to generate statistics on the free throws a player makes during practice. <Rest of algorithm description>

~~~~~

Example Prompt 3 (Providing General Solution Description):

I have written a Python script that calculates the mean, median, and standard deviation of a basketball player's successful free throws made over a week of practice. I was given a free throw attempt count and free throw success rate for each day of practice in the week.

I now need to write another version of my solution using modules other than random, math, and statistics. Can you suggest modules I can use? Don't provide solution code, only enough code to demonstrate the use of any recommended functions.

~~~~~

NOTE: Problem 2 may contain following variations suggested by GenAI:

- Calculating mean, median, and std dev using NumPy, SciPy, or Pandas instead of statistics

"""

# Import the statistics module

import statistics

def main():

# Specify the number of days being tracked

num\_practice\_days = int(input("Enter the number of practice days: "))

# Initialize arrays of free throw attempts and free throw successes

free\_throw\_attempt\_list = []

free\_throw\_success\_rate\_list = []

# Repeat num\_practice\_days times:

for \_ in range(num\_practice\_days):

# Request number of free throw attempts for a specific day

free\_throw\_attempts = int(input("Enter the number of free throw attempts for the day: "))

# Append number of free throw attempts to the free throw attempt array

free\_throw\_attempt\_list += [free\_throw\_attempts]

## CY105 AY26-1 WPS5

```
# Request the free throw success rate for the day
free_throw_success_rate = float(input("Enter the free throw success
rate for the day: "))
# Append the day's free throw success rate to the array
free_throw_success_rate_list += [free_throw_success_rate]

# Initialize the array of successful free throws made
free_throw_success_count_list = []

# Repeat length of free throw attempt list times
for index in range(len(free_throw_attempt_list)):
    # Calculate the free throw success count for the day by multiplying the
    value from free throw attempt list by value from free throw success rate list
    at the same index
    free_throw_success_count =
free_throw_attempt_list[index]*free_throw_success_rate_list[index]
    # Append the day's free throw success count to the array
    free_throw_success_count_list += [free_throw_success_count]

# Calculate the mean number of successful free throws
free_throw_mean = statistics.mean(free_throw_success_count_list)
# Calculate the median number of successful free throws
free_throw_median = statistics.median(free_throw_success_count_list)
# Calculate the standard deviation of successful free throws
free_throw_std_dev = statistics.stdev(free_throw_success_count_list)

# ~ EXAMPLE SOLUTION USING NUMPY ~
#import numpy as np
#free_throw_mean = np.mean(free_throw_success_count_list)
#free_throw_median = np.median(free_throw_success_count_list)
#free_throw_std_dev = np.std(free_throw_success_count_list, ddof=1)

# ~ EXAMPLE SOLUTION USING SCIPY ~
#from scipy import stats
#free_throw_mean = stats.tmean(free_throw_success_count_list)
#free_throw_median = stats.scoreatpercentile(free_throw_success_count_list,
50)
#free_throw_std_dev = stats.tstd(free_throw_success_count_list)

# ~ EXAMPLE SOLUTION USING PANDAS
#import pandas as pd
#data = pd.Series(free_throw_success_count_list)
#free_throw_mean = data.mean()
#free_throw_median = data.median()
#free_throw_std_dev = data.std()
```

## CY105 AY26-1 WPS5

```
# Print the mean number of successful free throws
print("Free throw mean:", round(free_throw_mean, 1))
# Print the median number of successful free throws
print("Free throw median:", round(free_throw_median, 1))
# Print the standard deviation of successful free throws
print("Free throw standard deviation:", round(free_throw_std_dev, 1))
```

main()

### *Test Cases:*

#### 2) Test Case Description 1

##### a) Inputs:

- i) 7
- ii) 20
- iii) 0.75
- iv) 25
- v) 0.8
- vi) 22
- vii) 0.7
- viii) 30
- ix) 0.85
- x) 28
- xi) 0.78
- xii) 24
- xiii) 0.82
- xiv) 26
- xv) 0.76

##### b) Expected Prompt(s):

- i) "Enter the number of practice days: "
- ii) "Enter the number of free throw attempts for the day: "
- iii) "Enter the free throw success rate for the day: "
- iv) "Enter the number of free throw attempts for the day: "
- v) "Enter the free throw success rate for the day: "
- vi) "Enter the number of free throw attempts for the day: "
- vii) "Enter the free throw success rate for the day: "
- viii) "Enter the number of free throw attempts for the day: "
- ix) "Enter the free throw success rate for the day: "
- x) "Enter the number of free throw attempts for the day: "
- xi) "Enter the free throw success rate for the day: "
- xii) "Enter the number of free throw attempts for the day: "
- xiii) "Enter the free throw success rate for the day: "

## CY105 AY26-1 WPS5

xiv) "Enter the number of free throw attempts for the day: "

xv) "Enter the free throw success rate for the day: "

c) Expected Output(s) (using provided input):

i) "Free throw mean: 19.6"

ii) "Free throw median: 19.8"

d) "Free throw standard deviation: 3.6"

### *Tested Concepts:*

- Iteration - For
- Strings and Lists
- Using Modules



UNITED STATES MILITARY ACADEMY  
**WEST POINT.**

**TAB N**

**PROBLEM SET 6**

---

## LESSONS COVERED

- Conditionals I
- Conditionals II
- Iteration II (While loops)

### Part A – Canvas Component: 5 pts

- 2-3 KC/CT questions on above lessons

### Part B – zyBooks Component: 10 pts

- 2x Code Writing Problems (no algorithm provided)

Legend: Updates to be made. Question Answers. Feedback Section. Configured in Canvas/zyBooks.

## Canvas Component (5 PTS)

### 1. Question Title: Conditionals I and II

Question Type: Dropdown

Question Points: 3

Question Stem:

# the values assigned to battery and obstacle\_detected are set here by the table below; the underscores are placeholders for those values

```
battery = __
obstacle_detected = __
if battery > 75: # line 5
    print("Continue mission")
elif battery > 25: # line 7
    if obstacle_detected: # line 8
        print("Pause and scan")
    else: # line 10
        print("Proceed with caution")
else: # line 12
    print("Return to base")
```

For the above lines of code, match the assignments for battery and obstacle\_detected with the correct output.

|                                         | Continue mission     |
|-----------------------------------------|----------------------|
| battery = 75, obstacle_detected = True  | Pause and scan       |
| battery = 30, obstacle_detected = False | Proceed with caution |
| battery = 25, obstacle_detected = True  | Return to base       |

Feedback:

Commented [TW1]: TW

Commented [DC2]: I'd add two lines above the first if saying something like "battery = ... # the value here is part of the question below" and another for obstacle\_detected. It was a little confusing looking for the definition of those variables but they weren't declared! (Or possibly update the description to say that they were declared before the above code.)

Commented [TW2R2]: Thanks, does that look good now? I think you're making a good point, and I welcome you to edit it directly if you can make it more clear.

Commented [DJ3]: Assume the four choices in this column are the options in a dropdown? It was unclear from the way this is formatted.

Commented [TW3R2]: The idea is drag-and-drop matching, where the three things on the left would get dragged over to one of the four possible things on the right. Does that make sense?

Commented [TW3R3]: We changed this to be a dropdown after meeting with Suzanne, to make sure it works in Canvas.

|                                         |                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| battery = 75, obstacle_detected = True  | The first if statement on line 5 only evaluates to True when the value of battery is strictly greater than 75, so this causes execution to continue with the next elif statement in the chain on line 7. The final output is then determined by the inner binary conditional on lines 8 and 10, depending on the value of obstacle_detected. |
| battery = 30, obstacle_detected = False | obstacle_detected being False causes the else portion of the nested binary conditional, on line 10, to execute.                                                                                                                                                                                                                              |
| battery = 25, obstacle_detected = True  | The second conditional on line 7 only evaluates to True when the value of battery is strictly greater than 25, so this causes execution to continue with the else statement on line 12.                                                                                                                                                      |

**Commented [KL4]:** I found the terms 'first conditional' and 'second conditional' a little ambiguous - is 'first conditional' the first test of the outer loop? Is 'second conditional' the 'elif' statement of the outer loop? Or the 'if' statement of the nested loop?

**Commented [TW4R2]:** I made a couple edits here. Is that better?

**Commented [WL4R3]:** I've added line numbers now for greater clarity.

## 2. Question Title: Iteration II

Question Type: Dropdown

Question Type: Multiple choice

Question Points: 2

Question Stem: Given the code snippet below, answer the next two multiple choice questions.

```
time_left = 10
while time_left > 0:
    print("Counting down:", time_left)
    time_left -= 3
```

What is the value of time\_left when the loop terminates?

|    |       |
|----|-------|
| 0  | False |
| -2 | True  |
| 1  | False |
| 3  | False |

How many times does the loop body execute?

|   |       |
|---|-------|
| 0 | False |
| 3 | False |
| 4 | True  |
| 5 | False |

### Feedback:

The block of code inside the loop executes four times. At the beginning of the fourth iteration, time\_left is 1. Then time\_left -= 3 makes it -2, so the condition for the while loop

**Commented [TW5]:** TW

**Commented [DJ6]:** I'd suggest just making this a multiple choice question with the 4 choices you have listed on the left side instead of T/F.

**Commented [TW6R2]:** That works, I changed it.

**Commented [TW6R3]:** We split this one into two multiple choice questions about the same code snippet after meeting with Suzanne.

becomes False.

## zyBooks Component (10 PTS)

### 1. CODING PROBLEM 1 – CUSTOMS AND COURTESIES SIMULATOR (5 PTS)

Write a program that simulates encounters with other officers and correctly tells a user when to render or return salutes. Begin by prompting the user for their officer pay grade [1-10]. Then randomly choose another pay grade and whether the user is indoors or outdoors. Print the random choices and the appropriate action.

#### Sample Input/Output 1:

What grade of officer are you? **3**

Meet an O-5 indoors: No salute

#### Sample Input/Output 2:

What grade of officer are you? **2**

Meet an O-1 outdoors: Return their salute

#### Sample Input/Output 3:

What grade of officer are you? **5**

Meet an O-6 outdoors: Render a salute

#### *Provided to Students:*

```
'''
Place problem header here
'''

import random
'''
Uncomment the line below to generate the sample output
'''

#random.seed(10)

def main():
    # enter code here
    # remember to include algorithm as in-line comments

main()
```

#### *Sample Solution:*

Commented [TW7]: TW

Commented [DJ8]: Should we include a "reminder" of the rules for saluting to be clear?

Commented [JH8R2]: I don't think they need a reminder of the rules. The rules should be fully inculcated by now.

Commented [JD9]: Include one more example with the third possible output so they see the exact wording.

Commented [TW9R2]: That makes sense; done.



```

'''
Problem: Simulate officer customs and courtesies
Input: A pay grade between 1 and 10, inclusive, and whether to go again after
each round.
Output: A random pay grade, indoors or outdoors, and the appropriate saluting
action.
'''

import random
'''
Uncomment the line below to generate the sample output
'''

#random.seed(10)

def main():
    SETTINGS = ['indoors', 'outdoors']
    # Prompt user for their pay grade
    my_grade = int(input("What grade of officer are you? "))
    # Get random choices for this round
    setting = random.choice(SETTINGS)
    grade = random.randint(1, 10)
    # Prepare a string for the choices
    interaction = 'Meet an O-' + str(grade) + ' ' + setting + ':'
    # First check if we're outdoors
    if setting == 'outdoors':
        # Check my_grade against their grade
        if grade > my_grade:
            response = 'Render a salute'
        elif grade < my_grade:
            response = 'Return their salute'
        else:
            response = 'No salute'
    # If we're indoors, grade doesn't matter
    else:
        response = 'No salute'

    # Print the resulting interaction
    print(interaction, response)

main()

```

#### Test Cases:

- 1) Test Case Description 1 (seed = 10)
  - a) Inputs: 5
  - b) Expected Prompt(s):
    - i) "What grade of officer are you? "

**Commented [JD10]:** This isn't something we teach - I would include the entire print statement in one line inside the conditionals.

**Commented [TW10R2]:** I leave this one for you @Hutton, John E LTC

**Commented [LP10R3]:** I agree with Denise

**Commented [JH10R4]:** Added 4<sup>th</sup> test case. Refactored code to eliminate the "end =".

- c) Expected Output(s):
  - i) Meet an O-7 indoors: No salute
- 2) Test Case Description 2 (seed = 5)
  - a) Inputs: 1
  - b) Expected Prompt(s):
    - i) "What grade of officer are you? "
  - c) Expected Output(s):
    - i) Meet an O-6 outdoors: Render a salute
- 3) Test Case Description 3 (seed = 23)
  - a) Inputs: 5
  - b) Expected Prompt(s):
    - i) "What grade of officer are you? "
  - c) Expected Output(s):
    - i) Meet an O-2 outdoors: Return their salute
- 4) Test Case Description 4 (seed = 23)
  - a) Inputs: 2
  - b) Expected Prompt(s):
    - i) "What grade of officer are you? "
  - c) Expected Output(s):
    - i) Meet an O-2 outdoors: No salute
- 5)

**Tested Concepts:**

- Conditionals I and II

## 2. CODING PROBLEM 2 – DRIVING SIMULATOR (5 PTS)

Write a program that simulates an irresponsible driver cruising through different speed zones until crashing or getting stopped by the police for speeding. On each iteration, randomly generate the driver's speed (an integer **between** 20 and 90 mph inclusive) and the speed limit (multiples of 10 between 30 and 70 mph, inclusive). When the driver is speeding, but going less than 20 mph over the limit, they have a 50-50 chance of getting caught by the police. When the driver speeds 20 mph or more over the limit, they instantly crash. The police will not stop the driver if they are going at or below the speed limit. Run iterations of this until reaching one of the stopping conditions (i.e. getting stopped by the police or crashing).

### Sample Input/Output 1:

Driver speed: 37  
 Speed limit: 70  
 The driver keeps going  
 Driver speed: 28  
 Speed limit: 50

**Commented [DJ11]:** Seems like there is a lot of overlap between the two coding problems. I actually like this one better. Understand this one focuses on while loops and conditionals, but maybe come up with something that doesn't use random?

**Commented [TW11R2]:** We just got some initial guidance from Suzanne on code writing problems for this problem set, and we'll make some significant changes to the first one (making it much simpler, with no iteration), so there will be much less overlap between the two. If you like this one more anyway, we'll keep this one as is for now.

**Commented [DJ11R3]:** Sounds good. Let me know when you have an update (or draft) to look at.

**Commented [TW11R4]:** We updated this problem and Coding Problem 1 after meeting with Suzanne.

**Commented [TW12]:** TW

**Commented [JD13]:** an integer between 20 and 90 mph inclusive

**Commented [TW13R2]:** Ok, done. During our discussion with Suzanne yesterday, there was some talk about whether this should really be a float, but I'm thinking now that keeping it as an integer is much easier for them.

The driver keeps going  
Driver speed: 35  
Speed limit: 60  
The driver keeps going  
Driver speed: 77  
Speed limit: 60  
Speeding  
The driver keeps going  
Driver speed: 46  
Speed limit: 30  
Speeding  
The driver keeps going  
Driver speed: 23  
Speed limit: 60  
The driver keeps going  
Driver speed: 75  
Speed limit: 70  
Speeding  
Stopped by the police

Sample Input/Output 2:

Driver speed: 71  
Speed limit: 60  
Speeding  
Stopped by the police

WARNING: the test cases are based on the most obvious way to answer this question. There are many different ways to reach a valid solution.

Commented [HT14]: Add WARNING

*Provided to Students:*

```
'''  
Place problem header here  
'''  
import random  
  
'''  
Uncomment the line below to generate the sample output 1  
'''  
#random.seed(1)  
'''  
Uncomment the line below to generate the sample output 2
```

```

'''
#random.seed(99)

def main():
    # enter code here
    # remember to include algorithm as in-line comments

main()

```

#### Sample Solution:

```

'''
Problem: Simulate driving speeds and outcomes
Input: None
Output: A random number of iterations showing the driver's speed, the speed
limit, and the outcome until they either crash or get stopped by the police.
'''
import random

'''
Uncomment the line below to generate the sample output 1
'''
#random.seed(1)
'''
Uncomment the line below to generate the sample output 2
'''
#random.seed(99)

def main():
    STOPPED = [True, False]
    # 1. Initialize keep_driving Boolean
    keep_driving = True
    # 2. Loop as long as keep_driving is True
    while keep_driving:
        # 3. Get random choices for this round
        speed = random.randint(20, 90)
        limit = random.randrange(30,80,10)
        # 4. Print the choices
        print('Driver speed:', speed)
        print('Speed limit:', limit)
        # 5. Check speed against the Limit
        if speed > limit:
            print('Speeding')
            # 5.1. Check if more than 20 over the limit, and set
            # keep_driving Boolean to False if so
            if speed >= limit + 20:

```

```

        print('Crashed')
        keep_driving = False
    else:
        # 5.2. Flip a coin if less than 20 over the limit
        if random.choice(STOPPED):
            # 5.3. If stopped, set keep_driving Boolean to False
            print('Stopped by the police')
            keep_driving = False
        else:
            # 5.4. If not stopped, keep going
            print('The driver keeps going')
    else:
        # 5.5. If not speeding, keep going
        print('The driver keeps going')

main()

```

#### Test Cases:

##### 6) Test Case Description 1

- a) Inputs: seed 1
- b) Expected prompt(s):
- c) Expected Output(s):
  - i) Driver speed: 37
  - ii) Speed limit: 70
  - iii) The driver keeps going.
  - iv) Driver speed: 28
  - v) Speed limit: 50
  - vi) The driver keeps going.
  - vii) Driver speed: 35
  - viii) Speed limit: 60
  - ix) The driver keeps going.
  - x) Driver speed: 77
  - xi) Speed limit: 60
  - xii) Speeding.
  - xiii) The driver keeps going.
  - xiv) Driver speed: 46
  - xv) Speed limit: 30
  - xvi) Speeding.
  - xvii) The driver keeps going.
  - xviii) Driver speed: 23
  - xix) Speed limit: 60
  - xx) The driver keeps going.

- xxi) Driver speed: 75
- xxii) Speed limit: 70
- xxiii) Speeding.
- xxiv) Stopped by the police.

7) Test Case Description 2

- a) Inputs: seed 99
- b) Expected prompt(s):
- c) Expected Output(s):
  - i) Driver speed: 71
  - ii) Speed limit: 60
  - iii) Speeding.
  - iv) Stopped by the police.

8) Test Case Description 3

- a) Inputs: seed 4
- b) Expected prompt(s):
- c) Expected Output(s):
  - i) Driver speed: 50
  - ii) Speed limit: 50
  - iii) The driver keeps going.
  - iv) Driver speed: 33
  - v) Speed limit: 60
  - vi) The driver keeps going.
  - vii) Driver speed: 81
  - viii) Speed limit: 40
  - ix) Speeding.
  - x) Crashed.

**Tested Concepts:**

- Conditionals I and II
- Iteration II



UNITED STATES MILITARY ACADEMY  
**WEST POINT.**

## **TAB O**

## **PROBLEM SET 7**

---

## LESSONS COVERED

- Iteration II (Review)
- Writing Functions (New)

### **Part A – Canvas Component: 5 pts**

- 2-3x knowledge check/code tracing

### **Part B – zyBooks Component: 10 pts**

- 1x Code Writing Problem – Write a Function
- 1x Code Writing Problem – use the function you wrote and write main.

Legend: Updates to be made. Question Answers. Feedback Section. Configured in Canvas/zyBooks

## Canvas Component (5 PTS)

### 1. Question Title: Lesson Title X

Question Type: Fill in the Blank, True/False

Question Points: 3 (0.5 each)

Question Stem: Consider the following code snippet:

```
def ammo_reload(max_ammo, current_ammo):  
    ammo_segment = 4  
    while current_ammo < max_ammo:  
        current_ammo += ammo_segment  
  
    print('Ammo reloaded!')  
ammo_reload(30, 0)
```

Commented [HT1]: Maybe use stripper\_clip instead of ammo\_segment so they are more clued into what this is doing

|                                                                              |                              |
|------------------------------------------------------------------------------|------------------------------|
| Is this a definite or an indefinite loop?                                    | <b>Definite</b> / Indefinite |
| How many times does the while loop execute?                                  | 7, <b>8</b> , 10, 0          |
| What is the value of current_ammo after the function has finished executing? | 0, 30, <b>32</b> , 34        |

### Feedback:

- This is an **indefinite** loop because it keeps running until current\_ammo reaches max\_ammo; the number of iterations isn't fixed ahead of time.



- The loop adds 4 each time starting at 0 until current\_ammo is no longer < 30. It runs 8 times before exceeding 30.
- The loop continues until current\_ammo reaches 32, not 30, because the condition checks for < 30, not <= 30.

2. Question Title: Lesson Title X

Question Type: Dropdown (True/False)

Question Points: 3 (1 pt each)

Question Stem: Consider the following Code snippet. Given the following values for target and random\_roll evaluate the return value of the variable destroyed after execution.

```
import random
def assess_hit(target):

    hit_rate = .80

    if (target == "Mi-8" or target == "Bunker"):
        hit_rate = 0.5

    elif (target == "T-90" or target == "T-80U"):
        hit_rate = 0.25

    random_roll = random.random()

    if random_roll <= hit_rate:
        destroyed = True
    else:
        destroyed = False

    return destroyed
```

| target   | random_roll | destroyed |
|----------|-------------|-----------|
| "T-90"   | 0.1345      | `True`    |
| "Bunker" | 0.8932      | `False`   |
| "BMP-3"  | 0.8000      | `True`    |

Feedback:

- Remember, T-90s only have a 25% hit chance. Because  $0.1345 \leq 0.25$ , the vehicle should be destroyed.
- Check the comparison:  $0.8932 > 0.5$ , so the Bunker survives—destroyed should be False.

Commented [JD2]: random\_roll

Commented [PL3]: I suggest evaluate the return value of destroyed after execution

Commented [MH3R2]: Corrected.

- The default hit rate is 0.8, and since random\_roll equals 0.8, that's still a hit—destroyed should be True.

## zyBooks Component (10 PTS)

### 1. CODING PROBLEM — WRITE A FUNCTION (5 PTS)

Write a function called `engage_target` that takes two parameters, `target` (a string), and `ammo` (a non-negative integer). While `ammo` is greater than zero and the target is **not** destroyed, this function will repeatedly call the provided function `assess_hit`, determine if the target is hit, and decrement `ammo`.

If the target is destroyed, return the remaining `ammo`. If the target is not destroyed, return -1.

#### Sample Input/Output 1:

//Should we provide output to the students?//

#### Provided to Students:

```
"""
assess_hit(target)
**NOTE: This function is provided and should not be modified!**
Purpose: Determine if a shot hits the target based on hit rates
Input: target (string)
Output: destroyed (boolean)
Algorithm Steps:
1. Set hit_rate based on target type:
    a. Default hit_rate = 0.80
    b. If target is "Mi-8" or "Bunker", hit_rate = 0.50
    c. If target is "T-90" or "T-80U", hit_rate = 0.25
2. Generate a random number between 0 and 1
3. If random number <= hit_rate, set destroyed = True
4. Else, set destroyed = False
5. Return destroyed
"""

import random
def assess_hit(target):

    random.seed(1) #Comment this line out if you want to explore other cases

    hit_rate = .80
    if (target == "Mi-8" or target == "Bunker"):
        hit_rate = 0.5
    elif (target == "T-90" or target == "T-80U"):
        hit_rate = 0.25

    random_roll = random.random()
    if random_roll <= hit_rate:
```

Commented [JH4]: Shouldn't this be an 'and'? Or should we avoid using 'and' or 'or' at all?

Commented [CL4R2]: corrected.

Commented [PL5]: I am having a problem following this logic. I can understand that I would call the provided function if I still have ammo but why would I call it if the target is destroyed. Shouldn't I call it if the target is NOT destroyed???

Commented [HT6]: Add as single line comments

```

        destroyed = True
    else:
        destroyed = False
    return destroyed

'''
engage_target(target, ammo)
Purpose: Fire at a target until it is destroyed or ammo runs out
Input: target (string), ammo (int)
Output:
    if target destroyed: return ammo_remaining (int)
    if target is not destroyed: return -1
Constraints: Use assess_hit(target) to determine if a shot hits
Algorithm Steps:
    1. Initialize target_destroyed
    2. While ammo > 0 and target is not destroyed:
        a. Call assess_hit(target) to see if the shot hits
        b. If hit, set target_destroyed to True
        c. Decrement ammo by 1
    3. If target_destroyed is True, return ammo remaining
    4. If target_destroyed is False, return -1
'''

def engage_target(target, ammo):
    # enter code here
    # remember to include algorithm as in-line comments

```

### Sample Solution:

```

'''
assess_hit(target)
**NOTE: This function is provided and should not be modified!**
Purpose: Determine if a shot hits the target based on hit rates
Input: target (string)
Output: destroyed (boolean)
Algorithm Steps:
    1. Set hit_rate based on target type:
        a. Default hit_rate = 0.80
        b. If target is "Mi-8" or "Bunker", hit_rate = 0.50
        c. If target is "T-90" or "T-80U", hit_rate = 0.25
    2. Generate a random number between 0 and 1
    3. If random number <= hit_rate, set destroyed = True
    4. Else, set destroyed = False
    5. Return destroyed
'''

import random
def assess_hit(target):

    random.seed(1) #Comment this line out if you want to explore other cases

```

```

hit_rate = .80
if (target == "Mi-8" or target == "Bunker"):
    hit_rate = 0.5
elif (target == "T-90" or target == "T-80U"):
    hit_rate = 0.25

random_roll = random.random()
if random_roll <= hit_rate:
    destroyed = True
else:
    destroyed = False
return destroyed
'''

engage_target(target, ammo)
Purpose: Fire at a target until it is destroyed or ammo runs out
Input: target (string), ammo (int)
Output:
    if target destroyed: return ammo_remaining (int)
    if target is not destroyed: return -1
Constraints: Use assess_hit(target) to determine if a shot hits
Algorithm Steps:
    1. Initialize target_destroyed
    2. While ammo > 0 and target is not destroyed:
        a. Call assess_hit(target) to see if the shot hits
        b. If hit, set target_destroyed to True
        c. Decrement ammo by 1
    3. If target_destroyed is True, return ammo remaining
    4. If target_destroyed is False, return -1
'''

def engage_target(target, ammo):
    # 1. Initialize target_destroyed
    target_destroyed = False
    # 2. While ammo > 0 and target is not destroyed:
    while ammo > 0 and not target_destroyed:
        # a. Call assess_hit(target) to see if the shot hits
        if assess_hit(target):
            # b. If hit, set target_destroyed to True
            target_destroyed = True
        # c. Decrement ammo by 1
        ammo -= 1
    # 3. If target_destroyed is True, return ammo remaining
    if target_destroyed == True:
        return ammo
    # 4. If target_destroyed is False, return -1
    else:

```

**Commented [PL7]:** For a sample solution we normally write something like `if target_destroyed == True:`

**Commented [MH7R2]:** ...grouchy noises.... Fixed.

```
return -1
```

#### Test Cases:

1) Test Case 1.

a) INPUTS:

1

2

3

4

2) Test Case Description 2

3) Test Case Description 3

4) Test Case Description 4

5) Etc.

#### Tested Concepts:

- ..

## 2. CODING PROBLEM – WRITE A PROGRAM (5 PTS)

Write a program that uses your function *engage\_target* to simulate a combat scenario for a M1A1 Abrams tank. The amount of ammo will be provided by the user and the target list is provided in the code. The tank will engage each target in sequence and report the results of each target engagement including the target identified, ammo remaining, and the results (target destroyed or disengaged).

Commented [PL8]: Can we shorten the list?

#### Sample Input/Output 1:

Enter initial ammo count: **20**

Engaging target: T-90 with 20 rounds of ammo.

Target T-90 destroyed! Ammo remaining: 18

Engaging target: T-80U with 18 rounds of ammo.

Target T-80U destroyed! Ammo remaining: 16

Engaging target: Mi-8 with 16 rounds of ammo.

Target Mi-8 destroyed! Ammo remaining: 14

Engaging target: Bunker with 14 rounds of ammo.

Target Bunker destroyed! Ammo remaining: 9

#### Provided to Students:

```
#Remember to include all of the previous functions!
```

```
'''
```

```
Problem: Simulate engaging a list of targets with limited ammo
```

```
Input: initial ammo (int)
```

```
Output: Print status of each target engagement
```

```
Algorithm Steps:
```

```

"""
#copy assess_hit() function here

#place engage_target() function here

def main():
    random.seed(15) #Comment this line out if you want to explore other cases
    target_list = ["T-90", "T-80U", "Mi-8", "Bunker"]
    # enter code here
    # remember to include algorithm steps as comments
main()

```

```

"""
Problem: Simulate engaging a list of targets with limited ammo
Input: None
Output: Print status of each target engagement
Algorithm Steps:
1. Prompt user for initial ammo_count
2. Initialize target_list = ["T-90", "T-80U", "Mi-8", "Bunker"]
3. For each target in target_list:
    a. Print "Engaging target: {target} with {ammo_count} rounds of ammo."
    b. Call engage_target(target, ammo_count) and store result in ammo_count
    c. If ammo_remaining >= 0:
        i. Print "Target {target} destroyed! Ammo remaining: {ammo_remaining}"
    d. Else:
        i. Print "Failed to destroy target {target}. Disengage!"
        ii. Set ammo_count to 0
"""

def main():
    random.seed(15) #Comment this line out if you want to explore other cases
    #1. Prompt user for initial ammo_count
    ammo_count = int(input("Enter initial ammo count: "))
    #2. Initialize target_list = ["T-90", "T-80U", "Mi-8", "Bunker"]
    target_list = ["T-90", "T-80U", "Mi-8", "Bunker"]
    #3. For each target in target_list:
    for target in target_list:
        # a. Print "Engaging target: {target} with {ammo_count} rounds of ammo."
        print("Engaging target: "+target+" with "+str(ammo_count)+" rounds of ammo.")
        # b. Call engage_target(target, ammo_count) and store result in ammo_count
        ammo_count = engage_target(target, ammo_count)
        # c. If ammo_remaining >= 0:
        if ammo_count >= 0:
            # i. Print "Target {target} destroyed! Ammo remaining: {ammo_count}"
            print("Target "+str(target)+" destroyed! Ammo remaining: "+str(ammo_count))
        else:
            # i. Print "Failed to destroy target {target}. Disengage!"
            print("Failed to destroy target "+target+". Disengage!")

```

```
# ii. Set ammo count to 0
ammo_count = 0

main()
```

***Test Cases:***

6) Test Case 1.

a) INPUTS:

- 1
- 2
- 3
- 4

7) Test Case Description 2

8) Test Case Description 3

9) Test Case Description 4

10) Etc.

***Tested Concepts:***

- ..



UNITED STATES MILITARY ACADEMY  
**WEST POINT.**

## **TAB P**

## **PROBLEM SET 8**

---



## LESSONS COVERED

- Iteration II (Review)
- Writing Functions (New)

### Part A – Canvas Component: 7 pts

- 2-3x knowledge check/code tracing

### Part B – zyBooks Component: 8 pts

- 1x Code Writing Problem – Write a Function
- 1x Code Writing Problem – use the function you wrote and write main.

Legend: Updates to be made. Question Answers. Feedback Section. Configured in Canvas/zyBooks.

## Canvas Component (7 PTS)

Configured already in Canvas (library module) + 5 1-point questions

### 1. Code writing problem 1 (4 PTS)

Write a function called `is_allowed_char` that takes a character as its only parameter. Your function should return True if the character is a lowercase letter, uppercase letter, or space. Otherwise, it should return False. For simplicity, you may assume the string that is passed into the function is always of length 1.

Sample invocations:

```
>>> is_allowed_char("a")
True
>>> is_allowed_char("2")
False
>>> is_allowed_char(" ")
True
>>> is_allowed_char("!")
False
```

Now, write a new function called `normalize_string` that takes a string as its only parameter. Your function must use the `is_allowed_char` function you created previously to generate and return a version of the provided string that has no special characters. ~~and where all the letters are lowercased.~~

Sample invocations:

```
>>> normalize_string("Hello, friends of the world!")
Hello friends of the world
```

Commented [RD1]: Change this to "must"?

Commented [JH2]: Removed requirement to avoid use of `.lower()`.

```
>>> normalize_string("na-na-na-na-na BATMAN!")
nanananana BATMAN
```

**Commented [LP3]:** The first example looks like the comma becomes a space but now I see that this is not the case as I was then thinking the - in second example should become a space. I am wrong.

#### *Provided to Students:*

```
'''
Place problem header here
'''

# define is_allowed_char here

# define normalize_string here

# DO NOT TOUCH BELOW THIS LINE
def main():
    print(normalize_string("!@#$$%"))
    print(normalize_string("This is a test"))
    print(normalize_string("@'A1B2c3d4/{~E'"))

main()
```

#### *Sample Solution:*

```
'''
Function: is_allowed_char
Parameter: a single character
Output: Returns True if it is an allowable character, False otherwise
'''

def is_allowed_char(c):
    if c == " ": # if it is space
        return True
    if c >= "A" and c <= "Z":
        return True
    if c >= "a" and c <= "z": # check for Letters
        return True
    return False

'''

Function: normalize_string
Parameter: a string
Output: Returns a new string with all the special characters removed
'''

def normalize_sting(string):
    newStr = "" # initialize string
    for char in string: # for every character in the string
        if is_allowed_char(char): # if it is not a special character
```

**Commented [RD4]:** String?

**Commented [JH4R2]:** Yes, I have made a few changes in testing.

```

        newStr += char
    return newStr # return the string

# DO NOT TOUCH BELOW THIS LINE
def main():
    print(normalize_string("!@#%"))
    print(normalize_string("This is a test"))
    print(normalize_string("@'A1B2c3d4/{~E'"))

main()

```

#### Test Cases:

- a) Inputs:
- b) Outputs:
  - i)
  - ii) This is a test
  - iii) ABcdE

#### Tested Concepts:

..

## 2. CODING PROBLEM – SIMILARITY DETECTOR (4 PTS)

Plagiarism detection software commonly uses subsequence detection to figure out if a piece of text is similar to another. A string  $X$  is considered a subsequence of another string  $Y$  if all the letters in  $X$  occur in order in  $Y$  (though the letters need not be contiguous to each other). So "ape" is a subsequence of "apple", but "pal" is not a subsequence of "apple".

Write a main function that asks the user to enter two strings, first a longer string, then a shorter string. You may assume the user will not enter upper case letters. Your main program should then normalize the string using the functions you wrote in problem 1, and then determine if the shorter string is a subsequence of the longer one. If it is, your program should output the percent overlap, rounded to one decimal place. If it is not, it should output "no subsequence detected".

#### Sample Input/Output 1:

Enter longer string: **hello, friends of the world!**  
 Enter shorter string: **hello world**  
 Strings are 42.3% similar

#### Provided to Students:

**Commented [RD5]:** Highlight/underline this or something.

**Commented [RD6]:** ..., even though all letters of "pal" appear in "apple".

**Commented [RD7]:** Is this necessary? It introduces a dependency (i.e., if they couldn't do the first problem, they can't do this one).

**Commented [MS7R2]:** that's the point!

**Commented [AK7R3]:** Recommend changing to "normalize the provided strings using the functions you wrote ..."  
 To align with verbiage used in Q1

**Commented [MS7R4]:** sorry, the verbiage was in flux. I have been going from meeting to meeting. The current verbiage should be correct.

**Commented [RD8]:** The code below assumes that the second string is longer than the first, but the instructions here don't make that clear.

**Commented [JH8R2]:** The way it's written, which is longer doesn't matter. However, that introduces another complexity, so it might be better to tell them which is longer.

**Commented [JH8R3]:** Or maybe the extra complexity is part of the intent.

**Commented [JH8R4]:** Updated to specify order of long/short in the instructions.

```
'''
```

```
Place problem header here
```

```
'''
```

```
# Place your function from problem 1 here.
```

```
# Place your main function here.
```

### *Sample Solution:*

```
'''
```

```
Function: is_allowed_char
```

```
Parameter: a single character
```

```
Output: Returns True if it is an allowable character, False otherwise
```

```
'''
```

```
def is_allowed_char(c):
```

```
    if c == " ": # if it is space
```

```
        return True
```

```
    if c >= "A" and c <= "Z":
```

```
        return True
```

```
    if c >= "a" and c <= "z": # check for Letters
```

```
        return True
```

```
    return False
```

```
'''
```

```
Function: normalize_string
```

```
Parameter: a string
```

```
Output: Returns a new string with all the special characters removed
```

```
'''
```

```
def normalize_string(string):
```

```
    newStr = "" # initialize list
```

```
    for char in string: # for every character in the string
```

```
        if is_allowed_char(char): # if it is not a special character
```

```
            newStr += char
```

```
    return newStr # return the string
```

```
'''
```

```
Function: is_subsequence
```

```
Parameter: two strings, where the first is shorter than the second
```

```
Output: Returns True if the shorter string is a subsequence of the other, False otherwise.
```

```
Note: While it was not required that a separate function was created for this problem, notice how creating a function greatly simplifies the solution
```

```
'''
```

```
def is_subsequence(shorter, longer):
```

```

counter = 0
for char in longer:
    if shorter[counter] == char:
        counter += 1
    if counter == len(shorter):
        return True
return False

```

**Commented [JH9]:** Note that this introduces a bug. If the last character in the short string appears in the long string again after being matched, you get an index out of range. I'll fix it, but it makes the solution a bit tougher.

**Commented [RD10]:** I'd recommend changing this to "return counter == len(shorter)" or at least mentioning that as a possibility. We have too many CS majors (for example) who still do the if <cond> return True else False idiom...

**Commented [JH10R2]:** done

```

"""
Problem: plagiarism detection
Input: two strings
Output: determine if the shorter string is a subsequence of the longer one. If
so, output percent similarity
"""

```

```

def main():
    # get strings from user
    long_string = input("Enter longer string: ")
    short_string = input("Enter shorter string: ")
    norm_long_string = normalize_string(long_string)
    norm_short_string = normalize_string(short_string)
    # compute the lengths of the inputted strings
    len1 = len(norm_long_string)
    len2 = len(norm_short_string)

    # check to see if a substring exists
    substring_detected = is_subsequence(norm_short_string, norm_long_string)
    if substring_detected:
        perc_similarity = round(100 * len2/len1, 1)
        print("strings are " + str(perc_similarity) + "% similar")
    else:
        print("no subsequence detected")

main()

```

### Test Cases:

- 1) Test Case 1
  - a) Inputs:
    - i) this is a test.
    - ii) hit
  - b) Outputs:
    - i) strings are 21.4% similar
- 2) Test Case 2
  - a) Inputs:

- i) hello, friends of the world!
  - ii) hello world
- b) Outputs:
  - i) strings are 42.3% similar
- 3) Test Case 3
  - a) Inputs:
    - i) don't panic.
    - ii) cry
  - b) Outputs:
    - i) no subsequence detected
- 4) Test Case 3
  - a) Inputs:
    - i) this is another test.
    - ii) nat
  - b) Outputs:
    - i) no subsequence detected

***Tested Concepts:***

- Function decomposition
- WPR level problem



UNITED STATES MILITARY ACADEMY  
**WEST POINT.**

**TAB Q**

**PROBLEM SET 9**

---

## LESSONS COVERED

- (Review) AI & Ethics
- List & String Methods

### zyBooks Component: 15 pts

- 1x Code Writing Problem – Write a Function (that uses list/string methods)
- 1x Code Writing Problem – use the function you wrote and write main.
- 1x Free Response

Legend: Updates to be made. Question Answers. Feedback Section. Configured in Canvas/zyBooks.

Scenario: You're serving as an information security officer, and your boss has tasked you with developing a machine learning model to detect potential insider threats by analyzing the communications of your people within your organization's Microsoft 365 instance. You have a dataset including years of Outlook emails, Teams messages, OneDrive files, and so on. The text of these messages and files may include greetings, emotional expressions, file references, and keywords that could indicate risk.

Before training a machine learning model, however, you'll need to prepare the dataset. You'll use multiple stages to clean it up and extract features from each sample. This assignment will take you through two programming tasks that would be part of this preparation.

In the final part of this assignment, you'll discuss the ethical considerations and potential pitfalls of using machine learning for the purpose of flagging individuals in the organization as potential insider threats.

### 1. CODING PROBLEM 1 – Clean and tokenize text (5 PTS)

Write a function `clean_tokenize(dirty_sentence)` that takes one parameter, which is a sentence, *dirty\_sentence*. The function should convert all the letters in *dirty\_sentence* to lowercase before splitting it into a list of words (hereafter called *tokens*). The function should then remove the following "stop words" that are both very common and usually unhelpful to analyze ('a', 'an', 'and', 'at', 'the', 'is', 'for', 'of', 'that', 'this', 'to') and returns the list of remaining tokens.

Your `main()` function should prompt the user for a sentence, call the `clean_tokenize()` function you wrote, and then print the resulting list of tokens.

Example Input/Output

Text: **John, take a look at the attached report, investigation\_findings.pdf, and tell me - doesn't this just make you want to quit too?**



```
['john', 'take', 'look', 'attached', 'report', 'investigation_findings.pdf', 'tell', 'me', '-', "doesn't", 'just', 'make', 'you', 'want', 'quit', 'too?']
```

#### *Provided to Students:*

```
'''
Place problem header here
'''

def clean_tokenize(dirty_sentence):
    #Place code here

def main():
    #Place code here

main()
```

Commented [PL1]: I think this is too long for the number of points. Maybe we already take out the punctuation for them or reduce the amount of code another way

#### *Sample Solution:*

```
# problem obj: clean and tokenize text
# input: string of text
# output: list of tokens, all in lowercase letters, with no stop words

def clean_tokenize(dirty_sentence):
    # convert to lowercase and split the words into a sentence
    tokens = dirty_sentence.lower().split()

    # remove stop words
    stop_words = ['a', 'an', 'and', 'at', 'the', 'is', 'for', 'that', 'this', 'to', 'of']
    filtered_tokens = []
    for word in tokens:
        if word not in stop_words:
            filtered_tokens.append(word)

    # return the cleaned list of tokens
    return filtered_tokens

def main():
    # prompt for the text of a dataset sample
    text = input('Text: ')

    # call the clean_tokenize() function
    tokens = clean_tokenize(text)
```

```
print(tokens)

main()
```

### *Test Cases:*

#### Test Case 1

##### Inputs:

Hello, Matt. What you are proposing is not only illegal it could potentially get both of us fired! V/R, Cal Bondson

##### Outputs:

['hello,', 'matt.', 'what', 'you', 'are', 'proposing', 'not', 'only', 'illegal', 'it', 'could', 'potentially', 'get', 'both', 'us', 'fired!', 'v/r,', 'cal', 'bondson']

#### Test Case 2

##### Inputs:

Janice, We had such a great time with you and Brad! I disagree with you on clowns; my heart about quit when they all jumped out of the car. For me, the best part of the night was when they fired the strong-man out of the cannon!

##### Outputs:

['janice,', 'we', 'had', 'such', 'great', 'time', 'with', 'you', 'brad!', 'i', 'disagree', 'with', 'you', 'on', 'clowns;', 'my', 'heart', 'about', 'quit', 'when', 'they', 'all', 'jumped', 'out', 'car.', 'me,', 'best', 'part', 'night', 'was', 'when', 'they', 'fired', 'strong-man', 'out', 'cannon!']

#### Test Case 3

##### Inputs:

For home use - confidential\_report.pdf; annual\_maintenance\_record.xlsx;  
SECRET\_REL\_FVEY\_OPLAN.pptx

##### Outputs:

['home', 'use', '-', 'confidential\_report.pdf;', 'annual\_maintenance\_record.xlsx;', 'secret\_rel\_fvey\_oplan.pptx']

### *Tested Concepts:*

- List & String Methods

## 2. CODING PROBLEM 2 – EXTRACTING FEATURES (5 PTS)

Write a function `extract_features(sentence)` that takes one parameter, which is a sentence, *sentence*, and returns a list of features found in that string. This list of features for each

sentence would be the actual input to a machine learning model. The list of features will be as follows:

[n\_tokens, suspicious\_found, file\_ref\_count]

Where:

- 'n\_tokens': number of tokens
- 'suspicious\_found': a Boolean that is True if any suspicious keyword is present (look for 'classified', 'illegal', 'disagree', 'fired', and 'quit')
- 'file\_ref\_count': number of file references in the list of tokens (look for 'pdf', 'docx', 'xlsx', and 'pptx')

- Copy all of your code from Problem 1
- Add extract\_features(sentence) as specified above
- Modify your main() function to prompt the user for a sentence
- Print the returned list of features from extract\_features(sentence)

#### Sample **Input/Output** 1:

Text: **John, take a look at the attached report, investigation\_findings.pdf, and tell me - doesn't this just make you want to quit too?**

[16, True, 1]

\*Note: 16 here is the number of tokens, True is stating that suspicious words were found, 1 means that one file reference was found.

#### *Provided to Students:*

```
'''
Place problem header here
'''

# copy clean_tokenize(dirty_sentence) here

# define extract_features(sentence) here

def main():
    # place code here

main()

# problem obj: extract features from text
# input: string of text
# output: list of features
```

Commented [PL2]: Same concern. I think too long

```

def extract_features(sentence):
    # count number of tokens
    tokens_count = len(sentence.split())
    # suspicious keyword detection
    suspicious_keywords = ['classified', 'illegal', 'disagree', 'fired',
'quit']
    file_extensions = ['.pdf', '.docx', '.xlsx', '.pptx']
    file_ref_count = 0
    suspicious_found = False

    for sus in suspicious_keywords:
        if sus in sentence:
            suspicious_found = True

    for ext in file_extensions:
        if ext in sentence:
            file_ref_count += sentence.count(ext)

    # return the feature list
    return [tokens_count, suspicious_found, file_ref_count]

def clean_tokenize(dirty_sentence):
    # convert to lowercase and split the words into a sentence
    tokens = dirty_sentence.lower().split()

    # remove stop words
    stop_words = ['a', 'an', 'and', 'at', 'the', 'is', 'for', 'that', 'this',
'to', 'of']
    filtered_tokens = []
    for word in tokens:
        if word not in stop_words:
            filtered_tokens.append(word)

    # return the cleaned list of tokens
    return filtered_tokens

def main():
    # prompt for the text of a dataset sample
    text = input('Text: ')

    # call the clean_tokenize() function
    tokens = clean_tokenize(text)
    sentence = " ".join(tokens)

```

```
# call the extract_features() function
features = extract_features(sentence)
print(features)

main()
```

#### *Test Cases:*

Test Case 1

Inputs:

Hello, Matt. What you are proposing is not only illegal it could potentially get both of us fired! V/R, Cal Bondson

Outputs:

[19, True, 0]

Test Case 2

Inputs:

Janice, We had such a great time with you and Brad! I disagree with you on clowns; my heart about quit when they all jumped out of the car. For me, the best part of the night was when they fired the strong-man out of the cannon!

Outputs:

[36, True, 0]

Test Case 3

Inputs:

For home use - confidential\_report.pdf; annual\_maintenance\_record.xlsx;  
SECRET\_REL\_FVEY\_OPLAN.pptx

Outputs:

[6, False, 3]

#### *Tested Concepts:*

- List & String Methods

### 3. FREE RESPONSE – ETHICAL REFLECTION (5 PTS)

In no more than 250 words, discuss the ethical considerations and potential pitfalls of using machine learning for the detection of insider threats. Discuss data privacy. Discuss the consequences of false positives (you use the test cases from Part 2 of this assignment as examples if you think they're helpful). Discuss the potential for misuse or abuse of this type of system. How would your answers change if the task was to analyze public posts from your employees' personal social media accounts instead of your organization's Microsoft 365 data?